Final Handsout



Virtual University of Pakistan

Real Analysis II (MTH631)

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2.6 Heine-Borel Theorem

We are going to state and prove the Heine-Borel theorem for $\mathbb{R}^n$.

This theorem concerns *compact* sets. As in $\mathbb{R}$, a compact set in $\mathbb{R}^n$ is a closed and bounded set.

Recall that a collection $\mathcal{H}$ of open sets is an open covering of a set S if

$$S \subset \cup\{H : H \in \mathcal{H}\}.$$

Theorem: If $\mathcal{H}$ is an open covering of a compact subset S, then S can be covered by finitely many sets from $\mathcal{H}$.

Proof: The proof is by contradiction. We first consider the case where $n = 2$, so that you can visualize the method.

Suppose that there is a covering $\mathcal{H}$ for S from which it is impossible to select a finite subcovering.

Since S is bounded, S is contained in a closed square

$$T = \{(x, y) | a_1 \leq x \leq a_1 + L, a_2 \leq x \leq a_2 + L\}$$

with sides of length L

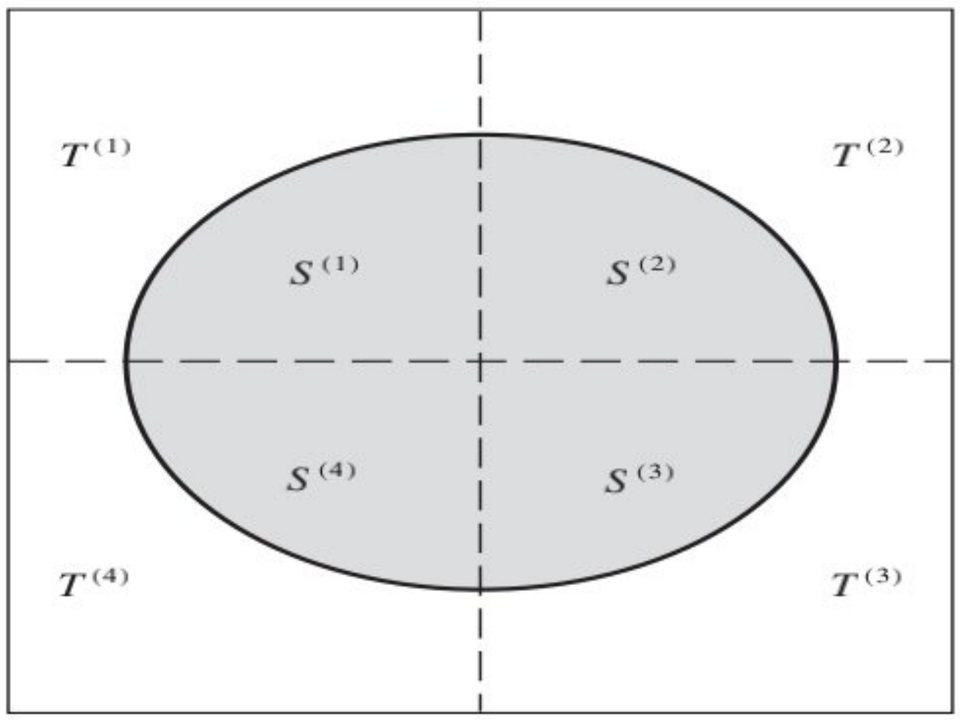


Figure 2.2: Heine-Borel Theorem for $n = 2$

Bisecting the sides of T leads to four closed squares, $T^{(1)}, T^{(2)}$, $T^{(3)}$, and $T^{(4)}$, with sides of length $L/2$. Let

$$S^{(i)} = S \cap T^{(i)}, \quad 1 \leq i \leq 4.$$

Each $S^{(i)}$, being the intersection of closed sets, is closed, and

$$S = \bigcup_{i=1}^{4} S^{(i)}.$$

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Moreover, $\mathcal{H}$ covers each $S^{(i)}$, but at least one $S^{(i)}$ cannot be covered by any finite subcollection of $\mathcal{H}$, since if all the $S^{(i)}$ could be, then so could S. Let S_1 be a set with this property, chosen from $S^{(1)}$, $S^{(2)}$, $S^{(3)}$, and $S^{(4)}$.

We are now back to the situation we started from: a compact set S_1 covered by $\mathcal{H}$, but not by any finite subcollection of $\mathcal{H}$. However, S_1 is contained in a square T_1 with sides of length $L/2$ instead of L. Bisecting the sides of T_1 and repeating the argument, we obtain a subset S_2 of S_1 that has the same properties as S, except that it is contained in a square with sides of length $L/4$. Continuing in this way produces a sequence of nonempty closed sets $S_0\,(=S)$, S_1, S_2, $\ldots$, such that $S_k \supset S_{k+1}$ and $d(S_k) \leq L/2^{k-1/2}\,(k \geq 0)$.

From Principle of Nested Sets Theorem, there is a point $\overline{\mathbf{X}}$ in $\bigcap_{k=1}^{\infty} S_k$.

Since $\overline{\mathbf{X}} \in S$, there is an open set H in $\mathcal{H}$ that contains $\overline{\mathbf{X}}$, and this H must also contain some ε-neighborhood of $\overline{\mathbf{X}}$. Since every $\mathbf{X}$ in S_k satisfies the inequality

$$|\mathbf{X} - \overline{\mathbf{X}}| \leq 2^{-k+1/2} L,$$

it follows that $S_k \subset H$ for k sufficiently large.

This contradicts our assumption on $\mathcal{H}$, which led us to believe that no S_k could be covered by a finite number of sets from $\mathcal{H}$.

Consequently, this assumption must be false: $\mathcal{H}$ must have a finite subcollection that covers S. This completes the proof for $n = 2$.

The idea of the proof is the same for $n > 2$. The counterpart of the square T is the *hypercube* with sides of length L:

$$T = \{(x_1, x_2, \ldots, x_n) : a_i \leq x_i \leq a_i + L, i = 1, 2, \ldots, n\}.$$

Halving the intervals of variation of the n coordinates x_1, x_2, $\ldots$, x_n divides T into 2^n closed hypercubes with sides of length $L/2$:

$$T^{(i)} = \{(x_1, x_2, \ldots, x_n) : b_i \leq x_i \leq b_i + L/2, 1 \leq i \leq n\},$$

where $b_i = a_i$ or $b_i = a_i + L/2$. If no finite subcollection of $\mathcal{H}$ covers S, then at least one of these smaller hypercubes must contain a subset of S that is not covered by any finite subcollection of S. Now the proof proceeds as for $n = 2$.

Remark: The Bolzano–Weierstrass theorem is valid in $\mathbb{R}^n$; its proof is the same as in $\mathbb{R}$.

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2.7 Connected Sets in $\mathbb{R}^n$

A subset S of $\mathbb{R}^n$ is *connected* if it is impossible to represent S as the union of two disjoint nonempty sets such that neither contains a limit point of the other.

If S cannot be expressed as $S = A \cup B$, where

$$A \neq \emptyset, \quad B \neq \emptyset, \quad \overline{A} \cap B = \emptyset, \quad \text{and} \quad A \cap \overline{B} = \emptyset. \tag{2.13}$$

If S can be expressed in this way, then S is *disconnected*.

Example: The empty set and singleton sets are connected, because they cannot be represented as the union of two disjoint nonempty sets.

Example: The space $\mathbb{R}^n$ is connected.

If $\mathbb{R}^n = A \cup B$ with $\overline{A} \cap B = \emptyset$ and $A \cap \overline{B} = \emptyset$, then $\overline{A} \subset A$ and $\overline{B} \subset B$.

That is, A and B are both closed and therefore are both open.

Since the only nonempty subset of $\mathbb{R}^n$ that is both open and closed is $\mathbb{R}^n$ itself, one of A and B is $\mathbb{R}^n$ and the other is empty.

2.7.1 Polygonal Path

If $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_k$ are points in $\mathbb{R}^n$

Let L_i is the line segment from $\mathbf{X}_i$ to $\mathbf{X}_{i+1}$, $1 \leq i \leq k-1$, we say that L_1, L_2, $\ldots$, L_{k-1} form a *polygonal path* from $\mathbf{X}_1$ to $\mathbf{X}_k$.

We say that $\mathbf{X}_1$ and $\mathbf{X}_k$ are *connected* by the polygonal path.

2.8 Polygonally Connected Set

A set S is *polygonally connected* if every pair of points in S can be connected by a polygonal path lying entirely in S.

Theorem: An open set S in $\mathbb{R}^n$ is connected if and only if it is polygonally connected.

Proof: For sufficiency, we will show that if S is disconnected, then S is not polygonally connected.

Let $S = A \cup B$, where A and B satisfy

$$A \neq \emptyset, \quad B \neq \emptyset, \quad \overline{A} \cap B = \emptyset, \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

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.

Suppose that $\mathbf{X}_1 \in A$ and $\mathbf{X}_2 \in B$, and assume that there is a polygonal path in S connecting $\mathbf{X}_1$ to $\mathbf{X}_2$. Then some line segment L in this path must contain a point $\mathbf{Y}_1$ in A and a point $\mathbf{Y}_2$ in B.

The line segment

$$\mathbf{X} = t\mathbf{Y}_2 + (1-t)\mathbf{Y}_1, \quad 0 \le t \le 1,$$

is part of L and therefore in S. Now define

$$\rho = \sup\{\tau : tY_2 + (1-t)\mathbf{Y}_1 \in A,\ 0 \le t \le \tau \le 1\}.$$

Let $\mathbf{X}_\rho = \rho\mathbf{Y}_2 + (1-\rho)\mathbf{Y}_1$. Then $\mathbf{X}_\rho \in \overline{A} \cap \overline{B}$.

However, since $\mathbf{X}_\rho \in A \cup B$ and $\overline{A} \cap B = A \cap \overline{B} = \emptyset$, this is impossible.

Therefore, the assumption that there is a polygonal path in S from $\mathbf{X}_1$ to $\mathbf{X}_2$ must be false.

For necessity, suppose that S is a connected open set and $\mathbf{X}_0 \in S$. Let A be the set consisting of $\mathbf{X}_0$ and the points in S can be connected to $\mathbf{X}_0$ by polygonal paths in S. Let B be set of points in S that cannot be connected to $\mathbf{X}_0$ by polygonal paths. If $\mathbf{Y}_0 \in S$, then S contains an ε-neighborhood $N_\varepsilon(\mathbf{Y}_0)$ of $\mathbf{Y}_0$, since S is open. Any point $\mathbf{Y}_1$ in $N_\varepsilon(\mathbf{Y}_0$ can be connected to $\mathbf{Y}_0$ by the line segment

$$\mathbf{X} = t\mathbf{Y}_1 + (1-t)\mathbf{Y}_0, \quad 0 \le t \le 1,$$

which lies in $N_\varepsilon(\mathbf{Y}_0)$ and therefore in S. This implies that $\mathbf{Y}_0$ can be connected to $\mathbf{X}_0$ by a polygonal path in S if and only if every member of $N_\varepsilon(\mathbf{Y}_0)$ can also. Thus, $N_\varepsilon(\mathbf{Y}_0) \subset A$ if $\mathbf{Y}_0 \in A$, and $N_\varepsilon(\mathbf{Y}_0) \in B$ if $\mathbf{Y}_0 \in B$. Therefore, A and B are open. Since $A \cap B = \emptyset$, this implies that $A \cap \overline{B} = \overline{A} \cap B = \emptyset$. Since A is nonempty ($\mathbf{X}_0 \in A$), it now follows that $B = \emptyset$, since if $B \neq \emptyset$, S would be disconnected. Therefore, $A = S$, which completes the proof of necessity.

Remark: Any polygonally connected set, open or not, is connected. The converse is false. A set (not open) may be connected but not polygonally connected.

Regions in $\mathbb{R}^n$: A *region* S in $\mathbb{R}^n$ is the union of an open connected set with some, all, or none of its boundary; thus, S^0 is connected, and every point of S is a limit point of S^0.

Example: Intervals are the only regions in $\mathbb{R}$. The n-ball $B_r(\mathbf{X}_0)$ is a region in $\mathbb{R}^n$, as is its closure $\overline{S}_r(\mathbf{X}_0)$. The set $S = \{(x,y) : x^2 + y^2 \le 1 \quad \text{or} \quad x^2 + y^2 \ge 4\}$ is not a region in $\mathbb{R}^2$, since it is not connected.

The set S_1 obtained by adding the line segment

$$L_1: \quad \mathbf{X} = t(0,2) + (1-t)(0,1), \quad 0 < t < 1,$$

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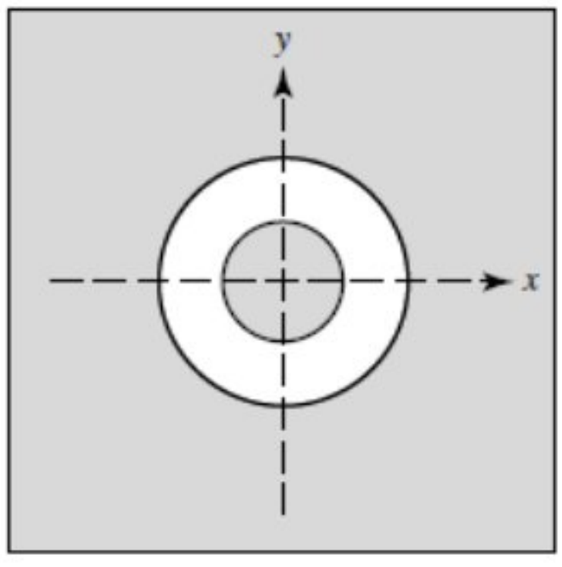


Figure 2.3: Disconnected set which is not a region

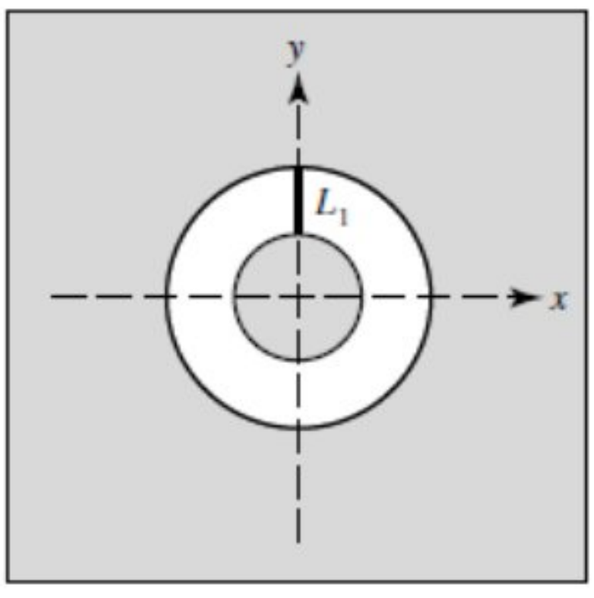


Figure 2.4: A connected set which is not a region

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to S is connected but is not a region, since points on the line segment are not limit points of S_1^0. The set S_2 obtained by adding to S_1 the points in the first quadrant bounded by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$ and the line segments L_1 and

$$L_2: \quad X = t(2,0) + (1-t)(1,0), \quad 0 < t < 1,$$

is a region.

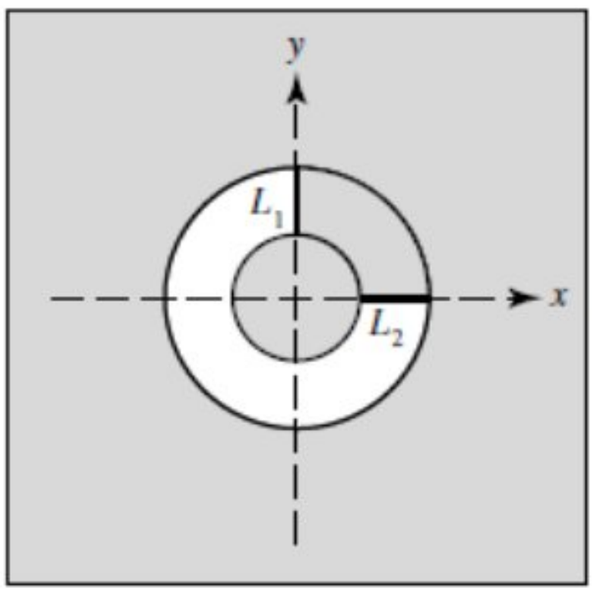


Figure 2.5: A region

2.9 Sequences in $\mathbb{R}^n$

A sequence $\{\mathbf{X}_r\}$ of points in $\mathbb{R}^n$ converges to a limit $\overline{\mathbf{X}}$ if and only if for every $\varepsilon > 0$ there is an integer K such that

$$|\mathbf{X}_r - \overline{\mathbf{X}}| < \varepsilon \quad \text{if} \quad r \geq K.$$

The $\mathbb{R}^n$ definitions of divergence, boundedness, subsequence, and sums, differences, and constant multiples of sequences are analogous to those we discussed in Analysis I.

Since $\mathbb{R}^n$ is not ordered for $n > 1$, monotonicity, limits inferior and superior of sequences in $\mathbb{R}^n$, and divergence to $\pm\infty$ are undefined for $n > 1$.

Products and quotients of members of $\mathbb{R}^n$ are also undefined if $n > 1$.

Several theorems from Analysis I remain valid for sequences in $\mathbb{R}^n$, with proofs unchanged, provided that $|\quad|$ is interpreted as distance in $\mathbb{R}^n$.

1. uniqueness of the limit.

2. Boundedness of a convergent sequence.

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3. Concerning limits of sums, differences, and constant multiples of convergent sequences.

4. Every subsequence of a convergent sequence converges to the limit of the sequence.

2.10 Domain of Function of n Variable

We denote the domain of a function f by D_f and the value of f at a point $\mathbf{X} = (x_1, x_2, \ldots, x_n)$ by $f(\mathbf{X})$ or $f(x_1, x_2, \ldots, x_n)$.

If a function is defined by a formula such as

$$f(\mathbf{X}) = \left(1 - x_1^2 - x_2^2 - \cdots - x_n^2\right)^{1/2} \tag{2.14}$$

$$g(\mathbf{X}) = \left(1 - x_1^2 - x_2^2 - \cdots - x_n^2\right)^{-1} \tag{2.15}$$

without specification of its domain, it is to be understood that its domain is the largest subset of $\mathbb{R}^n$ for which the formula defines a unique real number.

2.11 Limit at a Point of a Function of n Variables

A function $f(\mathbf{X})$ *approaches the limit* L *as* $\mathbf{X}$ *approaches* $\mathbf{X}_0$ and write

$$\lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X}) = L,$$

if $\mathbf{X}_0$ is a limit point of D_f and, for every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$|f(\mathbf{X}) - L| < \varepsilon$$

for all $\mathbf{X}$ in D_f such that

$$0 < |\mathbf{X} - \mathbf{X}_0| < \delta.$$

Example: If $g(x, y) = 1 - x^2 - 2y^2$, then

$$\lim_{(x,y)\to(x_0,y_0)} g(x, y) = 1 - x_0^2 - 2y_0^2 \tag{2.16}$$

for every (x_0, y_0).

To see this, we write

$$\begin{aligned} |g(x, y) - g(x_0 - y_0)| &= |(1 - x^2 - 2y^2) - (1 - x_0^2 - 2y_0^2)| \\ &\leq |x^2 - x_0^2| + 2|y^2 - y_0^2| \\ &= |(x + x_0)(x - x_0)| \\ &\quad +2|(y + y_0)(y - y_0)| \\ &\leq |\mathbf{X} - \mathbf{X}_0|(|x + x_0| + 2|y + y_0)|), \end{aligned} \tag{2.17}$$

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since

$$|x - x_0| \le |\mathbf{X} - \mathbf{X}_0| \quad \text{and} \quad |y - y_0| \le |\mathbf{X} - \mathbf{X}_0|.$$

If $|\mathbf{X} - \mathbf{X}_0| < 1$, then $|x| < |x_0| + 1$ and $|y| < |y_0| + 1$.

This and (2.17) imply that

$$|g(x, y) - g(x_0 - y_0)| < K|\mathbf{X} - \mathbf{X}_0| \quad \text{if} \quad |\mathbf{X} - \mathbf{X}_0| < 1,$$

where

$$K = (2|x_0| + 1) + 2(2|y_0| + 1).$$

Therefore, if $\varepsilon > 0$ and

$$|\mathbf{X} - \mathbf{X}_0| < \delta = \min\{1, \varepsilon/K\},$$

then

$$\left|g(x, y) - (1 - x_0^2 - 2y_0^2)\right| < \varepsilon.$$

Example: The function

$$h(x, y) = \frac{\sin\sqrt{1 - x^2 - 2y^2}}{\sqrt{1 - x^2 - 2y^2}}$$

is defined only on the interior of the region bounded by the ellipse

$$x^2 + 2y^2 = 1.$$

It is not defined at any point of the ellipse itself or on any deleted neighborhood of

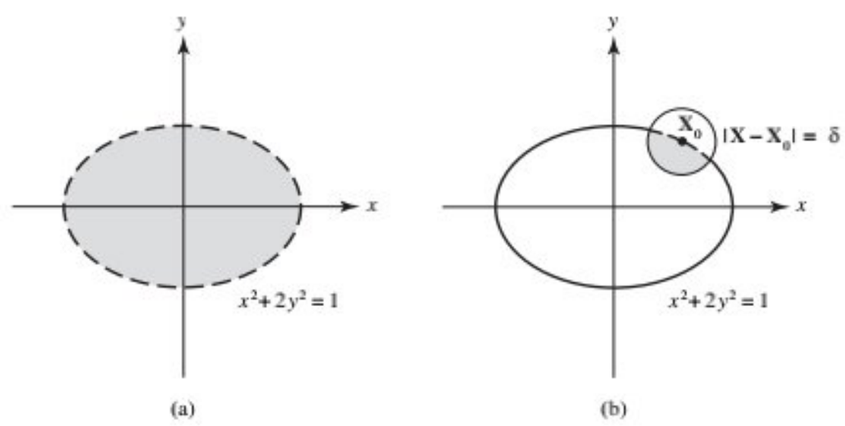


Figure 2.6: Domain of the function

such a point. Nevertheless,

$$\lim_{(x,y)\to(x_0,y_0)} h(x, y) = 1 \tag{2.18}$$

if

$$x_0^2 + 2y_0^2 = 1. \tag{2.19}$$

To see this, let

$$u(x, y) = \sqrt{1 - x^2 - 2y^2}.$$

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Then

$$h(x,y) = \frac{\sin u(x,y)}{u(x,y)}. \tag{2.20}$$

Recall that

$$\lim_{r\to 0} \frac{\sin r}{r} = 1.$$

Therefore, if $\varepsilon > 0$, there is a $\delta_1 > 0$ such that

$$\left|\frac{\sin u}{u} - 1\right| < \varepsilon \quad \text{if} \quad 0 < |u| < \delta_1. \tag{2.21}$$

From previous example, we have

$$\lim_{(x,y)\to(x_0,y_0)} (1 - x^2 - 2y^2) = 0.$$

If (2.19) holds, so there is a $\delta > 0$ such that

$$0 < u^2(x,y) = (1 - x^2 - 2y^2) < \delta_1^2.$$

if $\mathbf{X} = (x,y)$ is in the interior of the ellipse and $|\mathbf{X} - \mathbf{X}_0| < \delta$; that is, if $\mathbf{X}$ is in the shaded region.

Therefore,

$$0 < u = \sqrt{1 - x^2 - 2y^2} < \delta_1 \tag{2.22}$$

if $\mathbf{X}$ is in the interior of the ellipse and $|\mathbf{X} - \mathbf{X}_0| < \delta$; that is, if $\mathbf{X}$ is in the shaded region. This, (2.20), and (2.21) imply that

$$|h(x,y) - 1| < \varepsilon$$

for such $\mathbf{X}$, which is the required result.

Theorem: If $\lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X})$ exists, then it is unique.

Proof: See lecture.

Example: The function

$$f(x,y) = \frac{xy}{x^2 + y^2}$$

is defined everywhere in $\mathbb{R}^2$ except at $(0,0)$. Does $\lim_{(x,y)\to(0,0)} f(x,y)$ exist?

If we try to answer this question by letting (x,y) approach $(0,0)$ along the line $y = x$, we see the functional values

$$f(x,x) = \frac{x^2}{2x^2} = \frac{1}{2}$$

and conclude that the limit is $1/2$.

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However, if we let (x, y) approach $(0, 0)$ along the line $y = -x$, we see the functional values

$$f(x, -x) = -\frac{x^2}{2x^2} = -\frac{1}{2}$$

and conclude that the limit equals $-1/2$.

In fact, they are both incorrect. What we have shown is that

$$\lim_{x\to 0} f(x, x) = \frac{1}{2} \quad \text{and} \quad \lim_{x\to 0} f(x, -x) = -\frac{1}{2}.$$

Since $\lim_{x\to 0} f(x, x)$ and $\lim_{x\to 0} f(x, -x)$ must both equal $\lim_{(x,y)\to(0,0)} f(x, y)$.

Theorem: Suppose that f and g are defined on a set D, $\mathbf{X}_0$ is a limit point of D, and

$$\lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X}) = L_1, \quad \lim_{\mathbf{X}\to\mathbf{X}_0} g(\mathbf{X}) = L_2.$$

Then

$$\lim_{\mathbf{X}\to\mathbf{X}_0} (f + g)(\mathbf{X}) = L_1 + L_2, \tag{2.23}$$

$$\lim_{\mathbf{X}\to\mathbf{X}_0} (f - g)(\mathbf{X}) = L_1 - L_2, \tag{2.24}$$

$$\lim_{\mathbf{X}\to\mathbf{X}_0} (fg)(\mathbf{X}) = L_1 L_2, \tag{2.25}$$

$$if\ L_2 \neq 0,$$

$$\lim_{\mathbf{X}\to\mathbf{X}_0} \left(\frac{f}{g}\right)(\mathbf{X}) = \frac{L_1}{L_2}. \tag{2.26}$$

2.12 Infinite Limits and Limits at X → ∞

We say that $f(\mathbf{X})$ *approaches* ∞ *as* $\mathbf{X}$ *approaches* $\mathbf{X}_0$

$$\lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X}) = \infty$$

if $\mathbf{X}_0$ is a limit point of D_f and, M, there is a $\delta > 0$ such that

$$f(\mathbf{X}) > M \quad \text{whenever} \quad 0 < |\mathbf{X} - \mathbf{X}_0| < \delta \quad \text{and} \quad \mathbf{X} \in D_f.$$

We say that

$$\lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X}) = -\infty$$

$$if$$

$$\lim_{\mathbf{X}\to\mathbf{X}_0} (-f)(\mathbf{X}) = \infty.$$

Example: If

$$f(\mathbf{X}) = (1 - x_1^2 - x_2^2 - \cdots - x_n^2)^{-1/2},$$

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then

$$\lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X}) = \infty$$

if $|\mathbf{X}_0| = 1$, because

$$f(\mathbf{X}) = \frac{1}{|\mathbf{X} - \mathbf{X}_0|},$$

so

$$f(\mathbf{X}) > M \quad \text{if} \quad 0 < |\mathbf{X} - \mathbf{X}_0| < \delta = \frac{1}{M}.$$

Example: If

$$f(x, y) = \frac{1}{x + 2y + 1},$$

then $\lim_{(x,y)\to(1,-1)} f(x, y)$ does not exist (why not?).
But

$$\lim_{(x,y)\to(1,-1)} |f(x, y)| = \infty.$$

To see this, we observe that

$$\begin{aligned} |x + 2y + 1| &= |(x - 1) + 2(y + 1)| \\ &\leq \sqrt{5}|\mathbf{X} - \mathbf{X}_0| \quad \text{(by Schwarz's inequality)}, \end{aligned}$$

where $\mathbf{X}_0 = (1, -1)$. So

$$|f(x, y)| = \frac{1}{|x + 2y + 1|} \geq \frac{1}{\sqrt{5}|\mathbf{X} - \mathbf{X}_0|}.$$

Therefore,

$$|f(x, y)| > M \quad \text{if} \quad 0 < |\mathbf{X} - \mathbf{X}_0| < \frac{1}{M\sqrt{5}}.$$

Example: The function

$$f(x, y, z) = \frac{\left|\sin\left(\frac{1}{x^2+y^2+z^2}\right)\right|}{x^2 + y^2 + z^2}$$

assumes arbitrarily large values in every neighborhood of $(0, 0, 0)$.
For example, if $\mathbf{X}_k = (x_k, y_k, z_k)$, where

$$x_k = y_k = z_k = \frac{1}{\sqrt{3\left(k + \frac{1}{2}\right)\pi}},$$

then

$$f(\mathbf{X}_k) = \left(k + \frac{1}{2}\right)\pi.$$

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However, this does not imply that $\lim_{\mathbf{X}\to\mathbf{0}} f(\mathbf{X}) = \infty$. Since, for example, every neighborhood of $(0,0,0)$ also contains points

$$\overline{\mathbf{X}}_k = \left(\frac{1}{\sqrt{3k\pi}}, \frac{1}{\sqrt{3k\pi}}, \frac{1}{\sqrt{3k\pi}}\right).$$

For which $f(\overline{\mathbf{X}_k}) = 0$.

2.12.1 Limit at Infinity

If D_f is unbounded, we say that

$$\lim_{|\mathbf{X}|\to\infty} f(\mathbf{X}) = L \quad \text{(finite)}$$

if for every $\varepsilon > 0$, there is a number R such that

$$|f(\mathbf{X}) - L| < \varepsilon \quad \text{whenever} \quad |\mathbf{X}| \geq R \quad \text{and} \quad \mathbf{X} \in D_f.$$

Example: If

$$f(x,y,z) = \cos\left(\frac{1}{x^2+2y^2+z^2}\right),$$

then

$$\lim_{|\mathbf{X}|\to\infty} f(\mathbf{X}) = 1. \tag{2.27}$$

To see this, we recall that the continuity of $\cos u$ at $u = 0$ implies that for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|\cos u - 1| < \varepsilon \quad \text{if} \quad |u| < \delta.$$

Since

$$\frac{1}{x^2+2y^2+z^2} \leq \frac{1}{|\mathbf{X}|^2}.$$

It follows that if $|\mathbf{X}| > 1/\sqrt{\delta}$, then

$$\frac{1}{x^2+2y^2+z^2} < \delta.$$

Therefore,

$$|f(\mathbf{X}) - 1| < \varepsilon.$$

Example: Consider the function defined *only* on the domain

$$D = \{(x,y) : 0 < y \leq ax\}, \quad 0 < a < 1,$$

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by

$$f(x,y) = \frac{1}{x-y}.$$

We will show that

$$\lim_{|\mathbf{X}|\to\infty} f(x,y) = 0. \tag{2.28}$$

It is important to keep in mind that we need only consider (x,y) in D, since f is not defined elsewhere.

In D,

$$x - y \geq x(1-a) \tag{2.29}$$

and

$$|\mathbf{X}|^2 = x^2 + y^2 \leq x^2(1+a^2).$$

So

$$x \geq \frac{|\mathbf{X}|}{\sqrt{1+a^2}}.$$

This and (2.29) imply that

$$x - y \geq \frac{1-a}{\sqrt{1+a^2}}|\mathbf{X}|, \quad \mathbf{X} \in D.$$

So

$$|f(x,y)| \leq \frac{\sqrt{1+a^2}}{1-a}\frac{1}{|\mathbf{X}|}, \quad \mathbf{X} \in D.$$

This and (2.29) imply that

$$x - y \geq \frac{1-a}{\sqrt{1+a^2}}|\mathbf{X}|, \quad \mathbf{X} \in D.$$

So

$$|f(x,y)| \leq \frac{\sqrt{1+a^2}}{1-a}\frac{1}{|\mathbf{X}|}, \quad \mathbf{X} \in D.$$

Therefore,

$$|f(x,y)| < \varepsilon$$

if $\mathbf{X} \in D$ and

$$|\mathbf{X}| > \frac{\sqrt{1+a^2}}{1-a}\frac{1}{\varepsilon}.$$

Remarks: In the same manner we can define $\lim_{|\mathbf{X}|\to\infty} f(\mathbf{X}) = \infty$ and $\lim_{|\mathbf{X}|\to\infty} f(\mathbf{X}) = -\infty$. We will have the following notion $\lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X})$ exists means that $\lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X}) = L$, where L is finite; to leave open the possibility that $L = \pm\infty$.

We will say that $\lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X})$ exists in the extended reals. A similar convention applies to limits as $|\mathbf{X}| \to \infty$.

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2.13 Continuity

If $\mathbf{X}_0$ is in D_f and is a limit point of D_f, then we say that f is *continuous at* $\mathbf{X}_0$ if

$$\lim_{\mathbf{X}\to\mathbf{X}_0} f(\mathbf{X}) = f(\mathbf{X}_0).$$

Theorem: Suppose that $\mathbf{X}_0$ is in D_f and is a limit point of D_f. Then f is continuous at $\mathbf{X}_0$ if and only if for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(\mathbf{X}) - f(\mathbf{X}_0)| < \varepsilon$$

whenever

$$|\mathbf{X} - \mathbf{X}_0| < \delta \quad \text{and} \quad \mathbf{X} \in D_f.$$

Example: The function

$$f(x, y) = 1 - x^2 - 2y^2$$

is continuous on $\mathbb{R}^2$.

Solution: See lecture.

Example: Consider the function

$$h(x,y) = \begin{cases} \frac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}, & x^2+2y^2 < 1, \\ 1, & x^2+2y^2 = 1, \end{cases}$$

then it follows from the example we have discussed that h is continuous on the ellipse

$$x^2 + 2y^2 = 1.$$

Example: Can we redefine the function

$$f(x, y) = \frac{xy}{x^2 + y^2},$$

to make it continuous at $(0, 0)$.

The limit

$$\lim_{(x,y)\to(0,0)} f(x, y)$$

does not exist.

Consequently, it is impossible to define the function at origin to make it continuous.

Theorem: If f and g are continuous on a set S in $\mathbb{R}^n$, then so are $f + g$, $f - g$, and fg. Also, f/g is continuous at each $\mathbf{X}_0$ in S such that $g(\mathbf{X}_0) \neq 0$.

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2.14 Vector Valued Functions

Suppose that $g_1, g_2, \ldots, g_n$ are real-valued functions defined on a subset T of $\mathbb{R}^m$.

We define the *vector-valued function* $\mathbf{G}$ on T by

$$\mathbf{G}(\mathbf{U}) = (g_1(\mathbf{U}), g_2(\mathbf{U}), \ldots, g_n(\mathbf{U})), \quad \mathbf{U} \in T.$$

Then $g_1, g_2, \ldots, g_n$ are the *component functions* of $\mathbf{G} = (g_1, g_2, \ldots, g_n)$. We say that

$$\lim_{\mathbf{U} \to \mathbf{U}_0} \mathbf{G}(\mathbf{U}) = \mathbf{L} = (L_1, L_2, \ldots, L_n)$$

if

$$\lim_{\mathbf{U} \to \mathbf{U}_0} g_i(\mathbf{U}) = L_i, \quad 1 \leq i \leq n,$$

and that $\mathbf{G}$ is *continuous* at $\mathbf{U}_0$ if $g_1, g_2, \ldots, g_n$ are each continuous at $\mathbf{U}_0$.

Theorem: For a vector-valued function $\mathbf{G}$,

$$\lim_{\mathbf{U} \to \mathbf{U}_0} \mathbf{G}(\mathbf{U}) = \mathbf{L}$$

if and only if for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|\mathbf{G}(\mathbf{U}) - \mathbf{L}| < \varepsilon \quad \text{whenever} \quad 0 < |\mathbf{U} - \mathbf{U}_0| < \delta \quad \text{and} \quad \mathbf{U} \in D_{\mathbf{G}}.$$

Similarly, $\mathbf{G}$ is continuous at $\mathbf{U}_0$ if and only if for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|\mathbf{G}(\mathbf{U}) - \mathbf{G}(\mathbf{U}_0)| < \varepsilon \quad \text{whenever} \quad |\mathbf{U} - \mathbf{U}_0| < \delta \quad \text{and} \quad \mathbf{U} \in D_{\mathbf{G}}.$$

2.14.1 Composite Function

Let f be a real-valued function defined on a subset of $\mathbb{R}^n$, and let the vector-valued function $\mathbf{G} = (g_1, g_2, \ldots, g_n)$ be defined on a domain $D_{\mathbf{G}}$ in $\mathbb{R}^m$.

Let the set

$$T = \{\mathbf{U} : \mathbf{U} \in D_{\mathbf{G}} \quad \text{and} \quad \mathbf{G}(\mathbf{U}) \in D_f\},$$

be nonempty.

Composite function: Define the real-valued composite function

$$h = f \circ \mathbf{G}$$

on T by

$$h(\mathbf{U}) = f(\mathbf{G}(\mathbf{U})), \quad \mathbf{U} \in T.$$

$$T = \{\mathbf{U} : \mathbf{U} \in D_{\mathbf{G}} \quad \text{and} \quad \mathbf{G}(\mathbf{U}) \in D_f\},$$

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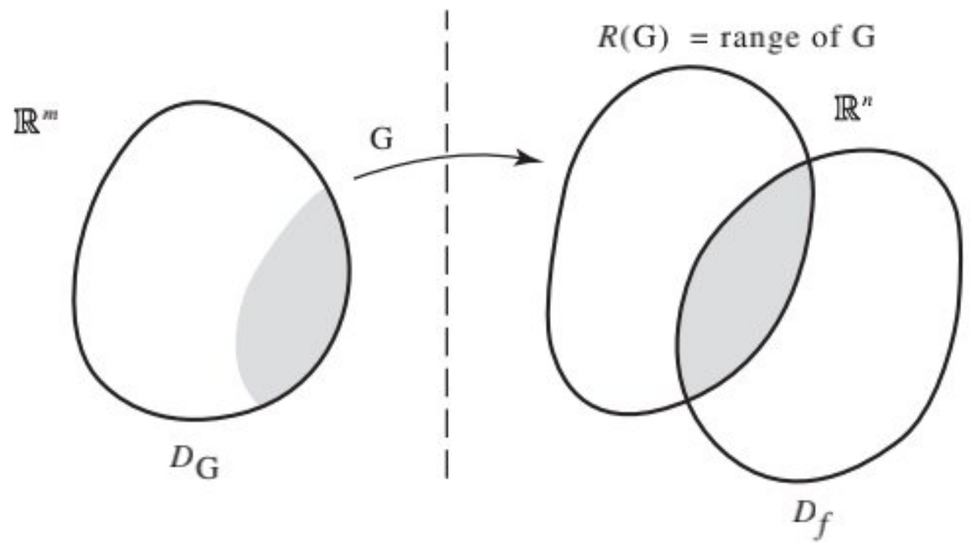


Figure 2.7: Composite of vector valued functions

Theorem: Suppose that $\mathbf{U}_0$ is in T and is a limit point of T, $\mathbf{G}$ is continuous at $\mathbf{U}_0$, and f is continuous at $\mathbf{X}_0 = \mathbf{G}(\mathbf{U}_0)$. Then $h = f \circ \mathbf{G}$ is continuous at $\mathbf{U}_0$.

Proof: Suppose that $\varepsilon > 0$. Since f is continuous at $\mathbf{X}_0 = \mathbf{G}(\mathbf{U}_0)$, there is an $\varepsilon_1 > 0$ such that

$$|f(\mathbf{X}) - f(\mathbf{G}(\mathbf{U}_0))| < \varepsilon \tag{2.30}$$

if

$$|\mathbf{X} - \mathbf{G}(\mathbf{U}_0)| < \varepsilon_1 \quad \text{and} \quad \mathbf{X} \in D_f. \tag{2.31}$$

Since $\mathbf{G}$ is continuous at $\mathbf{U}_0$, there is a $\delta > 0$ such that

$$|\mathbf{G}(\mathbf{U}) - \mathbf{G}(\mathbf{U}_0)| < \varepsilon_1 \quad \text{if} \quad |\mathbf{U} - \mathbf{U}_0| < \delta \quad \text{and} \quad \mathbf{U} \in D_{\mathbf{G}}.$$

By taking $\mathbf{X} = \mathbf{G}(\mathbf{U})$ in (2.30) and (2.31), we see that

$$|h(\mathbf{U}) - h(\mathbf{U}_0)| = |f(\mathbf{G}(\mathbf{U}) - f(\mathbf{G}(\mathbf{U}_0))| < \varepsilon$$

if

$$|\mathbf{U} - \mathbf{U}_0| < \delta \quad \text{and} \quad \mathbf{U} \in T.$$

Example: If

$$f(s) = \sqrt{s}$$

and

$$g(x, y) = 1 - x^2 - 2y^2,$$

then $D_f = [0, \infty]$, $D_g = \mathbb{R}^2$, and

$$T = \{(x, y) : x^2 + 2y^2 \leq 1\}.$$

We have proved that g is continuous on $\mathbb{R}^2$.

We can obtain the same conclusion by observing that the functions $p_1(x, y) = x$ and $p_2(x, y) = y$ are continuous on $\mathbb{R}^2$.

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Theorem: Suppose that $\mathbf{U}_0$ is in T and is a limit point of T, $\mathbf{G}$ is continuous at $\mathbf{U}_0$, and f is continuous at $\mathbf{X}_0 = \mathbf{G}(\mathbf{U}_0)$.

Then $h = f \circ \mathbf{G}$ is continuous at $\mathbf{U}_0$. Since f is continuous on D_f, the function

$$h(x, y) = f\left(g(x, y)\right) = \sqrt{1 - x^2 - 2y^2}$$

is continuous on T.

Example: If

$$g(x, y) = \sqrt{1 - x^2 - 2y^2}$$

and

$$f(s) = \begin{cases} \frac{\sin s}{s}, & s \neq 0, \\ 1, & s = 0, \end{cases}$$

then $D_f = (-\infty, \infty)$ and

$$D_g = T = \{(x, y) : x^2 + 2y^2 \leq 1\}.$$

We have proved that g is continuous on T. Since f is continuous on D_f, the composite function $h = f \circ g$ defined by

$$h(x, y) = \begin{cases} \frac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}, & x^2 + 2y^2 < 1, \\ 1, & x^2 + 2y^2 = 1, \end{cases}$$

is continuous on

$$D_g = T = \{(x, y) : x^2 + 2y^2 \leq 1\}.$$

2.15 Bounded Functions

The definitions of *bounded above, bounded below*, and *bounded* on a set S are the same for functions of n variables as for functions of one variable, as are the definitions of *supremum* and *infimum* of a function on a set S.

Theorem: If f is continuous on a compact set S in $\mathbb{R}^n$, then f is bounded on S.

Theorem: Let f be continuous on a compact set S in $\mathbb{R}^n$ and

$$\alpha = \inf_{\mathbf{X} \in S} f(\mathbf{X}), \quad \beta = \sup_{\mathbf{X} \in S} f(\mathbf{X}).$$

Then

$$f(\mathbf{X}_1) = \alpha \quad \text{and} \quad f(\mathbf{X}_2) = \beta$$

for some $\mathbf{X}_1$ and $\mathbf{X}_2$ in S.

Proof: See lecture.

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Theorem: Let f be continuous on a region S in $\mathbb{R}^n$.
Suppose that $\mathbf{A}$ and $\mathbf{B}$ are in S and

$$f(\mathbf{A}) < u < f(\mathbf{B}).$$

Then $f(\mathbf{C}) = u$ for some $\mathbf{C}$ in S.

Proof: If there is no such $\mathbf{C}$, then $S = R \cup T$, where

$$\begin{aligned} R &= \{\mathbf{X} : \mathbf{X} \in S \text{ and } f(\mathbf{X}) < u\} \\ T &= \{\mathbf{X} : \mathbf{X} \in S \text{ and } f(\mathbf{X}) > u\}. \end{aligned}$$

If $\mathbf{X}_0 \in R$, the continuity of f implies that there is a $\delta > 0$ such that

$$f(\mathbf{X}) < u \quad \text{if} \quad |\mathbf{X} - \mathbf{X}_0| < \delta$$

and $\mathbf{X} \in S$.

This means that $\mathbf{X}_0 \notin \overline{T}$. Therefore, $R \cap \overline{T} = \emptyset$. Similarly, $\overline{R} \cap T = \emptyset$. Therefore, S is disconnected, which contradicts the assumption that S is a region. Hence, we conclude that $f(\mathbf{C}) = u$ for some $\mathbf{C}$ in S.

Theorem: A function f is uniformly continuous on a subset S of its domain in $\mathbb{R}^n$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(\mathbf{X}) - f(\mathbf{X}')| < \varepsilon$$

whenever

$$|\mathbf{X} - \mathbf{X}'| < \delta$$

and $\mathbf{X}, \mathbf{X}' \in S$.

Remark: We emphasize that δ must depend only on ε and S, and not on the particular points $\mathbf{X}$ and $\mathbf{X}'$.

Theorem: If f is continuous on a compact set S in $\mathbb{R}^n$, then f is uniformly continuous on S.

Proof: See lecture.

2.16 Directional Derivative

Let $\mathbf{\Phi}$ be a unit vector and $\mathbf{X}$ a point in $\mathbb{R}^n$.

The directional derivative of f at $\mathbf{X}$ in the direction of $\mathbf{\Phi}$ is defined by

$$\frac{\partial f(\mathbf{X})}{\partial \mathbf{\Phi}} = \lim_{t \to 0} \frac{f(\mathbf{X} + t\mathbf{\Phi}) - f(\mathbf{X})}{t}$$

if the limit exists.

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That is, $\partial f(\mathbf{X})/\partial \mathbf{\Phi}$ is the ordinary derivative of the function

$$h(t) = f(\mathbf{X} + t\mathbf{\Phi})$$

at $t = 0$, if $h'(0)$ exists.

Example: Let $\mathbf{\Phi} = (\phi_1, \phi_2, \phi_3)$ and

$$f(x, y, z) = 3xyz + 2x^2 + z^2.$$

Then

$$\begin{aligned} h(t) &= f(x + t\phi_1, y + t\phi_2, z + t\phi_3), \\ &= 3(x + t\phi_1)(y + t\phi_2)(z + t\phi_3) + 2(x + t\phi_1)^2 \\ &\quad +(z + t\phi_3)^2. \end{aligned}$$

$$h(t) = 3(x + t\phi_1)(y + t\phi_2)(z + t\phi_3) + 2(x + t\phi_1)^2 + (z + t\phi_3)^2$$

Then we have

$$\begin{aligned} h'(t) &= 3\phi_1(y + t\phi_2)(z + t\phi_3) + 3\phi_2(x + t\phi_1)(z + t\phi_3) \\ &\quad + 3\phi_3(x + t\phi_1)(y + t\phi_2) + 4\phi_1(x + t\phi_1) \\ &\quad +2\phi_3(z + t\phi_3). \end{aligned}$$

Therefore,

$$\frac{\partial f(\mathbf{X})}{\partial \mathbf{\Phi}} = h'(0) = (3yz + 4x)\phi_1 + 3xz\phi_2 + (3xy + 2z)\phi_3.$$

2.16.1 Partial Derivative

Consider the unit vectors

$$\mathbf{E}_1 = (1, 0, \ldots, 0), \quad \mathbf{E}_2 = (0, 1, 0, \ldots, 0), \ldots, \quad \mathbf{E}_n = (0, \ldots, 0, 1).$$

Since $\mathbf{X}$ and $\mathbf{X} + t\mathbf{E}_i$ differ only in the ith coordinate, $\partial f(\mathbf{X})/\partial \mathbf{E}_i$ is called the *partial derivative of f with respect to x_i at $\mathbf{X}$*.

It is also denoted by $\partial f(\mathbf{X})/\partial x_i$ or $f_{x_i}(\mathbf{X})$; thus,

$$\frac{\partial f(\mathbf{X})}{\partial x_1} = f_{x_1}(\mathbf{X}) = \lim_{t \to 0} \frac{f(x_1 + t, x_2, \ldots, x_n) - f(x_1, x_2, \ldots, x_n)}{t},$$

$$f_{x_i}(\mathbf{X}) = \lim_{t \to 0} \frac{f(x_1, \ldots, x_{i-1}, x_i + t, x_{i+1}, \ldots, x_n) - f(x_1, x_2, \ldots, x_n)}{t}$$

if $2 \leq i \leq n$, and

$$\frac{\partial f(\mathbf{X})}{\partial x_n} = f_{x_n}(\mathbf{X}) = \lim_{t \to 0} \frac{f(x_1, \ldots, x_{n-1}, x_n + t) - f(x_1, \ldots, x_{n-1}, x_n)}{t},$$

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if the limits exist. If we write $\mathbf{X} = (x, y)$, then we denote the partial derivatives accordingly; thus,

$$\begin{aligned}\frac{\partial f(x,y)}{\partial x} &= f_x(x,y) = \lim_{h\to 0}\frac{f(x+h,y)-f(x,y)}{h}\\ \frac{\partial f(x,y)}{\partial y} &= f_y(x,y) = \lim_{h\to 0}\frac{f(x,y+h)-f(x,y)}{h}.\end{aligned}$$

It can be seen from these definitions that to compute $f_{x_i}(\mathbf{X})$ we simply differentiate f with respect to x_i according to the rules for ordinary differentiation, while treating the other variables as constants.

Example: Let

$$f(x,y,z) = 3xyz + 2x^2 + z^2.$$

Taking $\mathbf{\Phi} = \mathbf{E}_1$ (that is, setting $\phi_1 = 1$ and $\phi_2 = \phi_3 = 0$), we find that

$$\frac{\partial f(\mathbf{X})}{\partial x} = \frac{\partial f(\mathbf{X})}{\partial \mathbf{E}_1} = 3yz + 4x,$$

which is the result obtained by regarding y and z as constants in and taking the ordinary derivative with respect to x. Similarly,

$$\begin{aligned}\frac{\partial f(\mathbf{X})}{\partial y} &= \frac{\partial f(\mathbf{X})}{\partial \mathbf{E}_2} = 3xz\\ \frac{\partial f(\mathbf{X})}{\partial z} &= \frac{\partial f(\mathbf{X})}{\partial \mathbf{E}_3} = 3xy + 2z.\end{aligned}$$

Theorem: If $f_{x_i}(\mathbf{X})$ and $g_{x_i}(\mathbf{X})$ exist, then

$$\begin{aligned}\frac{\partial (f+g)(\mathbf{X})}{\partial x_i} &= f_{x_i}(\mathbf{X}) + g_{x_i}(\mathbf{X}),\\ \frac{\partial (fg)(\mathbf{X})}{\partial x_i} &= f_{x_i}(\mathbf{X})g(\mathbf{X}) + f(\mathbf{X})g_{x_i}(\mathbf{X}),\end{aligned}$$

and, if $g(\mathbf{X}) \neq 0$,

$$\frac{\partial (f/g)(\mathbf{X})}{\partial x_i} = \frac{g(\mathbf{X})f_{x_i}(\mathbf{X}) - f(\mathbf{X})g_{x_i}(\mathbf{X})}{[g(\mathbf{X})]^2}.$$

If $f_{x_i}(\mathbf{X})$ exists at every point of a set D, then it defines a function f_{x_i} on D.

If this function has a partial derivative with respect to x_j on a subset of D, we denote the partial derivative by

$$\frac{\partial}{\partial x_j}\left(\frac{\partial f}{\partial x_i}\right) = \frac{\partial^2 f}{\partial x_j \partial x_i} = f_{x_i x_j}.$$

Similarly,

$$\frac{\partial}{\partial x_k}\left(\frac{\partial^2 f}{\partial x_j \partial x_i}\right) = \frac{\partial^3 f}{\partial x_k \partial x_j \partial x_i} = f_{x_i x_j x_k}.$$

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The function obtained by differentiating f successively with respect to $x_{i_1}, x_{i_2}, \ldots, x_{i_r}$ is denoted by

$$\frac{\partial^r f}{\partial x_{i_r} \partial x_{i_{r-1}} \cdots \partial x_{i1}} = f_{x_{i_1}} \cdots x_{i_{r-1}} x_{i_r};$$

it is an *rth-order partial derivative of* f. The function

$$f(x, y) = 3x^2y^3 + xy$$

has partial derivatives everywhere. Its first-order partial derivatives are

$$f_x(x, y) = 6xy^3 + y, \quad f_y(x, y) = 9x^2y^2 + x.$$

Its second-order partial derivatives are

$$\begin{aligned} f_{xx}(x, y) &= 6y^3, & f_{yy}(x, y) &= 18x^2y, \\ f_{xy}(x, y) &= 18xy^2 + 1, & f_{yx}(x, y) &= 18xy^2 + 1. \end{aligned}$$

There are eight third-order partial derivatives. Some examples are

$$f_{xxy}(x, y) = 18y^2, \quad f_{xyx}(x, y) = 18y^2, \quad f_{yxx}(x, y) = 18y^2.$$

Compute $f_{xx}(0, 0)$, $f_{yy}(0, 0)$, $f_{xy}(0, 0)$, and $f_{yx}(0, 0)$ if

$$f(x, y) = \begin{cases} \frac{(x^2y+xy^2)\sin(x-y)}{x^2+y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

If $(x, y) \neq (0, 0)$, the ordinary rules for differentiation, applied separately to x and y, yield

$$\begin{aligned} f_x(x, y) &= \frac{(2xy+y^2)\sin(x-y)+(x^2y+xy^2)\cos(x-y)}{x^2+y^2} \\ &\quad - \frac{2x(x^2y+xy^2)\sin(x-y)}{(x^2+y^2)^2}, \quad (x, y) \neq (0, 0), \end{aligned} \tag{2.32}$$

and

$$\begin{aligned} f_y(x, y) &= \frac{(x^2+2xy)\sin(x-y)-(x^2y+xy^2)\cos(x-y)}{x^2+y^2} \\ &\quad - \frac{2y(x^2y+xy^2)\sin(x-y)}{(x^2+y^2)^2}, \quad (x, y) \neq (0, 0). \end{aligned} \tag{2.33}$$

These formulas do not apply if $(x, y) = (0, 0)$, so we find $f_x(0, 0)$ and $f_y(0, 0)$ from their definitions as difference quotients:

$$\begin{aligned} f_x(0, 0) &= \lim_{x\to 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x\to 0} \frac{0 - 0}{x} = 0, \\ f_y(0, 0) &= \lim_{y\to 0} \frac{f(0, y) - f(0, 0)}{y} = \lim_{y\to 0} \frac{0 - 0}{y} = 0. \end{aligned}$$

Setting $y = 0$ in (2.32) and (2.33) yields

$$f_x(x, 0) = 0, \quad f_y(x, 0) = \sin x, \quad x \neq 0,$$

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so

$$f_{xx}(0,0) = \lim_{x\to 0} \frac{f_x(x,0) - f_x(0,0)}{x} = \lim_{x\to 0} \frac{0-0}{x} = 0.$$

$$f_{yx}(0,0) = \lim_{x\to 0} \frac{f_y(x,0) - f_y(0,0)}{x} = \lim_{x\to 0} \frac{\sin x - 0}{x} = 1.$$

Setting $x = 0$ in (2.32) and (2.33) yields

$$f_x(0,y) = -\sin y, \quad f_y(0,y) = 0, \quad y \neq 0,$$

so

$$\begin{aligned} f_{xy}(0,0) &= \lim_{y\to 0} \frac{f_x(0,y) - f_x(0,0)}{y} = \lim_{y\to 0} \frac{-\sin y - 0}{y} = -1, \\ f_{yy}(0,0) &= \lim_{y\to 0} \frac{f_y(0,y) - f_y(0,0)}{y} = \lim_{y\to 0} \frac{0-0}{y} = 0. \end{aligned}$$

2.16.2 Equality of Mixed Partial Derivatives

Theorem: Suppose that f, f_x, f_y, and f_{xy} exist on a neighborhood N of (x_0, y_0), and f_{xy} is continuous at (x_0, y_0).

Then $f_{yx}(x_0, y_0)$ exists, and

$$f_{yx}(x_0, y_0) = f_{xy}(x_0, y_0). \tag{2.34}$$

Proof: Suppose that $\varepsilon > 0$. Choose $\delta > 0$ so that the open square

$$S_\delta = \{(x,y) : |x - x_0| < \delta, |y - y_0| < \delta\}$$

is in N.

$$|f_{xy}(\widehat{x}, \widehat{y}) - f_{xy}(x_0, y_0)| < \varepsilon \qquad \text{if} \quad (\widehat{x}, \widehat{y}) \in S_\delta. \tag{2.35}$$

This is possible because of the continuity of f_{xy} at (x_0, y_0). The function

$$A(h,k) = f(x_0 + h, y_0 + k) - f(x_0 + h, y_0) - f(x_0, y_0 + k) + f(x_0, y_0) \tag{2.36}$$

is defined if $-\delta < h, k < \delta$.

Moreover,

$$A(h,k) = \phi(x_0 + h) - \phi(x_0), \tag{2.37}$$

where

$$\phi(x) = f(x, y_0 + k) - f(x, y_0).$$

Since

$$\phi'(x) = f_x(x, y_0 + k) - f_x(x, y_0), \quad |x - x_0| < \delta,$$

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(2.37) and the mean value theorem imply that

$$A(h,k) = [f_x(\widehat{x}, y_0 + k) - f_x(\widehat{x}, y_0)]\, h. \tag{2.38}$$

where $\widehat{x}$ is between x_0 and $x_0 + h$.

The mean value theorem, applied to $f_x(\widehat{x}, y)$ (where $\widehat{x}$ is regarded as constant), also implies that

$$f_x(\widehat{x}, y_0 + k) - f_x(\widehat{x}, y_0) = f_{xy}(\widehat{x}, \widehat{y})k,$$

where $\widehat{y}$ is between y_0 and $y_0 + k$.

From this and (2.38),

$$A(h,k) = f_{xy}(\widehat{x}, \widehat{y})hk.$$

Now (2.35) implies that

$$\left|\frac{A(h,k)}{hk} - f_{xy}(x_0, y_0)\right| = |f_{xy}(\widehat{x}, \widehat{y}) - f_{xy}(x_0, y_0)| < \varepsilon \quad \text{if} \quad 0 < |h|, |k| < \delta.$$

Since (2.36) implies that

$$\begin{aligned}\lim_{k\to 0} \frac{A(h,k)}{hk} &= \lim_{k\to 0} \frac{f(x_0 + h, y_0 + k) - f(x_0 + h, y_0)}{hk} \\ &\quad - \lim_{k\to 0} \frac{f(x_0, y_0 + k) - f(x_0, y_0)}{hk} \\ &= \frac{f_y(x_0 + h, y_0) - f_y(x_0, y_0)}{h}.\end{aligned}$$

It follows from (2.39) that

$$\left|\frac{f_y(x_0 + h, y_0) - f_y(x_0, y_0)}{h} - f_{xy}(x_0, y_0)\right| \le \varepsilon \quad \text{if} \quad 0 < |h| < \delta.$$

Taking the limit as $h \to 0$ yields

$$|f_{yx}(x_0, y_0) - f_{xy}(x_0, y_0)| \le \varepsilon.$$

Since ε is an arbitrary positive number, this proves (2.34).

2.16.3 Generalization of Equality of Mixed Partial Derivative

Theorem: Suppose that f and all its partial derivatives of order $\le r$ are continuous on an open subset S of $\mathbb{R}^n$.

Then

$$f_{x_{i_1} x_{i_2}, \ldots, x_{i_r}}(\mathbf{X}) = f_{x_{j_1} x_{j_2}, \ldots, x_{j_r}}(\mathbf{X}), \quad \mathbf{X} \in S. \tag{2.39}$$

If each of the variables $x_1, x_2, \ldots, x_n$ appears the same number of times in

$$\{x_{i_1}, x_{i_2}, \ldots, x_{i_r}\} \quad \text{and} \quad \{x_{j_1}, x_{j_2}, \ldots, x_{j_r}\}.$$

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If this number is r_k, we denote the common value of the two sides of (2.39) by

$$\frac{\partial^r f(\mathbf{X})}{\partial x_1^{r_1} \partial x_2^{r_2} \cdots \partial x_n^{r_n}}. \tag{2.40}$$

It being understood that

$$0 \leq r_k \leq r, \quad 1 \leq k \leq n, \tag{2.41}$$

$$r_1 + r_2 + \cdots + r_n = r, \tag{2.42}$$

and, if $r_k = 0$, we omit the symbol ∂x_k^0 from the denominator of (2.40).

Remark: A function of several variables may have first-order partial derivatives at a point $\mathbf{X}_0$ but fail to be continuous at $\mathbf{X}_0$.

Example: Consider the function

$$f(x,y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases} \tag{2.43}$$

Then

$$\begin{aligned} f_x(0,0) &= \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{0-0}{h} = 0 \\ f_y(0,0) &= \lim_{k\to 0} \frac{f(0,k) - f(0,0)}{k} = \lim_{k\to 0} \frac{0-0}{k} = 0, \end{aligned}$$

but f is not continuous at $(0,0)$.

Remark: If differentiability of a function of several variables is to be a stronger property than continuity, as it is for functions of one variable, the definition of differentiability must require more than the existence of first partial derivatives.

A function f is differentiable at x_0 if and only if

$$\lim_{x\to x_0} \frac{f(x) - f(x_0) - m(x - x_0)}{x - x_0} = 0$$

for some constant m, in which case $m = f'(x_0)$.

2.17 Differentiability of Functions of Several Variables

A function f is *differentiable* at

$$\mathbf{X}_0 = (x_{10}, x_{20}, \ldots, x_{n0})$$

if $\mathbf{X}_0 \in D_f^0$ and there are constants m_1, m_2, ..., m_n such that

$$\lim_{\mathbf{X}\to\mathbf{X}_0} \frac{f(\mathbf{X}) - f(\mathbf{X}_0) - \sum_{i=1}^n m_i(x_i - x_{i0})}{|\mathbf{X} - \mathbf{X}_0|} = 0. \tag{2.44}$$

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Example: Show that the following function $f(x,y) = x^2 + 2xy$, is differentiable at any point (x_0, y_0).

$$\begin{aligned} f(x,y) - f(x_0,y_0) &= x^2 + 2xy - x_0^2 - 2x_0y_0 \\ &= x^2 - x_0^2 + 2(xy - x_0y_0) \\ &= (x - x_0)(x + x_0) + 2(xy - x_0y) \\ &\quad +2(x_0y - x_0y_0) \\ &= (x + x_0 + 2y)(x - x_0) + 2x_0(y - y_0) \\ &= 2(x_0 + y_0)(x - x_0) + 2x_0(y - y_0) \\ &\quad + (x - x_0)(x - x_0 + 2y - 2y_0) \\ &= m_1(x - x_0) + m_2(y - y_0) + (x - x_0)(x - x_0 + 2y - 2y_0), \end{aligned}$$

where

$$m_1 = 2(x_0 + y_0) = f_x(x_0, y_0) \quad \text{and} \quad m_2 = 2x_0 = f_y(x_0, y_0). \tag{2.45}$$

Therefore,

$$\begin{aligned} &\frac{|f(x,y) - f(x_0,y_0) - m_1(x - x_0) - m_2(y - y_0)|}{|\mathbf{X} - \mathbf{X}_0|} \\ &= \frac{|x - x_0||(x - x_0) + 2(y - y_0)|}{|\mathbf{X} - \mathbf{X}_0|} \\ &\leq \sqrt{5}|\mathbf{X} - \mathbf{X}_0|, \end{aligned}$$

by Schwarz's inequality. This implies that

$$\lim_{\mathbf{X} \to \mathbf{X}_0} \frac{f(x,y) - f(x_0,y_0) - m_1(x - x_0) - m_2(y - y_0)}{|\mathbf{X} - \mathbf{X}_0|} = 0,$$

so f is differentiable at (x_0, y_0).

Theorem: If f is differentiable at $\mathbf{X}_0 = (x_{10}, x_{20}, \ldots, x_{n0})$, then $f_{x_1}(\mathbf{X}_0)$, $f_{x_2}(\mathbf{X}_0)$, $\ldots$, $f_{x_n}(\mathbf{X}_0)$ exist and the constants m_1, m_2, $\ldots$, m_n in

$$\lim_{\mathbf{X} \to \mathbf{X}_0} \frac{f(\mathbf{X}) - f(\mathbf{X}_0) - \sum_{i=1}^{n} m_i(x_i - x_{i0})}{|\mathbf{X} - \mathbf{X}_0|} = 0,$$

are given by

$$m_i = f_{x_i}(\mathbf{X}_0), \quad 1 \leq i \leq n; \tag{2.46}$$

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that is,

$$\lim_{\mathbf{X}\to\mathbf{X}_0} \frac{f(\mathbf{X}) - f(\mathbf{X}_0) - \sum_{i=1}^{n} f_{x_i}(\mathbf{X}_0)(x_i - x_{i0})}{|\mathbf{X} - \mathbf{X}_0|} = 0.$$

Proof: Let i be a given integer in $\{1, 2, \ldots, n\}$. Let $\mathbf{X} = \mathbf{X}_0 + t\mathbf{E}_i$, so that $x_i = x_{i0} + t$, $x_j = x_{j0}$ if $j \neq i$, and $|\mathbf{X} - \mathbf{X}_0| = |t|$.

Then

$$\lim_{\mathbf{X}\to\mathbf{X}_0} \frac{f(\mathbf{X}) - f(\mathbf{X}_0) - \sum_{i=1}^{n} m_i(x_i - x_{i0})}{|\mathbf{X} - \mathbf{X}_0|} = 0.$$

and the differentiability of f at $\mathbf{X}_0$ imply that

$$\lim_{t\to 0} \frac{f(\mathbf{X}_0 + t\mathbf{E}_i) - f(\mathbf{X}_0) - m_i t}{t} = 0.$$

Hence,

$$\lim_{t\to 0} \frac{f(\mathbf{X}_0 + t\mathbf{E}_i) - f(\mathbf{X}_0)}{t} = m_i.$$

This proves (2.46), since the limit on the left is $f_{x_i}(\mathbf{X}_0)$, by definition.

2.17.1 Linear Function

A *linear function* is a function of the form

$$L(\mathbf{X}) = m_1 x_1 + m_2 x_2 + \cdots + m_n x_n, \tag{2.47}$$

where m_1, m_2, ..., m_n are constants. From definition of differentiability, f is differentiable at $\mathbf{X}_0$ if and only if there is a linear function L such that $f(\mathbf{X}) - f(\mathbf{X}_0)$ can be approximated so well near $\mathbf{X}_0$ by

$$L(\mathbf{X}) - L(\mathbf{X}_0) = L(\mathbf{X} - \mathbf{X}_0)$$

that

$$f(\mathbf{X}) - f(\mathbf{X}_0) = L(\mathbf{X} - \mathbf{X}_0) + E(\mathbf{X})(|\mathbf{X} - \mathbf{X}_0|), \tag{2.48}$$

where

$$\lim_{\mathbf{X}\to\mathbf{X}_0} E(\mathbf{X}) = 0. \tag{2.49}$$

Theorem: If f is differentiable at $\mathbf{X}_0$, then f is continuous at $\mathbf{X}_0$.

Proof: From $L(\mathbf{X}) = m_1 x_1 + m_2 x_2 + \cdots + m_n x_n$, and Schwarz's inequality,

$$|L(\mathbf{X} - \mathbf{X}_0)| \leq M|\mathbf{X} - \mathbf{X}_0|,$$

where

$$M = (m_1^2 + m_2^2 + \cdots + m_n^2)^{1/2}.$$

This and $f(\mathbf{X}) - f(\mathbf{X}_0) = L(\mathbf{X} - \mathbf{X}_0) + E(\mathbf{X})(|\mathbf{X} - \mathbf{X}_0|)$, imply that

$$|f(\mathbf{X}) - f(\mathbf{X}_0)| \leq (M + |E(\mathbf{X})|)|\mathbf{X} - \mathbf{X}_0|.$$

which, with (2.49), implies that f is continuous at $\mathbf{X}_0$.

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2.17.2 Differential

The linear function

$$L(\mathbf{X}) = f_{x_1}(\mathbf{X}_0)x_1 + f_{x_2}(\mathbf{X}_0)x_2 + \cdots + f_{x_n}(\mathbf{X}_0)x_n.$$

This function is called the *differential of f at $\mathbf{X}_0$*. We will denote it by $d_{\mathbf{X}_0}f$ and its value by $(d_{\mathbf{X}_0}f)(\mathbf{X})$.

Thus,

$$(d_{\mathbf{X}_0}f)(\mathbf{X}) = f_{x_1}(\mathbf{X}_0)x_1 + f_{x_2}(\mathbf{X}_0)x_2 + \cdots + f_{x_n}(\mathbf{X}_0)x_n. \tag{2.50}$$

In terms of the differential, differentiability can be rewritten as

$$\lim_{\mathbf{X}\to\mathbf{X}_0} \frac{f(\mathbf{X}) - f(\mathbf{X}_0) - (d_{\mathbf{X}_0}f)(\mathbf{X}-\mathbf{X}_0)}{|\mathbf{X}-\mathbf{X}_0|} = 0.$$

For convenience in writing $d_{\mathbf{X}_0}f$, and to conform with standard notation, we introduce the function dx_i, defined by

$$dx_i(\mathbf{X}) = x_i;$$

that is, dx_i is the function whose value at a point in $\mathbb{R}^n$ is the ith coordinate of the point.

It is the differential of the function $g_i(\mathbf{X}) = x_i$. From (2.50),

$$d_{\mathbf{X}_0}f = f_{x_1}(\mathbf{X}_0)\,dx_1 + f_{x_2}(\mathbf{X}_0\,dx_2 + \cdots + f_{x_n}(\mathbf{X}_0)\,dx_n. \tag{2.51}$$

If we write $\mathbf{X} = (x, y, \ldots,)$, then we write

$$d_{\mathbf{X}_0}f = f_x(\mathbf{X}_0)\,dx + f_y(\mathbf{X}_0)\,dy + \cdots,$$

where dx, dy, ... are the functions defined by

$$dx(\mathbf{X}) = x, \quad dy(\mathbf{X}) = y, \ldots$$

When it is not necessary to emphasize the specific point $\mathbf{X}_0$, (2.51) can be written more simply as

$$df = f_{x_1}\,dx_1 + f_{x_2}\,dx_2 + \cdots + f_{x_n}\,dx_n.$$

When dealing with a specific function at an arbitrary point of its domain, we may use the hybrid notation

$$df = f_{x_1}(\mathbf{X})\,dx_1 + f_{x_2}(\mathbf{X})\,dx_2 + \cdots + f_{x_n}(\mathbf{X})\,dx_n.$$

Example: The function

$$f(x, y) = x^2 + 2xy$$

is differentiable at every $\mathbf{X}$ in $\mathbb{R}^n$.

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The differential of the functions is

$$df = (2x + 2y)\,dx + 2x\,dy.$$

To find $d_{\mathbf{X}_0} f$ with $\mathbf{X}_0 = (1, 2)$, we set $x_0 = 1$ and $y_0 = 2$; thus,

$$\begin{aligned} d_{\mathbf{X}_0} f &= 6\,dx + 2\,dy \\ (d_{\mathbf{X}_0} f)(\mathbf{X} - \mathbf{X}_0) &= 6(x-1) + 2(y-2). \end{aligned}$$

Since $f(1, 2) = 5$, the differentiability of f at $(1, 2)$ implies that

$$\lim_{(x,y)\to(1,2)} \frac{f(x,y) - 5 - 6(x-1) - 2(y-2)}{\sqrt{(x-1)^2 + (y-2)^2}} = 0.$$

Example: The differential of a function $f = f(x)$ of one variable is given by

$$d_{x_0} f = f'(x_0)\,dx,$$

where dx is the identity function; that is,

$$dx(t) = t.$$

For example, if

$$f(x) = 3x^2 + 5x^3,$$

then

$$df = (6x + 15x^2)\,dx.$$

If $x_0 = -1$, then

$$d_{x_0} f = 9\,dx, \quad (d_{x_0} f)(x - x_0) = 9(x+1),$$

and, since $f(-1) = -2$,

$$\lim_{x\to -1} \frac{f(x) + 2 - 9(x+1)}{x+1} = 0.$$

Remark: Unfortunately, the notation for the differential is so complicated that it obscures the simplicity of the concept. The peculiar symbols df, dx, dy, etc., were introduced in the early stages of the development of calculus to represent very small (infinitesimal) increments in the variables. However, in modern usage they are not quantities at all, but linear functions. This meaning of the symbol dx differs from its meaning in $\int_a^b f(x)\,dx$, where it serves merely to identify the variable of integration; indeed, some authors omit it in the latter context and write simply $\int_a^b f$.

Lemma: If f is differentiable at $\mathbf{X}_0$, then

$$f(\mathbf{X}) - f(\mathbf{X}_0) = (d_{\mathbf{X}_0} f)(\mathbf{X} - \mathbf{X}_0) + E(\mathbf{X})|\mathbf{X} - \mathbf{X}_0|,$$

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where E is defined in a neighborhood of $\mathbf{X}_0$ and

$$\lim_{\mathbf{X}\to\mathbf{X}_0} E(\mathbf{X}) = E(\mathbf{X}_0) = 0.$$

Theorem: If f and g are differentiable at $\mathbf{X}_0$, then so are $f+g$ and fg. The same is true of f/g if $g(\mathbf{X}_0) \neq 0$. The differentials are given by

$$\begin{aligned} d_{\mathbf{X}_0}(f+g) &= d_{\mathbf{X}_0}f + d_{\mathbf{X}_0}g, \\ d_{\mathbf{X}_0}(fg) &= f(\mathbf{X}_0)d_{\mathbf{X}_0}g + g(\mathbf{X}_0)d_{\mathbf{X}_0}f, \end{aligned}$$

and

$$d_{\mathbf{X}_0}\left(\frac{f}{g}\right) = \frac{g(\mathbf{X}_0)d_{\mathbf{X}_0}f - f(\mathbf{X}_0)d_{\mathbf{X}_0}g}{[g(\mathbf{X}_0)]^2}.$$

2.17.3 A sufficient Condition for Differentiability

Theorem: If f_{x_1}, f_{x_2}, ..., f_{x_n} exist on a neighborhood of $\mathbf{X}_0$ and are continuous at $\mathbf{X}_0$, then f is differentiable at $\mathbf{X}_0$.

Proof: Let $\mathbf{X}_0 = (x_{10}, x_{20}, \ldots, x_{n0})$ and suppose that $\varepsilon > 0$. Our assumptions imply that there is a $\delta > 0$ such that $f_{x_1}, f_{x_2}, \ldots, f_{x_n}$ are defined in the n-ball

$$S_\delta(\mathbf{X}_0) = \{\mathbf{X} : |\mathbf{X} - \mathbf{X}_0| < \delta\}$$

and

$$|f_{x_j}(\mathbf{X}) - f_{x_j}(\mathbf{X}_0)| < \varepsilon \quad \text{if} \quad |\mathbf{X} - \mathbf{X}_0| < \delta, \quad 1 \leq j \leq n. \tag{2.52}$$

Let $\mathbf{X} = (x_1, x_, \ldots, x_n)$ be in $S_\delta(\mathbf{X}_0)$. Define

$$\mathbf{X}_j = (x_1, \ldots, x_j, x_{j+1,0}, \ldots, x_{n0}), \quad 1 \leq j \leq n-1,$$

and $\mathbf{X}_n = \mathbf{X}$. Thus, for $1 \leq j \leq n$, $\mathbf{X}_j$ differs from $\mathbf{X}_{j-1}$ in the jth component only, and the line segment from $\mathbf{X}_{j-1}$ to $\mathbf{X}_j$ is in $S_\delta(\mathbf{X}_0)$. Now write

$$f(\mathbf{X}) - f(\mathbf{X}_0) = f(\mathbf{X}_n) - f(\mathbf{X}_0) = \sum_{j=1}^{n} [f(\mathbf{X}_j) - f(\mathbf{X}_{j-1})], \tag{2.53}$$

and consider the auxiliary functions

$$\begin{aligned} g_1(t) &= f(t, x_{20}, \ldots, x_{n0}), \\ g_j(t) &= f(x_1, \ldots, x_{j-1}, t, x_{j+1,0}, \ldots, x_{n0}), \quad 2 \leq j \leq n-1, \\ g_n(t) &= f(x_1, \ldots, x_{n-1}, t), \end{aligned} \tag{2.54}$$

where, in each case, all variables except t are temporarily regarded as constants. Since

$$f(\mathbf{X}_j) - f(\mathbf{X}_{j-1}) = g_j(x_j) - g_j(x_{j0}),$$

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the mean value theorem implies that

$$f(\mathbf{X}_j) - f(\mathbf{X}_{j-1}) = g_j'(\tau_j)(x_j - x_{j0}),$$

where τ_j is between x_j and x_{j0}. From (2.54),

$$g_j'(\tau_j) = f_{x_j}(\widehat{\mathbf{X}}_j),$$

where $\widehat{\mathbf{X}}_j$ is on the line segment from $\mathbf{X}_{j-1}$ to $\mathbf{X}_j$. Therefore,

$$f(\mathbf{X}_j) - f(\mathbf{X}_{j-1}) = f_{x_j}(\widehat{\mathbf{X}}_j)(x_j - x_{j0}),$$

and (2.53) implies that

$$\begin{aligned} f(\mathbf{X}) - f(\mathbf{X}_0) &= \sum_{j=1}^{n} f_{x_j}(\widehat{\mathbf{X}}_j)(x_j - x_{j0}) \\ &= \sum_{j=1}^{n} f_{x_j}(\mathbf{X}_0)(x_j - x_{j0}) + \sum_{j=1}^{n} [f_{x_j}(\widehat{\mathbf{X}}_j) - f_{x_j}(\mathbf{X}_0)](x_j - x_{j0}). \end{aligned}$$

From this and (2.52),

$$\left| f(\mathbf{X}) - f(\mathbf{X}_0) - \sum_{j=1}^{n} f_{x_j}(\mathbf{X}_0)(x_j - x_{j0}) \right| \leq \varepsilon \sum_{j=1}^{n} |x_j - x_{j0}| \leq n\varepsilon |\mathbf{X} - \mathbf{X}_0|,$$

which implies that f is differentiable at $\mathbf{X}_0$.

2.17.4 Continuously Differentiable Function

We say that f is *continuously differentiable* on a subset S of $\mathbb{R}^n$ if S is contained in an open set on which $f_{x_1}, f_{x_2}, \ldots, f_{x_n}$ are continuous.

The above theorem implies that such a function is differentiable at each $\mathbf{X}_0$ in$S$.

Example: If

$$f(x, y) = \frac{x^2 + y^2}{x - y},$$

then

$$f_x(x, y) = \frac{2x}{x - y} - \frac{x^2 + y^2}{(x - y)^2}$$

$$f_y(x, y) = \frac{2y}{x - y} + \frac{x^2 + y^2}{(x - y)^2}.$$

Since f_x and f_y are continuous on

$$S = \{(x, y) : x \neq y\},$$

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f is continuously differentiable on S.

Remark: If f_{x_1}, f_{x_2}, ..., f_{x_n} exist on a neighborhood of $\mathbf{X}_0$ and are continuous at $\mathbf{X}_0$, then f is differentiable at $\mathbf{X}_0$. These conditions are not necessary for differentiability; that is, a function may be differentiable at a point $\mathbf{X}_0$ even if its first partial derivatives are not continuous at $\mathbf{X}_0$.

Example: let

$$f(x,y) = \begin{cases} (x-y)^2 \sin \frac{1}{x-y}, & x \neq y, \\ 0, & x = y. \end{cases}$$

Then

$$f_x(x,y) = 2(x-y)\sin\frac{1}{x-y} - \cos\frac{1}{x-y}, \quad x \neq y,$$

and

$$f_x(x,x) = \lim_{h\to 0} \frac{f(x+h,x) - f(x,x)}{h} = \lim_{h\to 0} \frac{h^2 \sin(1/h) - 0}{h} = 0,$$

so f_x exists for all (x,y), but is not continuous on the line $y = x$.

Example: Let

$$f(x,y) = \begin{cases} (x-y)^2 \sin \frac{1}{x-y}, & x \neq y, \\ 0, & x = y. \end{cases}$$

The same is true of f_y, since

$$f_y(x,y) = -2(x-y)\sin\frac{1}{x-y} + \cos\frac{1}{x-y}, \quad x \neq y,$$

and

$$f_y(x,x) = \lim_{k\to 0} \frac{f(x,x+k) - f(x,x)}{k} = \lim_{k\to 0} \frac{k^2 \sin(-1/k) - 0}{k} = 0.$$

Now,

$$\frac{f(x,y) - f(0,0) - f_x(0,0)x - f_y(0,0)y}{\sqrt{x^2+y^2}}$$

$$= \begin{cases} \frac{(x-y)^2}{\sqrt{x^2+y^2}} \sin \frac{1}{x-y}, & x \neq y, \\ 0, & x = y, \end{cases}$$

and Schwarz's inequality implies that

$$\left| \frac{(x-y)^2}{\sqrt{x^2+y^2}} \sin\frac{1}{x-y} \right| \leq \frac{2(x^2+y^2)}{\sqrt{x^2+y^2}} = 2\sqrt{x^2+y^2}, \quad x \neq y.$$

Therefore,

$$\lim_{(x,y)\to(0,0)} \frac{f(x,y) - f(0,0) - f_x(0,0)x - f_y(0,0)y}{\sqrt{x^2+y^2}} = 0,$$

so f is differentiable at $(0,0)$, but f_x and f_y are not continuous at $(0,0)$.

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2.17.5 Geometric Interpretation of Differentiability

If a function f of one variable is differentiable at x_0, then the curve $y = f(x)$ has a tangent line

$$y = T(x) = f(x_0) + f'(x_0)(x - x_0).$$

The tangent line approximates it so well near x_0 that

$$\lim_{x \to x_0} \frac{f(x) - T(x)}{x - x_0} = 0.$$

Moreover, the tangent line is the limit of the secant line through the points $(x_1, f(x_0))$ and $(x_0, f(x_0))$ as x_1 approaches x_0. Differentiability of a function of n variables has an analogous geometric interpretation. We will illustrate it for $n = 2$. If f is defined in a region D in $\mathbb{R}^2$, then the set of points (x, y, z) such that

$$z = f(x, y), \quad (x, y) \in D, \tag{2.55}$$

is a *surface* in $\mathbb{R}^3$ **Geometric interpretation of differentiability**:

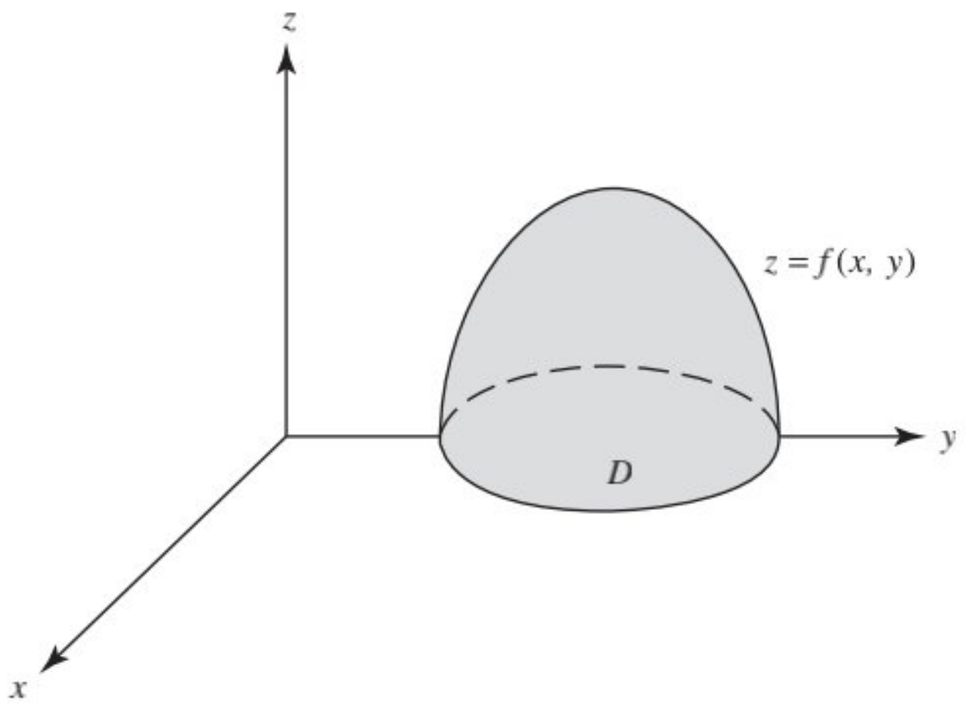


Figure 2.8: Domain of the function

If f is differentiable at $\mathbf{X}_0 = (x_0, y_0)$, then the plane

$$z = T(x, y) = f(\mathbf{X}_0) + f_x(\mathbf{X}_0)(x - x_0) + f_y(\mathbf{X}_0)(y - y_0) \tag{2.56}$$

intersects the surface $z = f(x, y)$ at $(x_0, y_0, f(x_0, y_0))$ and approximates the surface so well near (x_0, y_0) that

$$\lim_{(x,y) \to (x_0,y_0)} \frac{f(x, y) - T(x, y)}{\sqrt{(x - x_0)^2 + (y - y_0)^2}} = 0.$$

Moreover, (2.56) is the only plane in $\mathbb{R}^3$ with these properties.

We say that this plane is *tangent to the surface* $z = f(x, y)$ *at the point* $(x_0, y_0, f(x_0, y_0))$.

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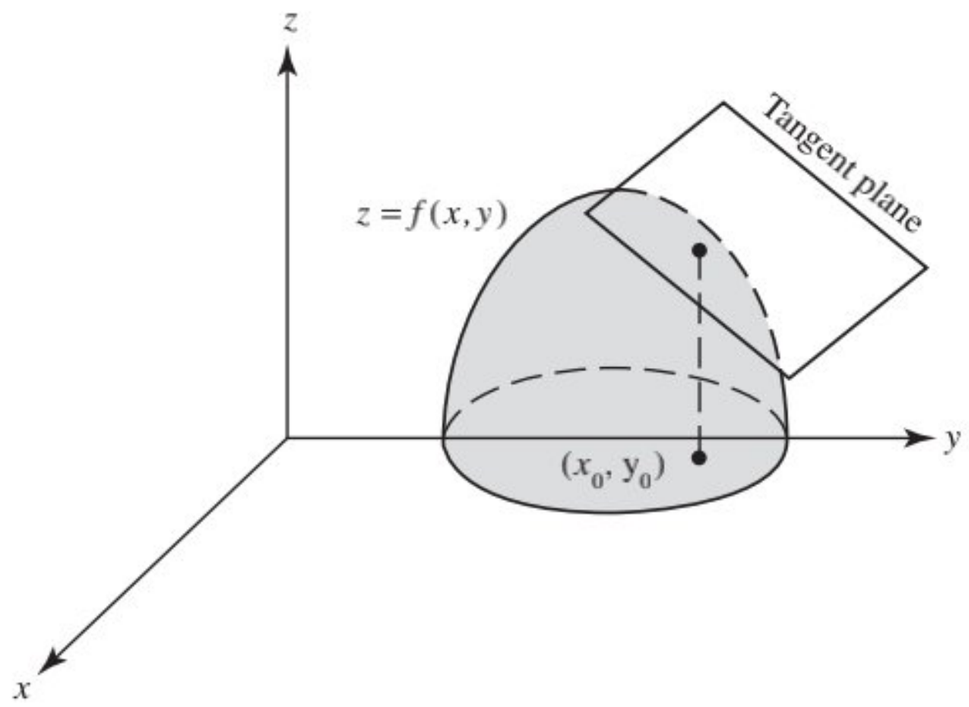


Figure 2.9: Geometric interpretation of differentiability

Show that the tangent plane to the surface $z = f(x, y)$ is the limit of the secant planes.

Let $\mathbf{X}_i = (x_i, y_i)$ $(i = 1, 2, 3)$. The equation of the secant plane through the points $(x_i, y_i, f(x_i, y_i))$ $(i = 1, 2, 3)$ on the surface $z = f(x, y)$ is of the form

$$z = f(\mathbf{X}_0) + A(x - x_0) + B(y - y_0), \tag{2.57}$$

where A and B satisfy the system

$$\begin{aligned} f(\mathbf{X}_1) &= f(\mathbf{X}_0) + A(x_1 - x_0) + B(y_1 - y_0), \\ f(\mathbf{X}_2) &= f(\mathbf{X}_0) + A(x_2 - x_0) + B(y_2 - y_0). \end{aligned}$$

Solving for A and B yields

$$A = \frac{(f(\mathbf{X}_1) - f(\mathbf{X}_0))(y_2 - y_0) - (f(\mathbf{X}_2) - f(\mathbf{X}_0))(y_1 - y_0)}{(x_1 - x_0)(y_2 - y_0) - (x_2 - x_0)(y_1 - y_0)} \tag{2.58}$$

$$B = \frac{(f(\mathbf{X}_2) - f(\mathbf{X}_0))(x_1 - x_0) - (f(\mathbf{X}_1) - f(\mathbf{X}_0))(x_2 - x_0)}{(x_1 - x_0)(y_2 - y_0) - (x_2 - x_0)(y_1 - y_0)} \tag{2.59}$$

if

$$(x_1 - x_0)(y_2 - y_0) - (x_2 - x_0)(y_1 - y_0) \neq 0, \tag{2.60}$$

which is equivalent to the requirement that $\mathbf{X}_0$, $\mathbf{X}_1$, and $\mathbf{X}_2$ do not lie on a line. If we write

$$\mathbf{X}_1 = \mathbf{X}_0 + t\mathbf{U} \quad \text{and} \quad \mathbf{X}_2 = \mathbf{X}_0 + t\mathbf{V},$$

where $\mathbf{U} = (u_1, u_2)$ and $\mathbf{V} = (v_1, v_2)$ are fixed nonzero vectors, then (2.58), (2.59), and (2.60) take the more convenient forms

$$A = \frac{\frac{f(\mathbf{X}_0 + t\mathbf{U}) - f(\mathbf{X}_0)}{t} v_2 - \frac{f(\mathbf{X}_0 + t\mathbf{V}) - f(\mathbf{X}_0)}{t} u_2}{u_1 v_2 - u_2 v_1}, \tag{2.61}$$

$$B = \frac{\frac{f(\mathbf{X}_0 + t\mathbf{V}) - f(\mathbf{X}_0)}{t} u_1 - \frac{f(\mathbf{X}_0 + t\mathbf{U}) - f(\mathbf{X}_0)}{t} v_1}{u_1 v_2 - u_2 v_1}, \tag{2.62}$$

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and

$$u_1v_2 - u_2v_1 \neq 0.$$

If f is differentiable at $\mathbf{X}_0$, then

$$f(\mathbf{X}) - f(\mathbf{X}_0) = f_x(\mathbf{X}_0)(x - x_0) + f_y(\mathbf{X}_0)(y - y_0) + \varepsilon(\mathbf{X})|\mathbf{X} - \mathbf{X}_0|, \tag{2.63}$$

where

$$\lim_{\mathbf{X}\to\mathbf{X}_0} \varepsilon(\mathbf{X}) = 0. \tag{2.64}$$

Substituting first $\mathbf{X} = \mathbf{X}_0 + t\mathbf{U}$ and then $\mathbf{X} = \mathbf{X}_0 + t\mathbf{V}$ in (2.63) and dividing by t yields

$$\frac{f(\mathbf{X}_0 + t\mathbf{U}) - f(\mathbf{X}_0)}{t} = f_x(\mathbf{X}_0)u_1 + f_y(\mathbf{X}_0)u_2 + E_1(t)|\mathbf{U}| \tag{2.65}$$

and

$$\frac{f(\mathbf{X}_0 + t\mathbf{V}) - f(\mathbf{X}_0)}{t} = f_x(\mathbf{X}_0)v_1 + f_y(\mathbf{X}_0)v_2 + E_2(t)|\mathbf{V}|, \tag{2.66}$$

where

$$E_1(t) = \varepsilon(\mathbf{X}_0 + t\mathbf{U})|t|/t \quad \text{and} \quad E_2(t) = \varepsilon(\mathbf{X}_0 + t\mathbf{V})|t|/t,$$

so

$$\lim_{t\to 0} E_i(t) = 0, \quad i = 1, 2, \tag{2.67}$$

because of (2.64). Substituting (2.65) and (2.66) into (2.61) and (2.62) yields

$$A = f_x(\mathbf{X}_0) + \Delta_1(t), \quad B = f_y(\mathbf{X}_0) + \Delta_2(t), \tag{2.68}$$

where

$$\Delta_1(t) = \frac{v_2|\mathbf{U}|E_1(t) - u_2|\mathbf{V}|E_2(t)}{u_1v_2 - u_2v_1}$$

and

$$\Delta_2(t) = \frac{u_1|\mathbf{V}|E_2(t) - v_1|\mathbf{U}|E_1(t)}{u_1v_2 - u_2v_1},$$

so

$$\lim_{t\to 0} \Delta_i(t) = 0, \quad i = 1, 2, \tag{2.69}$$

because of (2.67).

From (2.57) and (2.68), the equation of the secant plane is

$$z = f(\mathbf{X}_0) + [f_x(\mathbf{X}_0) + \Delta_1(t)](x - x_0) + [f_y(\mathbf{X}_0) + \Delta_2(t)](y - y_0).$$

Therefore, because of (2.69), the secant plane approaches the tangent plane (2.56) as t approaches zero.

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2.18 Maxima and Minima

We say that $\mathbf{X}_0$ is a *local extreme point* of f if there is a $\delta > 0$ such that

$$f(\mathbf{X}) - f(\mathbf{X}_0)$$

does not change sign in $S_\delta(\mathbf{X}_0) \cap D_f$.

More specifically, $\mathbf{X}_0$ is a *local maximum point* if

$$f(\mathbf{X}) \le f(\mathbf{X}_0)$$

or a *local minimum point* if

$$f(\mathbf{X}) \ge f(\mathbf{X}_0)$$

for all $\mathbf{X}$ in $S_\delta(\mathbf{X}_0) \cap D_f$.

Theorem: Suppose that f is defined in a neighborhood of $\mathbf{X}_0$ in $\mathbb{R}^n$ and $f_{x_1}(\mathbf{X}_0)$, $f_{x_2}(\mathbf{X}_0)$, ..., $f_{x_n}(\mathbf{X}_0)$ exist.

Let $\mathbf{X}_0$ be a local extreme point of f. Then

$$f_{x_i}(\mathbf{X}_0) = 0, \quad 1 \le i \le n. \tag{2.70}$$

Proof: Let $\mathbf{E}_1 = (1, 0, \ldots, 0)$, $\mathbf{E}_2 = (0, 1, 0, \ldots, 0)$, ..., $\mathbf{E}_n = (0, 0, \ldots, 1)$, and

$$g_i(t) = f(\mathbf{X}_0 + t\mathbf{E}_i), \quad 1 \le i \le n.$$

Then g_i is differentiable at $t = 0$, with

$$g_i'(0) = f_{x_i}(\mathbf{X}_0).$$

Since $\mathbf{X}_0$ is a local extreme point of f, $t_0 = 0$ is a local extreme point of g_i.

Remark: The converse of theorem is false, since (2.70) $f_{x_i}(\mathbf{X}_0) = 0, \quad 1 \le i \le n.$ may hold at a point $\mathbf{X}_0$ that is not a local extreme point of f.

For example, let $\mathbf{X}_0 = (0, 0)$ and

$$f(x, y) = x^3 + y^3.$$

We say that a point $\mathbf{X}_0$ where (2.70) holds is a *critical point* of f. Thus, if f is defined in a neighborhood of a local extreme point $\mathbf{X}_0$, then $\mathbf{X}_0$ is a critical point of f; however, a critical point need not be a local extreme point of f.

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2.19 Differentiable Vector Valued Function

A vector-valued function $\mathbf{G} = (g_1, g_2, \ldots, g_n)$ is *differentiable* at

$$\mathbf{U}_0 = (u_{10}, u_{20}, \ldots, u_{m0})$$

if its component functions g_1, g_2, $\ldots$, g_n are differentiable at $\mathbf{U}_0$.

Lemma: Suppose that $\mathbf{G} = (g_1, g_2, \ldots, g_n)$ is differentiable at

$$\mathbf{U}_0 = (u_{10}, u_{20}, \ldots, u_{m0}),$$

and define

$$M = \left(\sum_{i=1}^{n} \sum_{j=1}^{m} \left(\frac{\partial g_i(\mathbf{U}_0}{\partial u_j} \right)^2 \right)^{1/2}.$$

Then, if $\varepsilon > 0$, there is a $\delta > 0$ such that

$$\frac{|\mathbf{G}(\mathbf{U}) - \mathbf{G}(\mathbf{U}_0)|}{|\mathbf{U} - \mathbf{U}_0|} < M + \varepsilon \quad \text{if} \quad 0 < |\mathbf{U} - \mathbf{U}_0| < \delta.$$

Proof: Since g_1, g_2, $\ldots$, g_n are differentiable at $\mathbf{U}_0$ to g_i shows that

$$\begin{aligned} g_i(\mathbf{U}) - g_i(\mathbf{U}_0) &= (d_{\mathbf{U}_0} g_i)(\mathbf{U} - \mathbf{U}_0) + E_i(\mathbf{U})|(\mathbf{U} - \mathbf{U}_0| \\ &= \textstyle\sum_{j=1}^{m} \frac{\partial g_i(\mathbf{U}_0)}{\partial u_j}(u_j - u_{j0}) + E_i(\mathbf{U})|(\mathbf{U} - \mathbf{U}_0|, \end{aligned} \tag{2.71}$$

where

$$\lim_{\mathbf{U} \to \mathbf{U}_0} E_i(\mathbf{U}) = 0, \quad 1 \le i \le n. \tag{2.72}$$

From Schwarz's inequality,

$$|g_i(\mathbf{U}) - g_i(\mathbf{U}_0)| \le (M_i + |E_i(\mathbf{U})|)|\mathbf{U} - \mathbf{U}_0|,$$

where

$$M_i = \left(\sum_{j=1}^{m} \left(\frac{\partial g_i(\mathbf{U}_0)}{\partial u_j} \right)^2 \right)^{1/2}.$$

Therefore,

$$\frac{|\mathbf{G}(\mathbf{U}) - \mathbf{G}(\mathbf{U}_0)|}{|\mathbf{U} - \mathbf{U}_0|} \le \left(\sum_{i=1}^{n} (M_i + |E_i(\mathbf{U})|)^2 \right)^{1/2}.$$

From (2.72),

$$\lim_{\mathbf{U} \to \mathbf{U}_0} \left(\sum_{i=1}^{n} (M_i + |E_i(\mathbf{U})|)^2 \right)^{1/2} = \left(\sum_{i=1}^{n} M_i^2 \right)^{1/2} = M,$$

which implies the conclusion.

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2.20 The Chain Rule

Theorem: Suppose that the real-valued function f is differentiable at $\mathbf{X}_0$ in $\mathbb{R}^n$. The vector-valued function $\mathbf{G} = (g_1, g_2, \ldots, g_n)$ is differentiable at $\mathbf{U}_0$ in $\mathbb{R}^m$, and $\mathbf{X}_0 = \mathbf{G}(\mathbf{U}_0)$.

Then the real-valued composite function $h = f \circ \mathbf{G}$ defined by

$$h(\mathbf{U}) = f(\mathbf{G}(\mathbf{U})) \tag{2.73}$$

is differentiable at $\mathbf{U}_0$, and

$$d_{\mathbf{U}_0} h = f_{x_1}(\mathbf{X}_0) d_{\mathbf{U}_0} g_1 + f_{x_2}(\mathbf{X}_0) d_{\mathbf{U}_0} g_2 + \cdots + f_{x_n}(\mathbf{X}_0) d_{\mathbf{U}_0} g_n. \tag{2.74}$$

Proof: First we will show that $\mathbf{U}_0$ is an interior point of the domain of h. It is legitimate to ask if h is differentiable at $\mathbf{U}_0$. Let $\mathbf{X}_0 = (x_{10}, x_{20}, \ldots, x_{n0})$. Note that

$$x_{i0} = g_i(\mathbf{U}_0), \quad 1 \leq i \leq n,$$

by assumption.

Since f is differentiable at $\mathbf{X}_0$, which implies that

$$f(\mathbf{X}) - f(\mathbf{X}_0) = \sum_{i=1}^{n} f_{x_i}(\mathbf{X}_0)(x_i - x_{i0}) + E(\mathbf{X})|\mathbf{X} - \mathbf{X}_0|, \tag{2.75}$$

where

$$\lim_{\mathbf{X} \to \mathbf{X}_0} E(\mathbf{X}) = 0.$$

Substituting $\mathbf{X} = \mathbf{G}(\mathbf{U})$ and $\mathbf{X}_0 = \mathbf{G}(\mathbf{U}_0)$ in (2.75) and recalling (2.73) yields

$$\begin{aligned} h(\mathbf{U}) - h(\mathbf{U}_0) &= \sum_{i=1}^{n} f_{x_i}(\mathbf{X}_0)(g_i(\mathbf{U}) - g_i(\mathbf{U}_0)) \\ &\quad + E(\mathbf{G}(\mathbf{U}))|\mathbf{G}(\mathbf{U}) - \mathbf{G}(\mathbf{U}_0)|. \end{aligned} \tag{2.76}$$

Substituting $g_i(\mathbf{U}) - g_i(\mathbf{U}_0) = d_{\mathbf{U}_0} g_i)(\mathbf{U} - \mathbf{U}_0) + E_i(\mathbf{U})|\mathbf{U} - \mathbf{U}_0|$ into (2.76) yields

$$\begin{aligned} h(\mathbf{U}) - h(\mathbf{U}_0) &= \textstyle\sum_{i=1}^{n} f_{x_i}(\mathbf{X}_0)(d_{\mathbf{U}_0} g_i)(\mathbf{U} - \mathbf{U}_0) \\ &\quad + \left(\textstyle\sum_{i=1}^{n} f_{x_i}(\mathbf{X}_0) E_i(\mathbf{U})\right) |\mathbf{U} - \mathbf{U}_0| \\ &\quad + E(\mathbf{G}(\mathbf{U}))|\mathbf{G}(\mathbf{U}) - \mathbf{G}(\mathbf{U}_0|. \end{aligned}$$

Since

$$\lim_{\mathbf{U} \to \mathbf{U}_0} E(\mathbf{G}(\mathbf{U})) = \lim_{\mathbf{X} \to \mathbf{X}_0} E(\mathbf{X}) = 0.$$

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Due to Lemma we proved in previous module, imply that

$$\frac{h(\mathbf{U}) - h(\mathbf{U}_0) - \sum_{i=1}^{n} f_{x_i}(\mathbf{X}_0 d_{\mathbf{U}_0} g_i(\mathbf{U} - \mathbf{U}_0)}{|\mathbf{U} - \mathbf{U}_0|} = 0.$$

Therefore, h is differentiable at $\mathbf{U}_0$, and $d_{\mathbf{U}_0}h$ is given by (2.74).

Example: Let

$$f(x, y, z) = 2x^2 + 4xy + 3yz,$$

$$g_1(u, v) = u^2 + v^2, \quad g_2(u, v) = u^2 - 2v^2, \quad g_3(u, v) = uv,$$

and

$$h(u, v) = f(g_1(u, v), g_2(u, v), g_3(u, v)).$$

Let $\mathbf{U}_0 = (1, -1)$ and

$$\mathbf{X}_0 = (g_1(\mathbf{U}_0), g_2(\mathbf{U}_0), g_3(\mathbf{U}_0)) = (2, -1, -1).$$

Then

$$f_x(\mathbf{X}_0) = 4, \quad f_y(\mathbf{X}_0) = 5, \quad f_z(\mathbf{X}_0) = -3,$$

Since

$$g_1(u, v) = u^2 + v^2, \quad g_2(u, v) = u^2 - 2v^2, \quad g_3(u, v) = uv,$$

$$\begin{aligned}
\frac{\partial g_1(\mathbf{U}_0)}{\partial u} &= 2, & \frac{\partial g_1(\mathbf{U}_0)}{\partial v} &= -2, \\
\frac{\partial g_2(\mathbf{U}_0)}{\partial u} &= 2, & \frac{\partial g_2(\mathbf{U}_0)}{\partial v} &= 4, \\
\frac{\partial g_3(\mathbf{U}_0)}{\partial u} &= -1, & \frac{\partial g_3(\mathbf{U}_0)}{\partial v} &= 1.
\end{aligned}$$

Therefore,

$$d_{\mathbf{U}_0}g_1 = 2\,du - 2\,dv, \quad d_{\mathbf{U}_0}g_2 = 2\,du + 4\,dv, \quad d_{\mathbf{U}_0}g_3 = -du + dv.$$

According to chain rule we have

$$d_{\mathbf{U}_0}h = f_{x_1}(\mathbf{X}_0)d_{\mathbf{U}_0}g_1 + f_{x_2}(\mathbf{X}_0)d_{\mathbf{U}_0}g_2 + \cdots + f_{x_n}(\mathbf{X}_0)d_{\mathbf{U}_0}g_n.$$

$$\begin{aligned}
d_{\mathbf{U}_0}h &= f_x(\mathbf{X}_0)\, d_{\mathbf{U}_0}g_1 + f_y(\mathbf{X}_0)\, d_{\mathbf{U}_0}g_2 + f_z(\mathbf{X}_0)\, d_{\mathbf{U}_0}g_3 \\
&= 4(2\,du - 2\,dv) + 5(2\,du + 4\,dv) - 3(-du + dv) \\
&= 21\,du + 9\,dv.
\end{aligned}$$

Since

$$d_{\mathbf{U}_0}h = h_u(\mathbf{U}_0)\, du + h_v(\mathbf{U}_0)\, dv$$

we conclude that

$$h_u(\mathbf{U}_0) = 21 \quad \text{and} \quad h_v(\mathbf{U}_0) = 9. \tag{2.77}$$

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Alternatively: This can also be obtained by writing h explicitly in terms of (u, v) and differentiating; thus,

$$\begin{aligned} h(u,v) &= 2[g_1(u,v)]^2 + 4g_1(u,v)g_2(u,v) + 3g_2(u,v)g_3(u,v) \\ &= 2(u^2+v^2)^2 + 4(u^2+v^2)(u^2-2v^2) + 3(u^2-2v^2)uv \\ &= 6u^4 + 3u^3v - 6uv^3 - 6v^4. \end{aligned}$$

Hence,

$$h_u(u,v) = 24u^3 + 9u^2v - 6v^3 \quad \text{and} \quad h_v(u,v) = 3u^3 - 18uv^2 - 24v^3,$$

so $h_u(1,-1) = 21$ and $h_v(1,-1) = 9$, consistent with (2.77).

Corollary: Under the assumptions of the chain rule theorem

$$\frac{\partial h(\mathbf{U}_0)}{\partial u_i} = \sum_{j=1}^{n} \frac{\partial f(\mathbf{X}_0)}{\partial x_j} \frac{\partial g_j(\mathbf{U}_0)}{\partial u_i}, \quad 1 \le i \le m. \tag{2.78}$$

Proof:Substituting

$$d_{\mathbf{U}_0} g_i = \frac{\partial g_i(\mathbf{U}_0)}{\partial u_1}\, du_1 + \frac{\partial g_i(\mathbf{U}_0)}{\partial u_2}\, du_2 + \cdots + \frac{\partial g_i(\mathbf{U}_0)}{\partial u_m}\, du_m, \quad 1 \le i \le n,$$

into (2.74) and collecting multipliers of $du_1, du_2, \ldots, du_m$ yields

$$d_{\mathbf{U}_0} h = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} \frac{\partial f(\mathbf{X}_0)}{\partial x_j} \frac{\partial g_j(\mathbf{U}_0)}{\partial u_i} \right) du_i.$$

However, from Theorem **??**,

$$d_{\mathbf{U}_0} h = \sum_{i=1}^{m} \frac{\partial h(\mathbf{U}_0)}{\partial u_i}\, du_i.$$

Comparing the last two equations yields (2.78).

Remark: When it is not important to emphasize the particular point $\mathbf{X}_0$, we write

$$\frac{\partial h}{\partial u_i} = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} \frac{\partial g_j}{\partial u_i}, \quad 1 \le i \le m, \tag{2.79}$$

with the understanding that in calculating $\partial h(\mathbf{U}_0)/\partial u_i$, $\partial g_j/\partial u_i$ is evaluated at $\mathbf{U}_0$ and $\partial f/\partial x_j$ at $\mathbf{X}_0 = \mathbf{G}(\mathbf{U}_0)$.

$$\frac{\partial h}{\partial u_i} = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} \frac{\partial g_j}{\partial u_i}, \quad 1 \le i \le m, \tag{2.80}$$

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with the understanding that in calculating $\partial h(\mathbf{U}_0)/\partial u_i$, $\partial g_j/\partial u_i$ is evaluated at $\mathbf{U}_0$ and $\partial f/\partial x_j$ at $\mathbf{X}_0 = \mathbf{G}(\mathbf{U}_0)$. By replacing the symbol $\mathbf{G}$ with $\mathbf{X} = \mathbf{X}(\mathbf{U})$; then we write

$$h(\mathbf{U}) = f(\mathbf{X}(\mathbf{U}))$$

and

$$\frac{\partial h(\mathbf{U}_0)}{\partial u_i} = \sum_{j=1}^{n} \frac{\partial f(\mathbf{X}_0)}{\partial x_j} \frac{\partial x_j(\mathbf{U}_0)}{\partial u_i},$$

$$\text{or simply} \quad \frac{\partial h}{\partial u_i} = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} \frac{\partial x_j}{\partial u_i}. \tag{2.81}$$

2.21 Higher derivatives of composite functions

Higher derivatives of composite functions can be computed by repeatedly applying the chain rule.

For example, differentiating (2.81) with respect to u_k yields

$$\begin{aligned} \frac{\partial^2 h}{\partial u_k \partial u_i} &= \sum_{j=1}^{n} \frac{\partial}{\partial u_k} \left(\frac{\partial f}{\partial x_j} \frac{\partial x_j}{\partial u_i} \right) \\ &= \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} \frac{\partial^2 x_j}{\partial u_k \partial u_i} + \sum_{j=1}^{n} \frac{\partial x_j}{\partial u_i} \frac{\partial}{\partial u_k} \left(\frac{\partial f}{\partial x_j} \right). \end{aligned} \tag{2.82}$$

We must be careful finding

$$\frac{\partial}{\partial u_k} \left(\frac{\partial f}{\partial x_j} \right),$$

which really stands here for

$$\frac{\partial}{\partial u_k} \left(\frac{\partial f(\mathbf{X}(\mathbf{U}))}{\partial x_j} \right). \tag{2.83}$$

The safest procedure is to write temporarily

$$g(\mathbf{X}) = \frac{\partial f(\mathbf{X})}{\partial x_j};$$

then (2.83) becomes

$$\frac{\partial g(\mathbf{X}(\mathbf{U}))}{\partial u_k} = \sum_{s=1}^{n} \frac{\partial g(\mathbf{X}(\mathbf{U}))}{\partial x_s} \frac{\partial x_s(\mathbf{U})}{\partial u_k}.$$

Since

$$\frac{\partial g}{\partial x_s} = \frac{\partial^2 f}{\partial x_s \, \partial x_j},$$

this yields

$$\frac{\partial}{\partial u_k} \left(\frac{\partial f}{\partial x_k} \right) = \sum_{s=1}^{n} \frac{\partial^2 f}{\partial x_s \, \partial x_j} \frac{\partial x_s}{\partial u_k}.$$

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Substituting this into (2.82) yields

$$\frac{\partial^2 h}{\partial u_k\,\partial u_i} = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}\frac{\partial^2 x_j}{\partial u_k\,\partial u_i} + \sum_{j=1}^{n}\frac{\partial x_j}{\partial u_i}\sum_{s=1}^{n}\frac{\partial^2 f}{\partial x_s\,\partial x_j}\frac{\partial x_s}{\partial u_k}. \tag{2.84}$$

To compute $h_{u_i u_k}(\mathbf{U}_0)$ from this formula, we evaluate the partial derivatives of x_1, x_2, ..., x_n at $\mathbf{U}_0$ and those of f at $\mathbf{X}_0 = \mathbf{X}(\mathbf{U}_0)$. The formula is valid if x_1, x_2, ..., x_n and their first partial derivatives are differentiable at $\mathbf{U}_0$ and f, f_{x_i}, f_{x_2}, ..., f_{x_n} and their first partial derivatives are differentiable at $\mathbf{X}_0$.

Example: Let (r, θ) be polar coordinates in the xy-plane; that is,

$$x = r\cos\theta, \quad y = r\sin\theta.$$

Suppose that $f = f(x, y)$ is differentiable on a set S, and let

$$h(r, \theta) = f(r\cos\theta, r\sin\theta).$$

We have

$$\begin{aligned} \frac{\partial h}{\partial r} &= \frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial r} = \cos\theta\frac{\partial f}{\partial x} + \sin\theta\frac{\partial f}{\partial y} \\ \frac{\partial h}{\partial \theta} &= \frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta} = -r\sin\theta\frac{\partial f}{\partial x} + r\cos\theta\frac{\partial f}{\partial y}, \end{aligned} \tag{2.85}$$

where f_x and f_y are evaluated at $(x, y) = (r\cos\theta, r\sin\theta)$.

Example: Suppose that f_x and f_y just calculated are differentiable on an open set S in $\mathbb{R}^2$. Differentiating (2.85) with respect to r yields

$$\begin{aligned} \frac{\partial^2 h}{\partial r^2} &= \cos\theta\frac{\partial}{\partial r}\left(\frac{\partial f}{\partial x}\right) + \sin\theta\frac{\partial}{\partial r}\left(\frac{\partial f}{\partial y}\right) \\ &= \cos\theta\left(\frac{\partial^2 f}{\partial x^2}\frac{\partial x}{\partial r} + \frac{\partial^2 f}{\partial y\,\partial x}\frac{\partial y}{\partial r}\right) + \sin\theta\left(\frac{\partial^2 f}{\partial x\,\partial y}\frac{\partial x}{\partial r} + \frac{\partial^2 f}{\partial y^2}\frac{\partial y}{\partial r}\right). \end{aligned} \tag{2.86}$$

if $(x, y) \in S$. Since

$$\frac{\partial x}{\partial r} = \cos\theta, \quad \frac{\partial y}{\partial r} = \sin\theta, \quad \text{and} \quad \frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x}$$

if $(x, y) \in S$. The equation (2.86) yields

$$\frac{\partial^2 h}{\partial r^2} = \cos^2\theta\frac{\partial^2 f}{\partial x^2} + 2\sin\theta\cos\theta\frac{\partial^2 f}{\partial x\,\partial y} + \sin^2\theta\frac{\partial^2 f}{\partial y^2}.$$

Differentiating (2.85) with respect to θ yields

$$\begin{aligned} \frac{\partial^2 h}{\partial \theta\,\partial r} &= -\sin\theta\frac{\partial f}{\partial x} + \cos\theta\frac{\partial f}{\partial y} + \cos\theta\frac{\partial}{\partial \theta}\left(\frac{\partial f}{\partial x}\right) + \sin\theta\frac{\partial}{\partial \theta}\left(\frac{\partial f}{\partial y}\right) \\ &= -\sin\theta\frac{\partial f}{\partial x} + \cos\theta\frac{\partial f}{\partial y} + \cos\theta\left(\frac{\partial^2 f}{\partial x^2}\frac{\partial x}{\partial \theta} + \frac{\partial^2 f}{\partial y\,\partial x}\frac{\partial y}{\partial \theta}\right) \\ &\quad + \sin\theta\left(\frac{\partial^2 f}{\partial x\,\partial y}\frac{\partial x}{\partial \theta} + \frac{\partial^2 f}{\partial y^2}\frac{\partial y}{\partial \theta}\right). \end{aligned}$$

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Since

$$\frac{\partial x}{\partial \theta} = -r\sin\theta \quad \text{and} \quad \frac{\partial y}{\partial \theta} = r\cos\theta,$$

it follows that

$$\begin{aligned}\frac{\partial^2 h}{\partial\theta\,\partial r} &= -\sin\theta\frac{\partial f}{\partial x} + \cos\theta\frac{\partial f}{\partial y} - r\sin\theta\cos\theta\left(\frac{\partial^2 f}{\partial x^2} - \frac{\partial^2 f}{\partial y^2}\right)\\ &\quad + r(\cos^2\theta - \sin^2\theta)\frac{\partial^2 f}{\partial x\partial y}.\end{aligned}$$

Remark: For a composite function of the form

$$h(t) = f(x_1(t), x_2(t), \ldots, x_n(t))$$

where t is a real variable, $x_1, x_2, \ldots, x_n$ are differentiable at t_0, and f is differentiable at $\mathbf{X}_0 = \mathbf{X}(t_0)$. We have

$$h'(t_0) = \sum_{j=1}^{n} f_{x_j}(\mathbf{X}(t_0))x_j'(t_0). \tag{2.87}$$

Theorem: Let f be continuous at $\mathbf{X}_1 = (x_{11}, x_{21}, \ldots, x_{n1})$ and $\mathbf{X}_2 = (x_{12}, x_{22}, \ldots, x_{n2})$ and differentiable on the line segment L from $\mathbf{X}_1$ to $\mathbf{X}_2$.

Then

$$f(\mathbf{X}_2) - f(\mathbf{X}_1) = \sum_{i=1}^{n} f_{x_i}(\mathbf{X}_0)(x_{i2} - x_{i1}) = (d_{\mathbf{X}_0} f)(\mathbf{X}_2 - \mathbf{X}_1) \tag{2.88}$$

for some $\mathbf{X}_0$ on L distinct from $\mathbf{X}_1$ and $\mathbf{X}_2$.

Proof: An equation of L is

$$\mathbf{X} = \mathbf{X}(t) = t\mathbf{X}_2 + (1-t)\mathbf{X}_1, \quad 0 \le t \le 1.$$

Our hypotheses imply that the function

$$h(t) = f(\mathbf{X}(t))$$

is continuous on $[0,1]$ and differentiable on $(0,1)$.

Since

$$x_i(t) = tx_{i2} + (1-t)x_{i1},$$

We have

$$h'(t) = \sum_{i=1}^{n} f_{x_i}(\mathbf{X}(t))(x_{i2} - x_{i1}), \quad 0 < t < 1.$$

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From the mean value theorem for functions of one variable

$$h(1) - h(0) = h'(t_0)$$

for some $t_0 \in (0,1)$. Since $h(1) = f(\mathbf{X}_2)$ and $h(0) = f(\mathbf{X}_1)$, this implies (2.88) with $\mathbf{X}_0 = \mathbf{X}(t_0)$, i.e.,

$$f(\mathbf{X}_2) - f(\mathbf{X}_1) = \sum_{i=1}^{n} f_{x_i}(\mathbf{X}_0)(x_{i2} - x_{i1}) = (d_{\mathbf{X}_0} f)(\mathbf{X}_2 - \mathbf{X}_1).$$

Theorem: If $f_{x_1}, f_{x_2}, \ldots, f_{x_n}$ are identically zero in an open region S of $\mathbb{R}^n$, then f is constant in S.

Proof: We will show that if $\mathbf{X}_0$ and $\mathbf{X}$ are in S, then $f(\mathbf{X}) = f(\mathbf{X}_0)$.

Since S is an open region, S is polygonally connected.

Therefore, there are points

$$\mathbf{X}_0, \mathbf{X}_1, \ldots, \mathbf{X}_n = \mathbf{X}$$

such that the line segment L_i from $\mathbf{X}_{i-1}$ to $\mathbf{X}_i$ is in S, $1 \le i \le n$. From mean value theorem

$$f(\mathbf{X}_i) - f(\mathbf{X}_{i-1}) = \sum_{i=1}^{n} (d_{\widetilde{\mathbf{X}}_i} f)(\mathbf{X}_i - \mathbf{X}_{i-1}),$$

where $\widetilde{\mathbf{X}}$ is on L_i and therefore in S.

Therefore,

$$f_{x_i}(\widetilde{\mathbf{X}}_i) = f_{x_2}(\widetilde{\mathbf{X}}_i) = \cdots = f_{x_n}(\widetilde{\mathbf{X}}_i) = 0,$$

which means that $d_{\widetilde{\mathbf{X}}_i} f \equiv 0$. Hence,

$$f(\mathbf{X}_0) = f(\mathbf{X}_1) = \cdots = f(\mathbf{X}_n);$$

that is, $f(\mathbf{X}) = f(\mathbf{X}_0)$ for every $\mathbf{X}$ in S.

Motivation: Suppose that f is defined in an n-ball $B_\rho(\mathbf{X}_0)$, with $\rho > 0$.

If $\mathbf{X} \in B_\rho(\mathbf{X}_0)$, then

$$\mathbf{X}(t) = \mathbf{X}_0 + t(\mathbf{X} - \mathbf{X}_0) \in B_\rho(\mathbf{X}), \quad 0 \le t \le 1,$$

so the function

$$h(t) = f(\mathbf{X}(t))$$

is defined for $0 \le t \le 1$.

We know that

$$h'(t) = \sum_{i=1}^{n} f_{x_i}(\mathbf{X}(t)(x_i - x_{i0}).$$

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If f is differentiable in $B_\rho(\mathbf{X}_0)$, and

$$\begin{aligned} h''(t) &= \sum_{j=1}^{n} \frac{\partial}{\partial x_j} \left(\sum_{i=1}^{n} \frac{\partial f(\mathbf{X}(t))}{\partial x_i} (x_i - x_{i0}) \right) (x_j - x_{j0}) \\ &= \sum_{i,j=1}^{n} \frac{\partial^2 f(\mathbf{X}(t))}{\partial x_j \, \partial x_i} (x_i - x_{i0})(x_j - x_{j0}) \end{aligned}$$

If $f_{x_1}, f_{x_2}, \ldots, f_{x_n}$ are differentiable in $B_\rho(\mathbf{X}_0)$. Continuing in this way, we see that

$$\begin{aligned} h^{(r)}(t) &= \sum_{i_1, i_2, \ldots, i_r = 1}^{n} \frac{\partial^r f(\mathbf{X}(t))}{\partial x_{i_r} \, \partial x_{i_{r-1}} \cdots \partial x_{i_1}} (x_{i_1} - x_{i_1,0})(x_{i_2} - x_{i_2,0}) \\ &\quad \cdots (x_{i_r} - x_{i_r,0}) \end{aligned}$$

if all partial derivatives of f of order $\leq r-1$ are differentiable in $B_\rho(\mathbf{X_0})$.

2.22 rth Differential

Suppose that $r \geq 1$ and all partial derivatives of f of order $\leq r-1$ are differentiable in a neighborhood of $\mathbf{X}_0$.

Then the rth *differential of* f *at* $\mathbf{X}_0$, denoted by $d^{(r)}_{\mathbf{X}_0} f$, is defined by

$$d^{(r)}_{\mathbf{X}_0} f = \sum_{i_1, i_2, \ldots, i_r = 1}^{n} \frac{\partial^r f(\mathbf{X}_0)}{\partial x_{i_r} \partial x_{i_{r-1}} \cdots \partial x_{i_1}} dx_{i_1} dx_{i_2} \cdots dx_{i_r}, \tag{2.89}$$

where $dx_1, dx_2, \ldots, dx_n$ are the differentials, that is, dx_i is the function whose value at a point in $\mathbb{R}^n$ is the ith coordinate of the point. For convenience, we define

$$(d^{(0)}_{\mathbf{X}_0} f) = f(\mathbf{X}_0).$$

Notice that $d^{(1)}_{\mathbf{X}_0} f = d_{\mathbf{X}_0} f$.

Remark: Suppose that $r \geq 1$ and all partial derivatives of f of order $\leq r-1$ are differentiable in a neighborhood of $\mathbf{X}_0$, the value of

$$\frac{\partial^r f(\mathbf{X}_0)}{\partial x_{i_r} \partial x_{i_{r-1}} \cdots \partial x_{i_1}}$$

depends only on the number of times f is differentiated with respect to each variable, and not on the order in which the differentiations are performed.

Remark: The differential can be rewritten as

$$d^{(r)}_{\mathbf{X}_0} f = \sum_{r} \frac{r!}{r_1! r_2! \cdots r_n!} \frac{\partial^r f(\mathbf{X}_0)}{\partial x_1^{r_1} \partial x_2^{r_2} \cdots \partial x_n^{r_n}} (dx_1)^{r_1} (dx_2)^{r_2} \cdots (dx_n)^{r_n}, \tag{2.90}$$

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where $\sum_r$ indicates summation over all ordered n-tuples $(r_1, r_2, \ldots, r_n)$ of nonnegative integers such that

$$r_1 + r_2 + \cdots + r_n = r$$

and $\partial x_i^{r_i}$ is omitted from the "denominators" of all terms in (2.90) for which $r_i = 0$. In particular, if $n = 2$,

$$d^{(r)}_{\mathbf{X}_0} f = \sum_{j=0}^{r} \binom{r}{j} \frac{\partial^r f(x_0, y_0)}{\partial x^j \, \partial y^{r-j}} (dx)^j (dy)^{r-j}.$$

Example: Let

$$f(x, y) = \frac{1}{1 + ax + by},$$

where a and b are constants.

Then

$$\frac{\partial^r f(x, y)}{\partial x^j \, \partial y^{r-j}} = (-1)^r r! \frac{a^j b^{r-j}}{(1 + ax + by)^{r+1}},$$

so

$$\begin{aligned} d^{(r)}_{\mathbf{X}_0} f &= \frac{(-1)^r r!}{(1 + ax_0 + by_0)^{r+1}} \sum_{j=0}^{r} \binom{r}{j} a^j b^{r-j} (dx)^j (dy)^{r-j} \\ &= \frac{(-1)^r r!}{(1 + ax_0 + by_0)^{r+1}} (a\,dx + b\,dy)^r \end{aligned}$$

if $1 + ax_0 + by_0 \neq 0$.

Let

$$f(\mathbf{X}) = \exp\left(- \sum_{j=1}^{n} a_j x_j \right),$$

where $a_1, a_2, \ldots, a_n$ are constants. Then

$$\frac{\partial^r f(\mathbf{X})}{\partial x_1^{r_1} \partial x_2^{r_2} \cdots \partial x_n^{r_n}} = (-1)^r a_1^{r_1} a_2^{r_2} \cdots a_n^{r_n} \exp\left(- \sum_{j=1}^{n} a_j x_j \right).$$

Therefore,

$$\begin{aligned} (d^{(r)}_{\mathbf{X}_0} f)(\mathbf{\Phi}) &= (-1)^r \left(\sum_r \frac{r!}{r_1! r_2! \cdots r_n!} a_1^{r_1} a_2^{r_2} \cdots a_n^{r_n} (dx_1)^{r_1} (dx_2) r_2 \cdots (dx_n)^{r_n} \right) \\ &\quad \times \exp\left(- \sum_{j=1}^{n} a_j x_{j0} \right) \\ &= (-1)^r (a_1\,dx_1 + a_2\,dx_2 + \cdots + a_n\,dx_n)^r \exp\left(- \sum_{j=1}^{n} a_j x_{j0} \right) \end{aligned}$$

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2.23 Taylor's Theorem for Functions of n Variables

Theorem: Suppose that f and its partial derivatives of order $\leq k$ are differentiable at $\mathbf{X}_0$ and $\mathbf{X}$ in $\mathbb{R}^n$ and on the line segment L connecting them.

Then

$$f(\mathbf{X}) = \sum_{r=0}^{k} \frac{1}{r!}(d^{(r)}_{\mathbf{X}_0} f)(\mathbf{X} - \mathbf{X}) + \frac{1}{(k+1)!}(d^{(k+1)}_{\widetilde{\mathbf{X}}} f)(\mathbf{X} - \mathbf{X}_0) \tag{2.91}$$

for some $\widetilde{\mathbf{X}}$ on L distinct from $\mathbf{X}_0$ and $\mathbf{X}$.

Proof: Define

$$h(t) = f(\mathbf{X}_0 + t(\mathbf{X} - \mathbf{X}_0)). \tag{2.92}$$

With $\mathbf{\Phi} = \mathbf{X} - \mathbf{X}_0$, our assumptions and the discussion preceding Definition of differentials imply that $h, h', \ldots, h^{(k+1)}$ exist on $[0, 1]$.

From Taylor's theorem for functions of one variable,

$$h(1) = \sum_{r=0}^{k} \frac{h^{(r)}(0)}{r!} + \frac{h^{(k+1)}(\tau)}{(k+1)!}, \tag{2.93}$$

for some $\tau \in (0, 1)$.

From (2.92),

$$h(0) = f(\mathbf{X}_0) \quad \text{and} \quad h(1) = f(\mathbf{X}). \tag{2.94}$$

We have $\mathbf{\Phi} = \mathbf{X} - \mathbf{X}_0$,

$$h^{(r)}(0) = (d^{(r)}_{\mathbf{X}_0} f)(\mathbf{X} - \mathbf{X}_0), \quad 1 \leq r \leq k, \tag{2.95}$$

$$h^{(k+1)}(\tau) = \left(d^{k+1}_{\widetilde{\mathbf{X}}} f\right)(\mathbf{X} - \mathbf{X}_0) \tag{2.96}$$

where

$$\widetilde{\mathbf{X}} = \mathbf{X}_0 + \tau(\mathbf{X} - \mathbf{X}_0)$$

is on L and distinct from $\mathbf{X}_0$ and $\mathbf{X}$.

Substituting (2.94), (2.95), and (2.96) into (2.93) yields (2.91).

Let

$$f(x, y) = \frac{1}{1 + ax + by},$$

where a and b are constants.

Then

$$\frac{\partial^r f(x,y)}{\partial x^j \, \partial y^{r-j}} = (-1)^r r! \frac{a^j b^{r-j}}{(1 + ax + by)^{r+1}},$$

so

$$\begin{aligned} d^{(r)}_{\mathbf{X}_0} f &= \frac{(-1)^r r!}{(1 + ax_0 + by_0)^{r+1}} \sum_{j=0}^{r} \binom{r}{j} a^j b^{r-j} (dx)^j (dy)^{r-j} \\ &= \frac{(-1)^r r!}{(1 + ax_0 + by_0)^{r+1}} (a\,dx + b\,dy)^r \end{aligned}$$

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if $1 + ax_0 + by_0 \neq 0$.

Example: The Taylor series with $\mathbf{X}_0 = (0,0)$ and $\mathbf{\Phi} = (x,y)$ imply that if $1+ax+by > 0$, then

$$\frac{1}{1+ax+by} = \sum_{r=0}^{k}(-1)^r(ax+by)^r + (-1)^{k+1}\frac{(ax+by)^{k+1}}{(1+a\tau x+b\tau y)^{k+2}}$$

for some $\tau \in (0,1)$. (Note that τ depends on k as well as (x,y).)

Remark: By analogy with the situation for functions of one variable, we define the kth *Taylor polynomial of f about* $\mathbf{X}_0$ by

$$T_k(\mathbf{X}) = \sum_{r=0}^{k}\frac{1}{r!}(d^{(r)}_{\mathbf{X}_0}f)(\mathbf{X}-\mathbf{X}_0).$$

If the differentials exist; then we have

$$f(\mathbf{X}) = T_k(\mathbf{X}) + \frac{1}{(k+1)!}(d^{(k+1)}_{\widetilde{\mathbf{X}}}f)(\mathbf{X}-\mathbf{X}_0).$$

Theorem: Suppose that f and its partial derivatives of order $\leq k-1$ are differentiable in a neighborhood N of a point $\mathbf{X}_0$ in $\mathbb{R}^n$ and all kth-order partial derivatives of f are continuous at $\mathbf{X}_0$. Then

$$\lim_{\mathbf{X}\to\mathbf{X}_0}\frac{f(\mathbf{X})-T_k(\mathbf{X})}{|\mathbf{X}-\mathbf{X}_0|^k} = 0. \tag{2.97}$$

Proof: If $\varepsilon > 0$, there is a $\delta > 0$ such that $B_\delta(\mathbf{X}_0) \subset N$ and all kth-order partial derivatives of f satisfy the inequality

$$\left|\frac{\partial^k f(\widetilde{\mathbf{X}})}{\partial x_{i_k}\partial x_{i_{k-1}}\cdots\partial x_{i_1}} - \frac{\partial^k f(\mathbf{X}_0)}{\partial x_{i_k}\partial x_{i_{k-1}}\cdots\partial x_{i_1}}\right| < \varepsilon, \quad \widetilde{\mathbf{X}} \in B_\delta(\mathbf{X}_0). \tag{2.98}$$

Now suppose that $\mathbf{X} \in B_\delta(\mathbf{X}_0)$. From Taylor series expansion, with k replaced by $k-1$,

$$f(\mathbf{X}) = T_{k-1}(\mathbf{X}) + \frac{1}{k!}(d^{(k)}_{\widetilde{\mathbf{X}}}f)(\mathbf{X}-\mathbf{X}_0), \tag{2.99}$$

where $\widetilde{\mathbf{X}}$ is some point on the line segment from $\mathbf{X}_0$ to $\mathbf{X}$ and is therefore in $B_\delta(\mathbf{X}_0)$.

We can rewrite (2.99) as

$$f(\mathbf{X}) = T_k(\mathbf{X}) + \frac{1}{k!}\left[(d^{(k)}_{\widetilde{\mathbf{X}}}f)(\mathbf{X}-\mathbf{X}_0) - (d^{(k)}_{\mathbf{X}_0}f)(\mathbf{X}-\mathbf{X}_0)\right]. \tag{2.100}$$

But definition of differential and (2.98) imply that

$$\left|(d^{(k)}_{\widetilde{\mathbf{X}}}f)(\mathbf{X}-\mathbf{X}_0) - (d^{(k)}_{\mathbf{X}_0}f)(\mathbf{X}-\mathbf{X}_0)\right| < n^k\varepsilon|\mathbf{X}-\mathbf{X}_0|^k \tag{2.101}$$

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which implies that

$$\frac{|f(\mathbf{X}) - T_k(\mathbf{X})|}{|\mathbf{X} - \mathbf{X}_0|^k} < \frac{n^k \varepsilon}{k!}, \quad \mathbf{X} \in B_\delta(\mathbf{X}_0),$$

from (2.100). This implies (2.97).

2.23.1 Positive Definite

Let r be a positive integer and $\mathbf{X}_0 = (x_{10}, x_{20}, \ldots, x_{n0})$. A function of the form

$$p(\mathbf{X}) = \sum_r a_{r_1 r_2 \ldots r_n} (x_1 - x_{10})^{r_1} (x_2 - x_{20})^{r_2} \cdots (x_n - x_{n0})^{r_n}, \tag{2.102}$$

where the coefficients $\{a_{r_1 r_2 \ldots r_n}\}$ are constants and the summation is over all n-tuples of nonnegative integers $(r_1, r_2, \ldots, r_n)$ such that

$$r_1 + r_2 + \cdots + r_n = r,$$

is a *homogeneous polynomial of degree* r *in* $\mathbf{X} - \mathbf{X}_0$, provided that at least one of the coefficients is nonzero. For example, if f satisfies the conditions of rth differential, then the function

$$p(\mathbf{X}) = (d^{(r)}_{\mathbf{X}_0} f)(\mathbf{X} - \mathbf{X}_0)$$

is such a polynomial if at least one of the rth-order mixed partial derivatives of f at $\mathbf{X}_0$ is nonzero. Clearly, $p(\mathbf{X}_0) = 0$ if p is a homogeneous polynomial of degree $r \geq 1$ in $\mathbf{X} - \mathbf{X}_0$.

If $p(\mathbf{X}) \geq 0$ for all $\mathbf{X}$, we say that p is *positive semidefinite*; if $p(\mathbf{X}) > 0$ except when $\mathbf{X} = \mathbf{X}_0$, p is *positive definite*. Similarly, p is *negative semidefinite* if $p(\mathbf{X}) \leq 0$ or *negative definite* if $p(\mathbf{X}) < 0$ for all $\mathbf{X} \neq \mathbf{X}_0$. In all these cases, p is *semidefinite*. With p as in (2.102),

$$p(-\mathbf{X} + 2\mathbf{X}_0) = (-1)^r p(\mathbf{X}),$$

so p cannot be semidefinite if r is odd.

Example: The polynomial

$$p(x, y, z) = x^2 + y^2 + z^2 + xy + xz + yz$$

is homogeneous of degree 2 in $\mathbf{X} = (x, y, z)$. We can rewrite p as

$$p(x, y, z) = \frac{1}{2}\left[(x + y)^2 + (y + z)^2 + (z + x)^2\right].$$

so p is nonnegative, and $p(\overline{x}, \overline{y}, \overline{z}) = 0$ if and only if

$$\overline{x} + \overline{y} = \overline{y} + \overline{z} = \overline{z} + \overline{x} = 0,$$

which is equivalent to $(\overline{x}, \overline{y}, \overline{z}) = (0, 0, 0)$. Therefore, p is positive definite and $-p$ is negative definite.

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Example: The polynomial

$$\begin{aligned} p_1(x,y,z) &= x^2+y^2+z^2+2xy \\ p_1(x,y,z) &= (x+y)^2+z^2, \end{aligned}$$

so p_1 is nonnegative. Since $p_1(1,-1,0)=0$, p_1 is positive semidefinite and $-p_1$ is negative semidefinite.

The polynomial

$$p_2(x,y,z) = x^2-y^2+z^2$$

is not semidefinite, since, for example,

$$p_2(1,0,0)=1 \quad \text{and} \quad p_2(0,1,0)=1.$$

Theorem: Suppose that f and its partial derivatives of order $\leq k-1$ are differentiable in a neighborhood N of a point $\mathbf{X}_0$ in $\mathbb{R}^n$ and all kth-order partial derivatives of f are continuous at $\mathbf{X}_0$. with $k \geq 2$, and

$$d^{(r)}_{\mathbf{X}_0} f \equiv 0 \quad (1 \leq r \leq k-1), \quad d^{(k)}_{\mathbf{X}_0} f \not\equiv 0. \tag{2.103}$$

Then

- $\mathbf{X}_0$ is not a local extreme point of f unless $d^{(k)}_{\mathbf{X}_0} f$ is semi-definite as a polynomial in $\mathbf{X}-\mathbf{X}_0$.

  In particular, $\mathbf{X}_0$ is not a local extreme point of f if k is odd.

- $\mathbf{X}_0$ is a local minimum point of f if $d^{(k)}_{\mathbf{X}_0} f$ is positive definite, or a local maximum point if $d^{(k)}_{\mathbf{X}_0} f$ is negative definite.

- If $d^{(k)}_{\mathbf{X}_0} f$ is semidefinite, then $\mathbf{X}_0$ may be a local extreme point of f, but it need not be.

Corollary: Suppose that f, f_x, and f_y are differentiable in a neighborhood of a critical point $\mathbf{X}_0=(x_0,y_0)$ of f and f_{xx}, f_{yy}, and f_{xy} are continuous at (x_0,y_0).

Let

$$D = f_{xx}(x_0,y_0)f_{xy}(x_0,y_0) - f^2_{xy}(x_0,y_0).$$

Then

- (x_0,y_0) is a local extreme point of f if $D>0$; (x_0,y_0) is a local minimum point if $f_{xx}(x_0,y_0)>0$, or a local maximum point if $f_{xx}(x_0,y_0)<0$.

- (x_0,y_0) is not a local extreme point of f if $D<0$.

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Proof: Write $(x - x_0, y - y_0) = (u, v)$ and

$$p(u, v) = (d^{(2)}_{\mathbf{X}_0} f)(u, v) = Au^2 + 2Buv + Cv^2,$$

where $A = f_{xx}(x_0, y_0)$, $B = f_{xy}(x_0, y_0)$, and $C = f_{yy}(x_0, y_0)$, so

$$D = AC - B^2.$$

If $D > 0$, then $A \neq 0$, and we can write

$$\begin{aligned} p(u, v) &= A\left(u^2 + \frac{2B}{A}uv + \frac{B^2}{A^2}v^2\right) + \left(C - \frac{B^2}{A}\right)v^2 \\ &= A\left(u + \frac{B}{A}v\right)^2 + \frac{D}{A}v^2. \end{aligned}$$

This cannot vanish unless $u = v = 0$. Hence, $d^{(2)}_{\mathbf{X}_0} f$ is positive definite if $A > 0$ or negative definite if $A < 0$, and Theorem implies the first part of the corollary.

If $D < 0$, there are three possibilities:

1. $A \neq 0$; then $p(1, 0) = A$ and $p\left(-\frac{B}{A}, 1\right) = \frac{D}{A}$.

2. $C \neq 0$; then $p(0, 1) = C$ and $p\left(1, -\frac{B}{C}\right) = \frac{D}{C}$.

3. $A = C = 0$; then $B \neq 0$ and $p(1, 1) = 2B$ and $p(1, -1) = -2B$.

In each case the two given values of p differ in sign, so $\mathbf{X}_0$ is not a local extreme point of f, from Theorem part I.

Example: If

$$f(x, y) = e^{ax^2 + by^2}.$$

We have

$$f_x(x, y) = 2axf(x, y), \quad f_y(x, y) = 2byf(x, y),$$

so

$$f_x(0, 0) = f_y(0, 0) = 0,$$

and $(0, 0)$ is a critical point of f.

To apply Corollary, we calculate

$$\begin{aligned} f_{xx}(x, y) &= (2a + 4a^2x^2)f(x, y), \\ f_{yy}(x, y) &= (2b + 4b^2y^2)f(x, y), \\ f_{xy}(x, y) &= 4abxyf(x, y). \end{aligned}$$

Therefore,

$$D = f_{xx}(0, 0)f_{yy}(0, 0) - f^2_{xy}(0, 0) = (2a)(2b) - (0)(0) = 4ab.$$

Corollary implies that $(0, 0)$ is a local minimum point if a and b are positive. A local maximum if a and b are negative. Neither if one is positive and the other is negative. Corollary does not apply if a or b is zero.

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Chapter 3

Integral Calculus

Attempting to formulate definition of Riemann integral for a function defined on an infinite or semi-infinite interval would introduce questions concerning convergence of the resulting Riemann sums, which would be infinite series.

3.1 Locally Integrable Functions

We say f is *locally integrable* on an interval I if f is integrable on every finite closed subinterval of I.

For example,

$$f(x) = \sin x$$

is locally integrable on $(-\infty, \infty)$.

$$g(x) = \frac{1}{x(x-1)}$$

is locally integrable on $(-\infty, 0)$, $(0, 1)$, and $(1, \infty)$.

The function

$$h(x) = \sqrt{x}$$

is locally integrable on $[0, \infty)$.

If f is locally integrable on $[a, b)$, we define

$$\int_a^b f(x)\,dx = \lim_{c \to b-} \int_a^c f(x)\,dx \tag{3.1}$$

if the limit exists (finite). To include the case where $b = \infty$, we adopt the convention that $\infty - = \infty$.

Remarks:

- The limit in (3.1) always exists if $[a, b)$ is finite and f is locally integrable and bounded on $[a, b)$.
- In this case, the definition of Riemann integral and locally integrable function assign the same value to $\int_a^b f(x)\,dx$ no matter how $f(b)$ is defined. However, the limit may also exist in cases where $b = \infty$ or $b < \infty$ and f is unbounded as x approaches b from the left.

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- In these cases, the definition of locally integrable assigns a value to an integral that does not exist in the sense of Riemann integral, and $\int_a^b f(x)\,dx$ is said to be an *improper integral* that *converges* to the limit in (3.1).

Remarks:

- We also say in this case that f is *integrable on* $[a,b)$ and that $\int_a^b f(x)\,dx$ *exists*. If the limit in (3.1) does not exist (finite), we say that the improper integral $\int_a^b f(x)\,dx$ *diverges*, and f is *nonintegrable on* $[a,b)$.

- In particular, if $\lim_{c\to b-} \int_a^c f(x)\,dx = \pm\infty$, we say that $\int_a^b f(x)\,dx$ *diverges to* $\pm\infty$, and we write

$$\int_a^b f(x)\,dx = \infty \quad \text{or} \quad \int_a^b f(x)\,dx = -\infty,$$

  whichever the case may be.

If f is locally integrable on $(a,b]$, we define

$$\int_a^b f(x)\,dx = \lim_{c\to a+} \int_c^b f(x)\,dx$$

provided that the limit exists (finite).

To include the case where $a = -\infty$, we adopt the convention that $-\infty+ = -\infty$.

If f is locally integrable on (a,b), we define

$$\int_a^b f(x)\,dx = \int_a^\alpha f(x)\,dx + \int_\alpha^b f(x)\,dx,$$

where $a < \alpha < b$, provided that both improper integrals on the right exist (finite).

Remarks: The existence and value of $\int_a^b f(x)\,dx$ according to the above definition do not depend on the particular choice of α in (a,b).

When we wish to distinguish between improper integrals and integrals in the sense of definition of Riemann integral, we will call the latter *proper integrals*.

Example: The function

$$f(x) = 2x \sin\frac{1}{x} - \cos\frac{1}{x}$$

is locally integrable and the derivative of

$$F(x) = x^2 \sin\frac{1}{x}$$

on $[-2/\pi, 0)$.

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Hence,

$$\int_{-2/\pi}^{c} f(x)\,dx = x^2 \sin\frac{1}{x}\bigg|_{-2/\pi}^{c} = c^2 \sin\frac{1}{c} + \frac{4}{\pi^2}$$
$$\int_{-2/\pi}^{0} f(x)\,dx = \lim_{c\to 0-}\left(c^2 \sin\frac{1}{c} + \frac{4}{\pi^2}\right) = \frac{4}{\pi^2}.$$

However, this is not an improper integral, even though $f(0)$ is not defined and cannot be defined so as to make f continuous at 0. If we define $f(0)$ arbitrarily (say $f(0) = 10$), then f is bounded on the closed interval $[-2/\pi, 0]$ and continuous except at 0. Therefore, $\int_{-2/\pi}^{0} f(x)\,dx$ exists and equals $4/\pi^2$ as a proper integral, in the sense of definition of improper integral.

Example: The function

$$f(x) = (1-x)^{-p}$$

is locally integrable on $[0, 1)$.

If $p \neq 1$ and $0 < c < 1$,

$$\int_0^c (1-x)^{-p}\,dx = \frac{(1-x)^{-p+1}}{p-1}\bigg|_0^c = \frac{(1-c)^{-p+1}-1}{p-1}.$$

Hence,

$$\lim_{c\to 1-}\int_0^c (1-x)^{-p}\,dx = \begin{cases} (1-p)^{-1}, & p<1, \\ \infty, & p>1. \end{cases}$$

For $p = 1$,

$$\lim_{c\to 1-}\int_0^c (1-x)^{-1}\,dx = -\lim_{c\to 1-}\log(1-c) = \infty.$$

Hence,

$$\int_0^1 (1-x)^{-p}\,dx = \begin{cases} (1-p)^{-1}, & p<1, \\ \infty, & p\geq 1. \end{cases}$$

Example: The function

$$f(x) = x^{-p}$$

is locally integrable on $[1, \infty)$.

If $p \neq 1$ and $c > 1$,

$$\int_1^c x^{-p}\,dx = \frac{x^{-p+1}}{-p+1}\bigg|_1^c = \frac{c^{-p+1}-1}{-p+1}.$$

Hence,

$$\lim_{c\to\infty}\int_1^c x^{-p}\,dx = \begin{cases} (p-1)^{-1}, & p>1, \\ \infty, & p<1. \end{cases}$$

For $p = 1$,

$$\lim_{c\to\infty}\int_1^c x^{-1}\,dx = \lim_{c\to\infty}\log c = \infty.$$

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Hence,

$$\int_1^\infty x^{-p}\,dx = \begin{cases} (p-1)^{-1}, & p > 1, \\ \infty, & p \le 1. \end{cases}$$

Example: If $1 < c < \infty$, then

$$\int_1^c \frac{1}{x}\log\frac{1}{x}\,dx = -\int_1^c \frac{1}{x}\log x\,dx = -\frac{1}{2}(\log x)^2\Big|_1^c = -\frac{1}{2}(\log c)^2.$$

Hence,

$$\lim_{c\to\infty}\int_1^c \frac{1}{x}\log\frac{1}{x}\,dx = -\infty,$$

so

$$\int_1^\infty \frac{1}{x}\log\frac{1}{x}\,dx = -\infty.$$

The function $f(x) = \log x$ is locally integrable on $(0, 1]$, but unbounded as $x \to 0+$. Since

$$\lim_{c\to 0+}\int_c^1 \log x\,dx = \lim_{c\to 0+}(x\log x - x)\Big|_c^1 = -1 - \lim_{c\to 0+}(c\log c - c) = -1,$$

Definition ?? yields

$$\int_0^1 \log x\,dx = -1.$$

The function $f(x) = \cos x$ is locally integrable on $[0, \infty)$ and

$$\lim_{c\to\infty}\int_0^c \cos x\,dx = \lim_{c\to\infty}\sin c$$

does not exist; thus, $\int_0^\infty \cos x\,dx$ diverges, but not to $\pm\infty$.

In connection with Definition ??, it is important to recognize that the improper integrals $\int_a^\alpha f(x)\,dx$ and $\int_\alpha^b f(x)\,dx$ must converge *separately* for $\int_a^b f(x)\,dx$ to converge. For example, the existence of the symmetric limit

$$\lim_{R\to\infty}\int_{-R}^R f(x)\,dx,$$

which is called the *principal value of* $\int_{-\infty}^\infty f(x)\,dx$, does not imply that $\int_{-\infty}^\infty f(x)\,dx$ converges; thus,

$$\lim_{R\to\infty}\int_{-R}^R x\,dx = \lim_{R\to\infty} 0 = 0,$$

but $\int_0^\infty x\,dx$ and $\int_{-\infty}^0 x\,dx$ diverge and therefore so does $\int_{-\infty}^\infty x\,dx$.

Theorem: Suppose that $f_1, f_2, \ldots, f_n$ are locally integrable on $[a, b)$.
The integrals $\int_a^b f_1(x)\,dx$, $\int_a^b f_2(x)\,dx$, $\ldots$, $\int_a^b f_n(x)\,dx$ converge.
Let $c_1, c_2, \ldots, c_n$ be constants. Then $\int_a^b (c_1 f + c_2 f_1 + \cdots + c_n f_n)(x)\,dx$ converges.

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Furthermore,

$$\begin{aligned}\int_a^b (c_1 f_1 + c_2 f_2 + \cdots + c_n f_n)(x)\,dx &= c_1 \int_a^b f_1(x)\,dx \\ &\quad + c_2 \int_a^b f_2(x)\,dx \\ &\quad + \cdots + c_n \int_a^b f_n(x)\,dx.\end{aligned}$$

Proof: If $a < c < b$, then

$$\begin{aligned}\int_a^c (c_1 f_1 + c_2 f_2 + \cdots + c_n f_n)(x)\,dx &= c_1 \int_a^c f_1(x)\,dx \\ &\quad + c_2 \int_a^c f_2(x)\,dx \\ &\quad + \cdots + c_n \int_a^c f_n(x)\,dx.\end{aligned}$$

Letting $c \to b-$ yields the stated result.

Theorem: If f is nonnegative and locally integrable on $[a, b)$, then $\int_a^b f(x)\,dx$ converges if the function

$$F(x) = \int_a^x f(t)\,dt$$

is bounded on $[a, b)$, and $\int_a^b f(x)\,dx = \infty$ if it is not.

These are the only possibilities, and

$$\int_a^b f(t)\,dt = \sup_{a \le x < b} F(x)$$

in either case.

Proof: The function

$$F(x) = \int_a^x f(t)\,dt$$

is nondecreasing on $[a, b)$.

Recall: Suppose that f is monotonic on (a, b) and define

$$\alpha = \inf_{a<x<b} f(x), \qquad \beta = \sup_{a<x<b} f(x).$$

If f is nondecreasing, then $f(a+) = \alpha$ and $f(b-) = \beta$.

Remarks: We often write

$$\int_a^b f(x)\,dx < \infty$$

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to indicate that an improper integral of a nonnegative function converges.

Similarly, if f is nonpositive and $\int_a^b f(x)\,dx$ converges, we write

$$\int_a^b f(x)\,dx > -\infty$$

because a divergent integral of this kind can only diverge to $-\infty$.

- These conventions do not apply to improper integrals of functions that assume both positive and negative values in (a, b), since they may diverge without diverging to $\pm\infty$.

3.1.1 The Comparison Test

Theorem: If f and g are locally integrable on $[a, b)$ and

$$0 \le f(x) \le g(x), \quad a \le x < b, \tag{3.2}$$

then

1. $\int_a^b f(x)\,dx < \infty$ if $\int_a^b g(x)\,dx < \infty$

2. $\int_a^b g(x)\,dx = \infty$ if $\int_a^b f(x)\,dx = \infty$.

Proof: Since

$$0 \le f(x) \le g(x), \quad a \le x < b,$$

we have

$$\int_a^x f(t)\,dt \le \int_a^x g(t)\,dt, \quad a \le x < b.$$

So

$$\sup_{a\le x<b} \int_a^x f(t)\,dt \le \sup_{a\le x\le b} \int_a^x g(t)\,dt.$$

If $\int_a^b g(x)\,dx < \infty$, the right side of this inequality is finite by the previous Theorem, so the left side is also.

This implies that $\int_a^b f(x)\,dx < \infty$.

The proof is by contradiction. If $\int_a^b g(x)\,dx < \infty$, then (1) implies that $\int_a^b f(x)\,dx < \infty$, contradicting the assumption that $\int_a^b f(x)\,dx = \infty$.

Example: Determine the convergence of the improper integral

$$I = \int_0^1 \frac{2 + \sin \pi x}{(1-x)^p}\,dx.$$

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Solution: We are going to show that the improper integral converges if $p < 1$. Since

$$0 < \frac{2 + \sin \pi x}{(1-x)^p} \leq \frac{3}{(1-x)^p}, \quad 0 \leq x < 1.$$

We have

$$\int_0^1 \frac{3\,dx}{(1-x)^p} < \infty, \quad p < 1.$$

Example: Determine the convergence of the improper integral

$$I = \int_0^1 \frac{2 + \sin \pi x}{(1-x)^p}\,dx.$$

Solution: However, I diverges if $p \geq 1$, since

$$0 < \frac{1}{(1-x)^p} \leq \frac{2 + \sin \pi x}{(1-x)^p}, \quad 0 \leq x < 1,$$

and

$$\int_0^1 \frac{dx}{(1-x)^p} = \infty, \quad p \geq 1.$$

Remark: If f is any function (not necessarily nonnegative) locally integrable on $[a, b)$. If a_1 and c are in $[a, b)$, then

$$\int_a^c f(x)\,dx = \int_a^{a_1} f(x)\,dx + \int_{a_1}^c f(x)\,dx.$$

Since $\int_a^{a_1} f(x)\,dx$ is a proper integral, on letting $c \to b-$ we conclude that if either of the improper integrals $\int_a^b f(x)\,dx$ and $\int_{a_1}^b f(x)\,dx$ converges then so does the other, and in this case

$$\int_a^b f(x)\,dx = \int_a^{a_1} f(x)\,dx + \int_{a_1}^b f(x)\,dx.$$

Remark: This means that any theorem implying convergence or divergence of an improper integral $\int_a^b f(x)\,dx$ remains valid if its hypotheses are satisfied on a subinterval $[a_1, b)$ of $[a, b)$ rather than on all of $[a, b)$.

For example, the comparison test remains valid if we have

$$0 \leq f(x) \leq g(x), \quad a_1 \leq x < b,$$

where a_1 is any point in $[a, b)$.

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From this, you can see that if $f(x) \geq 0$ on some subinterval $[a_1, b)$ of $[a, b)$, but not necessarily for all x in $[a, b)$, we can still use the convention introduced earlier for positive functions; that is, we can write $\int_a^b f(x)\,dx < \infty$ if the improper integral converges or $\int_a^b f(x)\,dx = \infty$ if it diverges.

Theorem: Suppose that f and g are locally integrable on $[a, b)$, $g(x) > 0$ and $f(x) \geq 0$ on some subinterval $[a_1, b)$ of $[a, b)$, and

$$\lim_{x \to b-} \frac{f(x)}{g(x)} = M. \tag{3.3}$$

- If $0 < M < \infty$, then $\int_a^b f(x)\,dx$ and $\int_a^b g(x)\,dx$ converge or diverge together.
- If $M = \infty$ and $\int_a^b g(x)\,dx = \infty$, then $\int_a^b f(x)\,dx = \infty$.
- If $M = 0$ and $\int_a^b g(x)\,dx < \infty$, then $\int_a^b f(x)\,dx < \infty$.

Proof: From (3.3), there is a point a_2 in $[a_1, b)$ such that

$$0 < \frac{M}{2} < \frac{f(x)}{g(x)} < \frac{3M}{2}, \quad a_2 \leq x < b,$$

and therefore

$$\frac{M}{2} g(x) < f(x) < \frac{3M}{2} g(x), \quad a_2 \leq x < b. \tag{3.4}$$

The first inequality in (3.4) imply that

$$\int_{a_2}^b g(x)\,dx < \infty \quad \text{if} \quad \int_{a_2}^b f(x)\,dx < \infty.$$

The second inequality in (3.4) imply that

$$\int_{a_2}^b f(x)\,dx < \infty \quad \text{if} \quad \int_{a_2}^b g(x)\,dx < \infty.$$

Therefore, $\int_{a_2}^b f(x)\,dx$ and $\int_{a_2}^b g(x)\,dx$ converge or diverge together, and in the latter case they must diverge to ∞, since their integrands are nonnegative. If $M = \infty$, there is a point a_2 in $[a_1, b)$ such that

$$f(x) \geq g(x), \quad a_2 \leq x \leq b,$$

We have $\int_a^b f(x)\,dx = \infty$. If $M = 0$, there is a point a_2 in $[a_1, b)$ such that

$$f(x) \leq g(x), \quad a_2 \leq x \leq b,$$

so we have $\int_a^b f(x)\,dx < \infty$.

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3.2 Absolute integrability

We say that f is *absolutely integrable on* $[a,b)$ if f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|\,dx < \infty$. In this case we also say that $\int_a^b f(x)\,dx$ *converges absolutely* or *is absolutely convergent*.

Remark: If f is nonnegative and integrable on $[a,b)$, then f is absolutely integrable on $[a,b)$, since $|f| = f$.

Example: Since

$$\left|\frac{\sin x}{x^p}\right| \le \frac{1}{x^p}$$

and $\int_1^\infty x^{-p}\,dx < \infty$ if $p > 1$.

The comparison theorem implies that

$$\int_1^\infty \frac{|\sin x|}{x^p}\,dx < \infty, \quad p > 1.$$

The function

$$f(x) = \frac{\sin x}{x^p}$$

is absolutely integrable on $[1,\infty)$ if $p > 1$.

Example: It is not absolutely integrable on $[1,\infty)$ if $p \le 1$.

To see this, we first consider the case where $p = 1$.

Let k be an integer greater than 3. Then

$$\begin{aligned} \int_1^{k\pi} \frac{|\sin x|}{x}\,dx &> \int_\pi^{k\pi} \frac{|\sin x|}{x}\,dx \\ &= \sum_{j=1}^{k-1} \int_{j\pi}^{(j+1)\pi} \frac{|\sin x|}{x}\,dx \qquad (3.5) \\ &> \sum_{j=1}^{k-1} \frac{1}{(j+1)\pi} \int_{j\pi}^{(j+1)\pi} |\sin x|\,dx. \end{aligned}$$

But

$$\int_{j\pi}^{(j+1)\pi} |\sin x|\,dx = \int_0^\pi \sin x\,dx = 2,$$

so (3.5) implies that

$$\int_1^{k\pi} \frac{|\sin x|}{x}\,dx > \frac{2}{\pi} \sum_{j=1}^{k-1} \frac{1}{j+1}. \qquad (3.6)$$

However,

$$\frac{1}{j+1} \ge \int_{j+1}^{j+2} \frac{dx}{x}, \quad j = 1, 2, \ldots,$$

so (3.6) implies that

$$\begin{aligned} \int_1^{k\pi} \frac{|\sin x|}{x} &> \frac{2}{\pi} \sum_{j=1}^{k-1} \int_{j+1}^{j+2} \frac{dx}{x} \\ &= \frac{2}{\pi} \int_2^{k+1} \frac{dx}{x} = \frac{2}{\pi} \log \frac{k+1}{2}. \end{aligned}$$

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Since $\lim_{k\to\infty} \log[(k+1)/2] = \infty$, implies that

$$\int_1^\infty \frac{|\sin x|}{x}\, dx = \infty.$$

Now implies that

$$\int_1^\infty \frac{|\sin x|}{x^p}\, dx = \infty, \quad p \le 1. \tag{3.7}$$

Theorem: If f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|\, dx < \infty$, then $\int_a^b f(x)\, dx$ converges; that is, an absolutely convergent integral is convergent.

Proof: If

$$g(x) = |f(x)| - f(x).$$

Then

$$0 \le g(x) \le 2|f(x)|$$

and $\int_a^b g(x)\, dx < \infty$, because of comparison theorem and the absolute integrability of f. Since

$$f = |f| - g,$$

Due to comparison test, we can conclude that $\int_a^b f(x)\, dx$ converges.

3.3 Nonoscillatory and Oscillatory Functions

A function f is *nonoscillatory* at $b-$ $(= \infty \quad \text{if} \quad b = \infty)$ if f is defined on $[a,b)$ and does not change sign on some subinterval $[a_1, b)$ of $[a,b)$.

If f changes sign on every such subinterval, f is *oscillatory* at $b-$.

Remark: For a function that is locally integrable on $[a,b)$ and nonoscillatory at $b-$, convergence and absolute convergence of $\int_a^b f(x)\, dx$ amount to the same thing, so absolute convergence is not an interesting concept in connection with such functions.

However, an oscillatory function may be integrable, but not absolutely integrable, on $[a,b)$, as the next example shows. We then say that f is *conditionally* integrable on $[a,b)$, and that $\int_a^b f(x)\, dx$ converges *conditionally*.

3.4 Conditional convergence

An oscillatory function may be integrable, but not absolutely integrable, on $[a,b)$, as the next example shows. We then say that f is *conditionally* integrable on $[a,b)$, and that $\int_a^b f(x)\, dx$ converges *conditionally*.

Example: The integral

$$I(p) = \int_1^\infty \frac{\sin x}{x^p}\, dx$$

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is not absolutely convergent if $0 < p \le 1$.

We will show that it converges conditionally for these values of p.

Integration by parts yields

$$\int_1^c \frac{\sin x}{x^p}\,dx = \frac{-\cos c}{c^p} + \cos 1 - p\int_1^c \frac{\cos x}{x^{p+1}}\,dx. \tag{3.8}$$

Since

$$\left|\frac{\cos x}{x^{p+1}}\right| \le \frac{1}{x^{p+1}}$$

and $\int_1^\infty x^{-p-1}\,dx < \infty$ if $p > 0$, the comparison theorem implies that $x^{-p-1}\cos x$ is absolutely integrable $[1,\infty)$ if $p > 0$.

Therefore, we have an absolutely convergent integral, this implies that $x^{-p-1}\cos x$ is integrable $[1,\infty)$ if $p > 0$.

Letting $c \to \infty$ in (3.8), we find that $I(p)$ converges, and

$$I(p) = \cos 1 - p\int_1^\infty \frac{\cos x}{x^{p+1}}\,dx \quad \text{if} \quad p > 0.$$

This and $\int_1^\infty \frac{|\sin x|}{x^p}\,dx = \infty, \quad p \le 1$, imply that $I(p)$ converges conditionally if $0 < p \le 1$.

3.5 Dirichlet's Test

Theorem: Suppose that f is continuous and its antiderivative $F(x) = \int_a^x f(t)\,dt$ is bounded on $[a,b)$.

Let g' be absolutely integrable on $[a,b)$, and suppose that

$$\lim_{x\to b-} g(x) = 0. \tag{3.9}$$

Then $\int_a^b f(x)g(x)\,dx$ converges.

Proof: The continuous function fg is locally integrable on $[a,b)$. Integration by parts yields

$$\int_a^c f(x)g(x)\,dx = F(c)g(c) - \int_a^c F(x)g'(x)\,dx, a \le c < b. \tag{3.10}$$

The comparison test implies that the integral on the right converges absolutely as $c \to b-$, since $\int_a^b |g'(x)|\,dx < \infty$ by assumption, and

$$|F(x)g'(x)| \le M|g'(x)|,$$

where M is an upper bound for $|F|$ on $[a,b)$. Moreover, (3.9) and the boundedness of F imply that $\lim_{c\to b-} F(c)g(c) = 0$.

Letting $c \to b-$ in (3.10) yields

$$\int_a^b f(x)g(x)\,dx = -\int_a^b F(x)g'(x)\,dx,$$

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where the integral on the right converges absolutely.

Remark: Dirichlet's test is useful only if f is oscillatory at $b-$, since it can be shown that if f is nonoscillatory at $b-$ and F is bounded on $[a, b)$, then $\int_a^b |f(x)g(x)|\,dx < \infty$ if only g is locally integrable and bounded on $[a, b)$.

Remark: Dirichlet's test can also be used to show that certain integrals diverge.

Example: For example,

$$\int_1^\infty x^q \sin x\,dx$$

diverges if $q > 0$, but none of the other tests that we have studied so far implies this. It is not enough to argue that the integrand does not approach zero as $x \to \infty$ (a common mistake), since this does not imply divergence. To see that the integral diverges, we observe that if it converged for some $q > 0$, then $F(x) = \int_1^x x^q \sin x\,dx$ would be bounded on $[1, \infty)$.

We could let

$$f(x) = x^q \sin x \quad \text{and} \quad g(x) = x^{-q}$$

in Dirichlet's test and conclude that

$$\int_1^\infty \sin x\,dx$$

also converges. This is false.

3.6 Rectangles in $\mathbb{R}^n$

The

$$S_1 \times S_2 \times \cdots \times S_n$$

of subsets S_1, S_2, ..., S_n of $\mathbb{R}$ is the set of points $(x_1, x_2, \ldots, x_n)$ in $\mathbb{R}^n$ such that $x_1 \in S_1, x_2 \in S_2, \ldots, x_n \in S_n$. For example, the Cartesian product of the two closed intervals

$$[a_1, b_1] \times [a_2, b_2] = \{(x, y) : a_1 \le x \le b_1,\ a_2 \le y \le b_2\}$$

is a rectangle in $\mathbb{R}^2$ with sides parallel to the x- and y-axes.

The Cartesian product of three closed intervals

$$[a_1, b_1] \times [a_2, b_2] \times [a_3, b_3] = \{(x, y, z) : a_1 \le x \le b_1, \ a_2 \le y \le b_2,\ a_3 \le z \le b_3\}$$

is a rectangular parallelepiped in $\mathbb{R}^3$ with faces parallel to the coordinate axes. A *coordinate rectangle* R in $\mathbb{R}^n$ is the Cartesian product of n closed intervals; that is,

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n].$$

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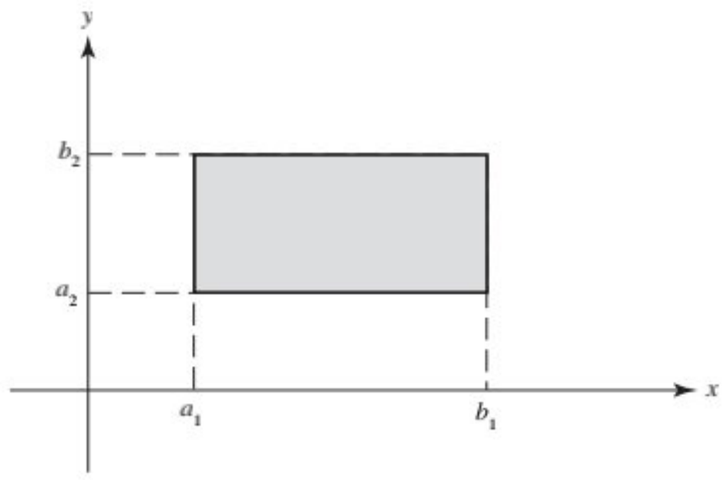


Figure 3.1: Rectangle in $\mathbb{R}^2$

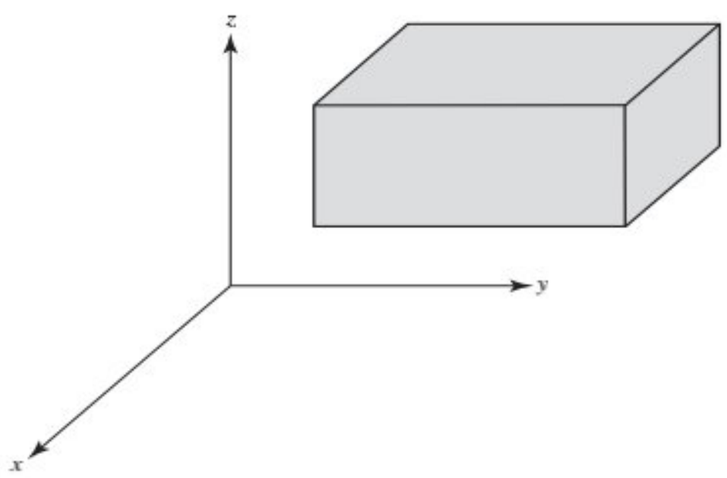


Figure 3.2: Rectangular parallelepiped in $\mathbb{R}^3$

The *content* of R is

$$V(R) = (b_1 - a_1)(b_2 - a_2) \cdots (b_n - a_n).$$

The numbers $b_1 - a_1$, $b_2 - a_2$, ..., $b_n - a_n$ are the *edge lengths* of R. If they are equal, then R is a *coordinate cube*. If $a_r = b_r$ for some r, then $V(R) = 0$ and we say that R is *degenerate*; otherwise, R is *nondegenerate*.

If $n = 1$, 2, or 3, then $V(R)$ is, respectively, the length of an interval, the area of a rectangle, or the volume of a rectangular parallelepiped. Henceforth, "rectangle" or "cube" will always mean "coordinate rectangle" or "coordinate cube" unless it is stated otherwise. If

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$$

and

$$P_r : \ a_r = a_{r0} < a_{r1} < \cdots < a_{rm_r} = b_r$$

is a partition of $[a_r, b_r]$, $1 \le r \le n$, then the set of all rectangles in $\mathbb{R}^n$ that can be written as

$$[a_{1,j_1-1}, a_{1j_1}] \times [a_{2,j_2-1}, a_{2j_2}] \times \cdots \times [a_{n,j_n-1}, a_{nj_n}],$$
$$1 \le j_r \le m_r, \quad 1 \le r \le n,$$

is a *partition* of R. We denote this partition by

$$\mathbf{P} = P_1 \times P_2 \times \cdots \times P_n. \tag{3.11}$$

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We define its *norm* to be the maximum of the norms of P_1, P_2, ..., P_n, thus,

$$\|\mathbf{P}\| = \max\{\|P_1\|, \|P_2\|, \ldots, \|P_n\|\}.$$

Put another way, $\|\mathbf{P}\|$ is the largest of the edge lengths of all the subrectangles in $\mathbf{P}$. Geometrically, a rectangle in $\mathbb{R}^2$ is partitioned by drawing horizontal and vertical lines through it; in $\mathbb{R}^3$, by drawing planes through it parallel to the coordinate axes. Partitioning divides a rectangle R into finitely many subrectangles that we can number in arbitrary order as R_1, R_2, ..., R_k. Sometimes it is convenient to write

$$\mathbf{P} = \{R_1, R_2, \ldots, R_k\}$$

rather than (3.11).

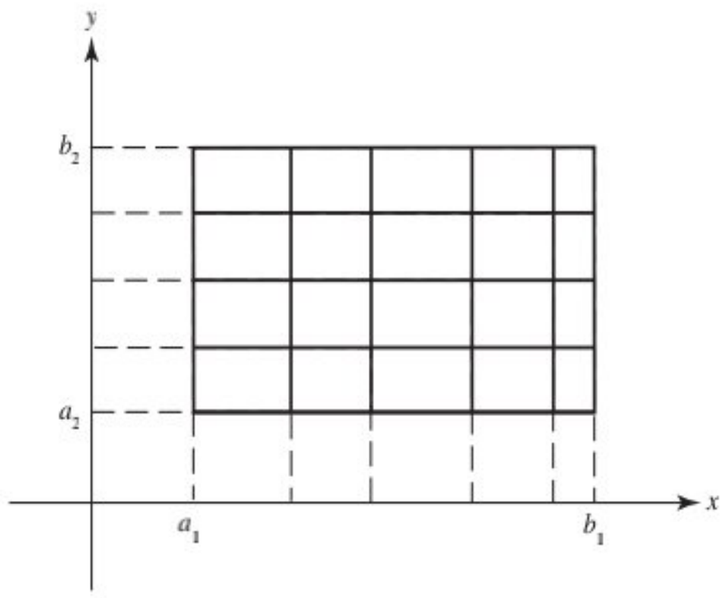


Figure 3.3: Partitioning of a rectangle in $\mathbb{R}^2$

- If $\mathbf{P} = P_1 \times P_2 \times \cdots \times P_n$ and $\mathbf{P}' = P_1' \times P_2' \times \cdots \times P_n'$ are partitions of the same rectangle, then $\mathbf{P}'$ is a *refinement* of $\mathbf{P}$ if P_i' is a refinement of P_i, $1 \le i \le n$.

3.7 Riemann Sum in $\mathbb{R}^n$

Suppose that f is a real-valued function defined on a rectangle R in $\mathbb{R}^n$, $\mathbf{P} = \{R_1, R_2, \ldots, R_k\}$ is a partition of R.

Let $\mathbf{X}_j$ is an arbitrary point in R_j, $1 \le j \le k$.

Then

$$\sigma = \sum_{j=1}^{k} f(\mathbf{X}_j)V(R_j)$$

is a *Riemann sum of f over* $\mathbf{P}$.

Since $\mathbf{X}_j$ can be chosen arbitrarily in R_j, there are infinitely many Riemann sums for a given function f over any partition $\mathbf{P}$ of R.

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3.8 Riemann Integral in $\mathbb{R}^n$

: Let f be a real-valued function defined on a rectangle R in $\mathbb{R}^n$.

We say that f is *Riemann integrable on* R if there is a number L with the following property:

For every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$|\sigma - L| < \varepsilon.$$

If σ is any Riemann sum of f over a partition $\mathbf{P}$ of R such that $\|\mathbf{P}\| < \delta$.
In this case, we say that L is the *Riemann integral of* f *over* R, and write

$$\int_R f(\mathbf{X})\, d\mathbf{X} = L.$$

Remarks: The integral $\int_R f(\mathbf{X}) d\mathbf{X}$ is also written as

$$\int_R f(x,y)\, d(x,y) \quad (n=2), \qquad \int_R f(x,y,z)\, d(x,y,z) \quad (n=3),$$

or

$$\int_R f(x_1, x_2, \ldots, x_n)\, d(x_1, x_2, \ldots, x_n) \quad (n \text{ arbitrary}).$$

Here $d\mathbf{X}$ does not stand for the differential of $\mathbf{X}$.

It merely identifies x_1, x_2, ..., x_n, the components of $\mathbf{X}$, as the variables of integration. To avoid this minor inconsistency, some authors write simply $\int_R f$ rather than $\int_R f(\mathbf{X})\, d\mathbf{X}$.

As in the case where $n = 1$, we will say simply "integrable" or "integral" when we mean "Riemann integrable" or "Riemann integral." If $n \geq 2$, we call the integral of above definition a *multiple integral*; for $n = 2$ and $n = 3$ we also call them *double* and *triple integrals*, respectively. When we wish to distinguish between multiple integrals and the integral we studied in Chapter $(n = 1)$, we will call the latter an *ordinary* integral.

Example: Find $\int_R f(x,y)\, d(x,y)$, where

$$R = [a,b] \times [c,d]$$

and

$$f(x,y) = x + y.$$

Solution: Let P_1 and P_2 be partitions of $[a,b]$ and $[c,d]$; thus,

$$P_1 : a = x_0 < x_1 < \cdots < x_r = b$$

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and

$$P_2 : c = y_0 < y_1 < \cdots < y_s = d.$$

A typical Riemann sum of f over $\mathbf{P} = P_1 \times P_2$ is given by

$$\sigma = \sum_{i=1}^{r} \sum_{j=1}^{s} (\xi_{ij} + \eta_{ij})(x_i - x_{i-1})(y_j - y_{j-1}), \tag{3.12}$$

$$\text{where} \quad x_{i-1} \le \xi_{ij} \le x_i \quad \text{and} \quad y_{j-1} \le \eta_{ij} \le y_j. \tag{3.13}$$

The midpoints of $[x_{i-1}, x_i]$ and $[y_{j-1}, y_j]$ are

$$\overline{x}_i = \frac{x_i + x_{i-1}}{2} \quad \text{and} \quad \overline{y}_j = \frac{y_j + y_{j-1}}{2}, \tag{3.14}$$

and (3.13) implies that

$$|\xi_{ij} - \overline{x}_i| \le \frac{x_i - x_{i-1}}{2} \le \frac{\|P_1\|}{2} \le \frac{\|\mathbf{P}\|}{2} \tag{3.15}$$

$$|\eta_{ij} - \overline{y}_j| \le \frac{y_j - y_{j-1}}{2} \le \frac{\|P_2\|}{2} \le \frac{\|\mathbf{P}\|}{2}. \tag{3.16}$$

Now we rewrite (3.12) as

$$\begin{aligned} \sigma = {} & \textstyle\sum_{i=1}^{r} \sum_{j=1}^{s} (\overline{x}_i + \overline{y}_j)(x_i - x_{i-1})(y_j - y_{j-1}) \\ & + \textstyle\sum_{i=1}^{r} \sum_{j=1}^{s} \left[(\xi_{ij} - \overline{x}_i) + (\eta_{ij} - \overline{y}_j)\right] \\ & (x_i - x_{i-1})(y_j - y_{j-1}). \end{aligned} \tag{3.17}$$

To find $\int_R f(x,y)\, d(x,y)$ from (3.17), we recall that

$$\sum_{i=1}^{r} (x_i - x_{i-1}) = b - a, \quad \sum_{j=1}^{s} (y_j - y_{j-1}) = d - c \tag{3.18}$$

and

$$\sum_{i=1}^{r} (x_i^2 - x_{i-1}^2) = b^2 - a^2, \quad \sum_{j=1}^{s} (y_j^2 - y_{j-1}^2) = d^2 - c^2. \tag{3.19}$$

Because of (3.15) and (3.16) the absolute value of the second sum in (3.17) does not exceed

$$\begin{aligned} \|\mathbf{P}\| \sum_{j=1}^{r} \sum_{j=1}^{s} (x_i - x_{i-1})(y_j - y_{j-1}) &= \|\mathbf{P}\| \left[\sum_{i=1}^{r} (x_i - x_{i-1})\right] \\ & \left[\sum_{j=1}^{s} (y_j - y_{j-1})\right] \\ = \|\mathbf{P}\|(b-a)(d-c) & \end{aligned}$$

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(see (3.18)), so (3.17) implies that

$$\left|\sigma - \sum_{i=1}^{r}\sum_{j=1}^{s}(\overline{x}_i + \overline{y}_j)(x_i - x_{i-1})(y_j - y_{j-1})\right| \leq \|\mathbf{P}\|(b-a)(d-c). \tag{3.20}$$

It now follows that

$$\begin{aligned}
& \textstyle\sum_{i=1}^{r}\sum_{j=1}^{s}\overline{x}_i(x_i - x_{i-1})(y_j - y_{j-1}) \\
= \; & \left[\textstyle\sum_{i=1}^{r}\overline{x}_i(x_i - x_{i-1})\right]\left[\textstyle\sum_{j=1}^{s}(y_j - y_{j-1})\right] \\
= \; & (d-c)\textstyle\sum_{i=1^r}\overline{x}_i(x_i - x_{i-1}) \quad \text{(from (3.18))} \\
= \; & \tfrac{d-c}{2}\textstyle\sum_{i=1}^{r}(x_i^2 - x_{i-1}^2) \qquad \text{(from (3.14))} \\
= \; & \tfrac{d-c}{2}(b^2 - a^2) \qquad\qquad \text{(from (3.19))}.
\end{aligned}$$

Similarly,

$$\sum_{i=1}^{r}\sum_{j=1}^{s}\overline{y}_j(x_i - x_{i-1})(y_j - y_{j-1}) = \frac{b-a}{2}(d^2 - c^2).$$

Therefore, (3.20) can be written as

$$\left|\sigma - \frac{d-c}{2}(b^2 - a^2) - \frac{b-a}{2}(d^2 - c^2)\right| \leq \|\mathbf{P}\|(b-a)(d-c).$$

Since the right side can be made as small as we wish by choosing $\|\mathbf{P}\|$ sufficiently small,

$$\int_R (x+y)\,d(x,y) = \frac{1}{2}\left[(d-c)(b^2-a^2) + (b-a)(d^2-c^2)\right].$$

Theorem: If f is unbounded on the nondegenerate rectangle R in $\mathbb{R}^n$, then f is not integrable on R.

Proof: We will show that if f is unbounded on R, $\mathbf{P} = \{R_1, R_2, \ldots, R_k\}$ is any partition of R, and $M > 0$, then there are Riemann sums σ and σ' of f over $\mathbf{P}$ such that

$$|\sigma - \sigma'| \geq M. \tag{3.21}$$

This implies that f cannot satisfy definition of Riemann integral. (Why?)

Let

$$\sigma = \sum_{j=1}^{k} f(\mathbf{X}_j)V(R_j)$$

be a Riemann sum of f over $\mathbf{P}$. Let

$$\sigma = \sum_{j=1}^{k} f(\mathbf{X}_j)V(R_j)$$

be a Riemann sum of f over $\mathbf{P}$.

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There must be an integer i in $\{1, 2, \ldots, k\}$ such that

$$|f(\mathbf{X}) - f(\mathbf{X}_i)| \geq \frac{M}{V(R_i)} \tag{3.22}$$

for some $\mathbf{X}$ in R_i, because if this were not so, we would have

$$|f(\mathbf{X}) - f(\mathbf{X}_j)| < \frac{M}{V(R_j)}, \quad \mathbf{X} \in R_j, \qquad 1 \leq j \leq k.$$

If this is so, then

$$\begin{aligned} |f(\mathbf{X})| &= |f(\mathbf{X}_j) + f(\mathbf{X}) - f(\mathbf{X}_j)| \leq |f(\mathbf{X}_j)| + |f(\mathbf{X}) - f(\mathbf{X}_j)| \\ &\leq |f(\mathbf{X}_j)| + \frac{M}{V(R_j)}, \quad \mathbf{X} \in R_j, \quad 1 \leq j \leq k. \end{aligned}$$

However, this implies that

$$|f(\mathbf{X})| \leq \max |f(\mathbf{X}_j)| + \frac{M}{V(R_j)} 1 \leq j \leq k, \quad \mathbf{X} \in R,$$

which contradicts the assumption that f is unbounded on R.

Now suppose that $\mathbf{X}$ satisfies (3.22).

Consider the Riemann sum

$$\sigma' = \sum_{j=1}^{n} f(\mathbf{X}'_j) V(R_j)$$

over the same partition $\mathbf{P}$, where

$$\mathbf{X}'_j = \begin{cases} \mathbf{X}_j, & j \neq i, \\ \mathbf{X}, & j = i. \end{cases}$$

Since

$$|\sigma - \sigma'| = |f(\mathbf{X}) - f(\mathbf{X}_i)| V(R_i),$$

(3.22) implies (3.21).

3.9 Upper and Lower Integrals

If f is bounded on a rectangle R in $\mathbb{R}^n$ and $\mathbf{P} = \{R_1, R_2, \ldots, R_k\}$ is a partition of R.

Let

$$M_j = \sup_{\mathbf{X} \in R_j} f(\mathbf{X}), \quad m_j = \inf_{\mathbf{X} \in R_j} f(\mathbf{X}).$$

The *upper sum* of f over $\mathbf{P}$ is

$$S(\mathbf{P}) = \sum_{j=1}^{k} M_j V(R_j).$$

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The *upper integral of* f *over* R, denoted by

$$\overline{\int_R} f(\mathbf{X})\, d\mathbf{X},$$

is the infimum of all upper sums.

Upper and Lower Integrals: The *lower sum of* f *over* $\mathbf{P}$ is

$$s(\mathbf{P}) = \sum_{j=1}^{k} m_j V(R_j).$$

The *lower integral of* f *over* R, denoted by

$$\underline{\int_R} f(\mathbf{X})\, d\mathbf{X},$$

is the supremum of all lower sums.

Theorem: Let f be bounded on a rectangle R and let $\mathbf{P}$ be a partition of R.

Then

1. The upper sum $S(\mathbf{P})$ of f over $\mathbf{P}$ is the supremum of the set of all Riemann sums of f over $\mathbf{P}$.
2. The lower sum $s(\mathbf{P})$ of f over $\mathbf{P}$ is the infimum of the set of all Riemann sums of f over $\mathbf{P}$.

Remarks: If

$$m \le f(\mathbf{X}) \le M \quad \text{for } \mathbf{X} \text{ in } R,$$

then

$$mV(R) \le s(\mathbf{P}) \le S(\mathbf{P}) \le MV(R);$$

therefore, $\overline{\int_R} f(\mathbf{X})\, d\mathbf{X}$ and $\underline{\int_R} f(\mathbf{X})\, d\mathbf{X}$ exist, are unique, and satisfy the inequalities

$$mV(R) \le \overline{\int_R} f(\mathbf{X})\, d\mathbf{X} \le MV(R)$$

and

$$mV(R) \le \underline{\int_R} f(\mathbf{X})\, d\mathbf{X} \le MV(R).$$

Remarks: The upper and lower integrals are also written as

$$\overline{\int_R} f(x,y)\, d(x,y) \quad \text{and} \quad \underline{\int_R} f(x,y)\, d(x,y) \quad (n=2),$$

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$$\overline{\int_R} f(x,y,z)\,d(x,y,z) \quad \text{and} \quad \underline{\int_R} f(x,y,z)\,d(x,y,z) \quad (n=3),$$

or

$$\overline{\int_R} f(x_1,x_2,\ldots,x_n)\,d(x_1,x_2,\ldots,x_n)$$

and

$$\underline{\int_R} f(x_1,x_2,\ldots,x_n)\,d(x_1,x_2,\ldots,x_n) \qquad (n \text{ arbitrary}).$$

Example: Find $\underline{\int_R} f(x,y)\,d(x,y)$ and $\overline{\int_R} f(x,y)\,d(x,y)$, with $R=[a,b]\times[c,d]$ and

$$f(x,y)=x+y.$$

Solution: Let P_1 and P_2 be partitions of $[a,b]$ and $[c,d]$; thus,

$$P_1 : a=x_0<x_1<\cdots<x_r=b$$

$$P_2 : c=y_0<y_1<\cdots<y_s=d.$$

The maximum and minimum values of f on the rectangle $[x_{i-1},x_i]\times[y_{j-1},y_j]$ are x_i+y_j and $x_{i-1}+y_{j-1}$, respectively.

Therefore,

$$S(\mathbf{P}) = \sum_{i=1}^{r}\sum_{j=1}^{s}(x_i+y_j)(x_i-x_{i-1})(y_j-y_{j-1}) \tag{3.23}$$

$$s(\mathbf{P}) = \sum_{i=1}^{r}\sum_{j=1}^{s}(x_{i-1}+y_{j-1})(x_i-x_{i-1})(y_j-y_{j-1}). \tag{3.24}$$

By substituting

$$x_i+y_j=\frac{1}{2}[(x_i+x_{i-1})+(y_j+y_{j-1})+(x_i-x_{i-1})+(y_j-y_{j-1})]$$

into (3.23). We find that

$$S(\mathbf{P})=\frac{1}{2}(\Sigma_1+\Sigma_2+\Sigma_3+\Sigma_4), \tag{3.25}$$

where

$$\begin{aligned}
\Sigma_1 &= \textstyle\sum_{i=1}^{r}(x_i^2-x_{i-1}^2)\sum_{j=1}^{s}(y_j-y_{j-1}) &&= (b^2-a^2)(d-c),\\
\Sigma_2 &= \textstyle\sum_{i=1}^{r}(x_i-x_{i-1})\sum_{j=1}^{s}(y_j^2-y_{j-1}^2) &&= (b-a)(d^2-c^2),\\
\Sigma_3 &= \textstyle\sum_{i=1}^{r}(x_i-x_{i-1})^2\sum_{j=1}^{s}(y_j-y_{j-1}) &&\le \|\mathbf{P}\|(b-a)(d-c),\\
\Sigma_4 &= \textstyle\sum_{i=1}^{r}(x_i-x_{i-1})\sum_{j=1}^{s}(y_j-y_{j-1})^2 &&\le \|\mathbf{P}\|(b-a)(d-c).
\end{aligned}$$

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Substituting these four results into (3.25) shows that

$$I < S(\mathbf{P}) < I + \|\mathbf{P}\|(b-a)(d-c),$$

where

$$I = \frac{(d-c)(b^2-a^2) + (b-a)(d^2-c^2)}{2}.$$

From this, we see that

$$\overline{\int_R} (x+y)\, d(x,y) = I.$$

After substituting

$$x_{i-1} + y_{j-1} = \frac{1}{2}[(x_i + x_{i-1}) + (y_j + y_{j-1}) - (x_i - x_{i-1}) - (y_j - y_{j-1})]$$

into (3.24), a similar argument shows that

$$I - \|\mathbf{P}\|(b-a)(d-c) < s(\mathbf{P}) < I.$$

So

$$\underline{\int_R} (x+y)\, d(x,y) = I.$$

Theorem: Suppose that $|f(\mathbf{X})| \le M$ if $\mathbf{X}$ is in the rectangle

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n].$$

Let $\mathbf{P} = P_1 \times P_2 \times \cdots \times P_n$ and $\mathbf{P}' = P_1' \times P_2' \times \cdots \times P_n'$ be partitions of R, where P_j' is obtained by adding r_j partition points to P_j, $1 \le j \le n$. Then

$$S(\mathbf{P}) \ge S(\mathbf{P}') \ge S(\mathbf{P}) - 2MV(R)\left(\sum_{j=1}^{n} \frac{r_j}{b_j - a_j}\right)\|\mathbf{P}\| \tag{3.26}$$

and

$$s(\mathbf{P}) \le s(\mathbf{P}') \le s(\mathbf{P}) + 2MV(R)\left(\sum_{j=1}^{n} \frac{r_j}{b_j - a_j}\right)\|\mathbf{P}\|. \tag{3.27}$$

Theorem: If f is bounded on a rectangle R, then

$$\underline{\int_R} f(\mathbf{X})\, d\mathbf{X} \le \overline{\int_R} f(\mathbf{X})\, d\mathbf{X}.$$

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Theorem: If f is integrable on a rectangle R, then

$$\underline{\int_R} f(\mathbf{X})\,d\mathbf{X} = \overline{\int_R} f(\mathbf{X})\,d\mathbf{X} = \int_R f(\mathbf{X})\,d\mathbf{X}.$$

Theorem: If f is bounded on a rectangle R and $\varepsilon > 0$, there is a $\delta > 0$ such that

$$\overline{\int_R} f(\mathbf{X})\,d\mathbf{X} \leq S(\mathbf{P}) < \overline{\int_R} f(\mathbf{X})\,d\mathbf{X} + \varepsilon$$

and

$$\underline{\int_R} f(\mathbf{X})\,d\mathbf{X} \geq s(\mathbf{P}) > \underline{\int_R} f(\mathbf{X})\,d\mathbf{X} - \varepsilon$$

if $\|\mathbf{P}\| < \delta$.

Theorem: A bounded function f is integrable on a rectangle R if and only if

$$\underline{\int_R} f(\mathbf{X})\,d\mathbf{X} = \overline{\int_R} f(\mathbf{X})\,d\mathbf{X}.$$

Theorem: If f is bounded on a rectangle R, then f is integrable on R if and only if for every $\varepsilon > 0$ there is a partition $\mathbf{P}$ of R such that

$$S(\mathbf{P}) - s(\mathbf{P}) < \varepsilon.$$

Theorem: If f is bounded on a rectangle R and

$$\underline{\int_R} f(\mathbf{X})\,d\mathbf{X} = \overline{\int_R} f(\mathbf{X})\,d\mathbf{X} = L,$$

then f is integrable on R, and

$$\int_R f(\mathbf{X})\,d\mathbf{X} = L.$$

Theorem: If f is continuous on a rectangle R in $\mathbb{R}^n$, then f is integrable on R.

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3.10 Sets with Zero Content

A subset E of $\mathbb{R}^n$ has zero content if for each $\varepsilon > 0$ there is a finite set of rectangles $T_1, T_2, \ldots, T_m$ such that

$$E \subset \bigcup_{j=1}^{m} T_j \tag{3.28}$$

and

$$\sum_{j=1}^{m} V(T_j) < \varepsilon. \tag{3.29}$$

Example: Since the empty set is contained in every rectangle, the empty set has zero content.

If E consists of finitely many points $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_m$, then $\mathbf{X}_j$ can be enclosed in a rectangle T_j such that

$$V(T_j) < \frac{\varepsilon}{m}, \quad 1 \le j \le m.$$

Then $E \subset \bigcup_{j=1}^{m} T_j$ and $\sum_{j=1}^{m} V(T_j) < \varepsilon$ hold, so E has zero content.

Example: Any bounded set E with only finitely many limit points has zero content.

To see this, we first observe that if E has no limit points, then it must be finite, by the Bolzano–Weierstrass theorem, and therefore must have zero content.

Now suppose that the limit points of E are $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_m$. Let $R_1, R_2, \ldots, R_m$ be rectangles such that $\mathbf{X}_i \in R_i^0$ and

$$V(R_i) < \frac{\varepsilon}{2m}, \quad 1 \le i \le m. \tag{3.30}$$

The set of points of E that are not in $\cup_{j=1}^{m} R_j$ has no limit points (why?) and, being bounded, must be finite (again by the Bolzano–Weierstrass theorem).

If this set contains p points, then it can be covered by rectangles $R_1', R_2', \ldots, R_p'$ with

$$V(R_j') < \frac{\varepsilon}{2p}, \quad 1 \le j \le p. \tag{3.31}$$

Now,

$$E \subset \left(\bigcup_{i=1}^{m} R_i \right) \bigcup \left(\bigcup_{j=1}^{p} R_j' \right)$$

From (3.30) and (3.31),

$$\sum_{i=1}^{m} V(R_i) + \sum_{j=1}^{p} V(R_j') < \varepsilon.$$

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Example: If f is continuous on $[a, b]$, then the curve

$$y = f(x), \quad a \le x \le b \tag{3.32}$$

(that is, the set $\{(x, y) : y = f(x),\ a \le x \le b\}$), has zero content in $\mathbb{R}^2$.

Lemma: The union of finitely many sets with zero content has zero content.

Theorem: Suppose that f is bounded on a rectangle

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n] \tag{3.33}$$

and continuous except on a subset E of R with zero content. Then f is integrable on R.

Example: The function

$$f(x, y) = \begin{cases} x + y, & 0 \le x < y \le 1, \\ 5, & 0 \le y \le x \le 1, \end{cases}$$

is continuous on $R = [0, 1] \times [0, 1]$ except on the line segment

$$y = x, \quad 0 \le x \le 1$$

Since the line segment has zero content, f is integrable on R.

3.11 Integral Over Bounded Set

Suppose that f is bounded on a bounded subset of S of $\mathbb{R}^n$. Let

$$f_S(\mathbf{X}) = \begin{cases} f(\mathbf{X}), & \mathbf{X} \in S, \\ 0, & \mathbf{X} \notin S. \end{cases} \tag{3.34}$$

Let R be a rectangle containing S. Then *the integral of f over S* is defined to be

$$\int_S f(\mathbf{X})\, d\mathbf{X} = \int_R f_S(\mathbf{X})\, d\mathbf{X}$$

if $\int_R f_S(\mathbf{X})\, d\mathbf{X}$ exists.

Area and volume as integrals: If S is a bounded subset of $\mathbb{R}^n$ and the integral $\int_S d\mathbf{X}$ (with integrand $f \equiv 1$) exists.

We call $\int_S d\mathbf{X}$ the *content* (also, *area* if $n = 2$ or *volume* if $n = 3$) of S, and denote it by $V(S)$.

Thus,

$$V(S) = \int_S d\mathbf{X}.$$

Theorem: Suppose that f is bounded on a bounded set S and continuous except on a subset E of S with zero content.

Suppose also that ∂S has zero content. Then f is integrable on S.

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3.12 Differentiable Surfaces

A *differentiable surface* S in $\mathbb{R}^n$ $(n > 1)$ is the image of a compact subset D of $\mathbb{R}^m$, where $m < n$, under a continuously differentiable transformation $\mathbf{G} : \mathbb{R}^m \to \mathbb{R}^n$. If $m = 1$, S is also called a *differentiable curve*.

Example: The circle

$$\{(x, y) : x^2 + y^2 = 9\}$$

is a differentiable curve in $\mathbb{R}^2$.

Since it is the image of $D = [0, 2\pi]$ under the continuously differentiable transformation $\mathbf{G} : \mathbb{R} \to \mathbb{R}^2$ defined by

$$\mathbf{X} = \mathbf{G}(\theta) = \begin{bmatrix} 3\cos\theta \\ 3\sin\theta \end{bmatrix}.$$

Example: The sphere

$$\{(x, y, z) : x^2 + y^2 + z^2 = 4\}$$

is a differentiable surface in $\mathbb{R}^3$.

Since it is the image of

$$D = \{(\theta, \phi) : 0 \le \theta \le 2\pi, -\pi/2 \le \phi \le \pi/2\}$$

under the continuously differentiable transformation $\mathbf{G} : \mathbb{R}^2 \to \mathbb{R}^3$ defined by

$$\mathbf{X} = \mathbf{G}(\theta, \phi) = \begin{bmatrix} 2\cos\theta\cos\phi \\ 2\sin\theta\cos\phi \\ 2\sin\phi \end{bmatrix}.$$

Theorem: A differentiable surface in $\mathbb{R}^n$ has zero content.

Let S, D, and $\mathbf{G}$ be as in Definition **??**. From Lemma **??**, there is a constant M such that

$$|\mathbf{G}(\mathbf{X}) - \mathbf{G}(\mathbf{Y})| \le M|\mathbf{X} - \mathbf{Y}| \quad \text{if} \quad \mathbf{X}, \mathbf{Y} \in D. \tag{3.35}$$

Since D is bounded, D is contained in a cube

$$C = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_m, b_m],$$

where

$$b_i - a_i = L, \quad 1 \le i \le m.$$

Suppose that we partition C into N^m smaller cubes by partitioning each of the intervals $[a_i, b_i]$ into N equal subintervals. Let $R_1, R_2, \ldots, R_k$ be the smaller cubes so produced that contain points of D, and select points $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_k$ such that $\mathbf{X}_i \in D \cap R_i$, $1 \le i \le k$. If $\mathbf{Y} \in D \cap R_i$, then (3.35) implies that

$$|\mathbf{G}(\mathbf{X}_i) - \mathbf{G}(\mathbf{Y})| \le M|\mathbf{X}_i - \mathbf{Y}|. \tag{3.36}$$

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Since $\mathbf{X}_i$ and $\mathbf{Y}$ are both in the cube R_i with edge length L/N,

$$|\mathbf{X}_i - \mathbf{Y}| \leq \frac{L\sqrt{m}}{N}.$$

This and (3.36) imply that

$$|\mathbf{G}(\mathbf{X}_i) - \mathbf{G}(\mathbf{Y})| \leq \frac{ML\sqrt{m}}{N},$$

which in turn implies that $\mathbf{G}(\mathbf{Y})$ lies in a cube $\widetilde{R}_i$ in $\mathbb{R}^n$ centered at $\mathbf{G}(\mathbf{X}_i)$, with sides of length $2ML\sqrt{m}/N$. Now

$$\sum_{i=1}^{k} V(\widetilde{R}_i) = k\left(\frac{2ML\sqrt{m}}{N}\right)^n \leq N^m\left(\frac{2ML\sqrt{m}}{N}\right)^n = (2ML\sqrt{m})^n N^{m-n}.$$

Since $n > m$, we can make the sum on the left arbitrarily small by taking N sufficiently large. Therefore, S has zero content.

Theorem: Suppose that S is a bounded set in $\mathbb{R}^n$, with boundary consisting of a finite number of differentiable surfaces.

Let f be bounded on S and continuous except on a set of zero content. Then f is integrable on S.

Example: Let

$$S = \{(x, y) : x^2 + y^2 = 1,\ x \geq 0\}.$$

The set S is bounded by a semicircle and a line segment, both differentiable curves in $\mathbb{R}^2$.

Let

$$f(x, y) = \begin{cases} (1 - x^2 - y^2)^{1/2}, & (x, y) \in S,\ y \geq 0, \\ -(1 - x^2 - y^2)^{1/2}, & (x, y) \in S,\ y < 0. \end{cases}$$

Then f is continuous on S except on the line segment

$$y = 0, \quad 0 \leq x < 1,$$

which has zero content.

Hence, from the theorem we just stated implies that f is integrable on S.

Theorem: If f and g are integrable on S, then so is $f + g$, and

$$\int_S (f + g)(\mathbf{X})\, d\mathbf{X} = \int_S f(\mathbf{X})\, d\mathbf{X} + \int_S g(\mathbf{X})\, d\mathbf{X}.$$

Theorem: If f is integrable on S and c is a constant, then cf is integrable on S, and

$$\int_S (cf)(\mathbf{X})\, d\mathbf{X} = c\int_S f(\mathbf{X})\, d\mathbf{X}.$$

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Theorem: If f and g are integrable on S and $f(\mathbf{X}) \leq g(\mathbf{X})$ for $\mathbf{X}$ in S, then

$$\int_S f(\mathbf{X})\, d\mathbf{X} \leq \int_S g(\mathbf{X})\, d\mathbf{X}.$$

Theorem: If f is integrable on S, then so is $|f|$, and

$$\left| \int_S f(\mathbf{X})\, d\mathbf{X} \right| \leq \int_S |f(\mathbf{X})|\, d\mathbf{X}.$$

Theorem: If f and g are integrable on S, then so is the product fg.

Theorem: Suppose that u is continuous and v is integrable and nonnegative on a rectangle R.

Then

$$\int_R u(\mathbf{X})v(\mathbf{X})\, d\mathbf{X} = u(\mathbf{X}_0) \int_R v(\mathbf{X})\, d\mathbf{X}$$

for some $\mathbf{X}_0$ in R.

Theorem: Suppose that S is contained in a bounded set T and f is integrable on S.

Then f_S is integrable on T, and

$$\int_T f_S(\mathbf{X})\, d\mathbf{X} = \int_S f(\mathbf{X})\, d\mathbf{X}.$$

Theorem: If f is integrable on disjoint sets S_1 and S_2, then f is integrable on $S_1 \cup S_2$, and

$$\int_{S_1 \cup S_2} f(\mathbf{X})\, d\mathbf{X} = \int_{S_1} f(\mathbf{X})\, d\mathbf{X} + \int_{S_2} f(\mathbf{X})\, d\mathbf{X}. \tag{3.37}$$

Theorem: Suppose that f is integrable on sets S_1 and S_2 such that $S_1 \cap S_2$ has zero content. Then f is integrable on $S_1 \cup S_2$, and

$$\int_{S_1 \cup S_2} f(\mathbf{X})\, d\mathbf{X} = \int_{S_1} f(\mathbf{X})\, d\mathbf{X} + \int_{S_2} f(\mathbf{X})\, d\mathbf{X}.$$

Example: Let

$$\begin{aligned} S_1 &= \{(x, y) : 0 \leq x \leq 1,\ 0 \leq y \leq 1 + x\} \\ S_2 &= \{(x, y) : -1 \leq x \leq 0,\ 0 \leq y \leq 1 - x\} \end{aligned}$$

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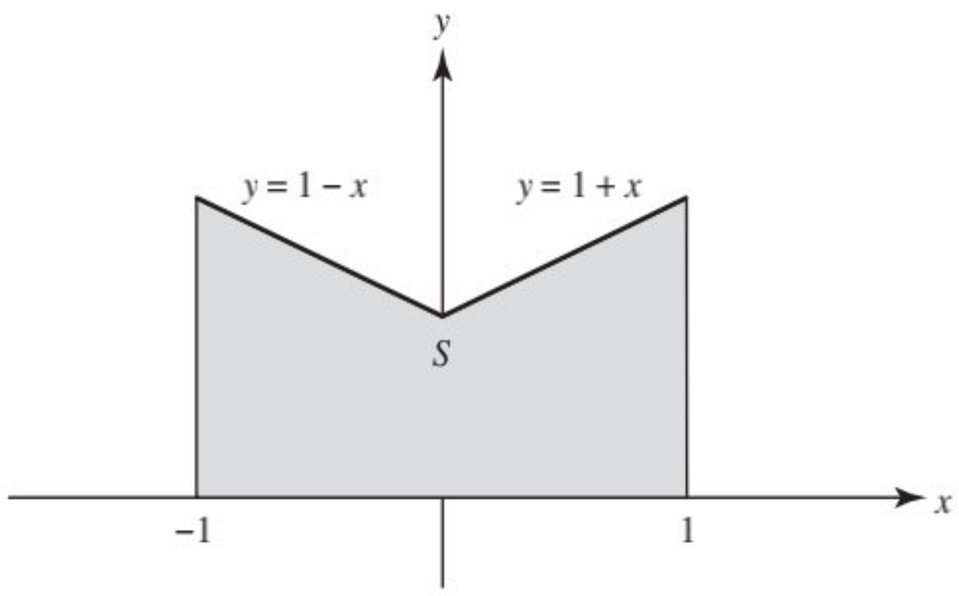


Figure 3.4: S_1 and S_2

Then

$$S_1 \cap S_2 = \{(0, y) : 0 \leq y \leq 1\}$$

has zero content.

Hence, by using corollary implies that if f is integrable on S_1 and S_2, then f is also integrable over

$$S = S_1 \cup S_2 = \{(x, y) : -1 \leq x \leq 1,\ 0 \leq y \leq 1 + |x|\}$$

and

$$\int_{S_1 \cup S_2} f(\mathbf{X})\, d\mathbf{X} = \int_{S_1} f(\mathbf{X})\, d\mathbf{X} + \int_{S_2} f(\mathbf{X})\, d\mathbf{X}.$$

3.13 Iterated Integrals

Let us first assume that f is continuous on $R = [a, b] \times [c, d]$.

Then, for each y in $[c, d]$, $f(x, y)$ is continuous with respect to x on $[a, b]$, so the integral

$$F(y) = \int_a^b f(x, y)\, dx$$

exists.

Moreover, the uniform continuity of f on R implies that F is continuous and therefore integrable on $[c, d]$.

We say that

$$I_1 = \int_c^d F(y)\, dy = \int_c^d \left(\int_a^b f(x, y)\, dx \right) dy$$

is an *iterated integral* of f over R.

Iterated integrals: We will usually write it as

$$I_1 = \int_c^d dy \int_a^b f(x, y)\, dx.$$

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Another iterated integral can be defined by writing

$$G(x) = \int_c^d f(x,y)\,dy, \quad a \le x \le b,$$

Defining

$$I_2 = \int_a^b G(x)\,dx = \int_a^b \left(\int_c^d f(x,y)\,dy \right) dx,$$

which we usually write as

$$I_2 = \int_a^b dx \int_c^d f(x,y)\,dy.$$

Example: Let

$$f(x,y) = x + y$$

and $R = [0,1] \times [1,2]$. Then

$$\begin{aligned} F(y) &= \int_0^1 f(x,y)\,dx = \int_0^1 (x+y)\,dx = \frac{1}{2} + y \\ I_1 &= \int_1^2 F(y)\,dy = \int_1^2 \left(\frac{1}{2} + y\right)\,dy = \left(\frac{y}{2} + \frac{y^2}{2}\right)\Big|_1^2 = 2. \end{aligned}$$

Also,

$$\begin{aligned} G(x) &= \int_1^2 (x+y)\,dy = \left(xy + \frac{y^2}{2}\right)\Big|_{y=1}^2 = x + \frac{3}{2}, \\ I_2 &= \int_0^1 G(x)\,dx = \int_0^1 \left(x + \frac{3}{2}\right)\,dx = \left(\frac{x^2}{2} + \frac{3x}{2}\right)\Big|_0^1 = 2. \end{aligned}$$

Theorem: Suppose that f is integrable on $R = [a,b] \times [c,d]$ and

$$F(y) = \int_a^b f(x,y)\,dx$$

exists for each y in $[c,d]$.

Then F is integrable on $[c,d]$, and

$$\int_c^d F(y)\,dy = \int_R f(x,y)\,d(x,y); \tag{3.38}$$

that is,

$$\int_c^d dy \int_a^b f(x,y)\,dx = \int_R f(x,y)\,d(x,y). \tag{3.39}$$

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Theorem: If f is integrable on $[a,b] \times [c,d]$, then

$$\int_a^b dx \int_c^d f(x,y)\,dy = \int_c^d dy \int_a^b f(x,y)\,dx,$$

provided that $\int_c^d f(x,y)\,dy$ exists for $a \le x \le b$ and $\int_a^b f(x,y)\,dx$ exists for $c \le y \le d$.

In particular, these hypotheses hold if f is continuous on $[a,b] \times [c,d]$.

Example: The function

$$f(x,y) = x + y$$

is continuous everywhere.

For example, let $R = [0,1] \times [1,2]$.

Then we have

$$\begin{aligned}
\int_R (x+y)\,d(x,y) &= \int_1^2 dy \int_0^1 (x+y)\,dx \\
&= \int_1^2 \left[\left(\frac{x^2}{2} + xy \right) \Big|_{x=0}^{1} \right] dy \\
&= \int_1^2 \left(\frac{1}{2} + y \right) dy = \left(\frac{y}{2} + \frac{y^2}{2} \right) \Big|_1^2 = 2.
\end{aligned}$$

Since f also satisfies the hypotheses of Fubini's Theorem with x and y interchanged, we can calculate the double integral from the iterated integral in which the integrations are performed in the opposite order.

Thus,

$$\begin{aligned}
\int_R (x+y)\,d(x,y) &= \int_0^1 dx \int_1^2 (x+y)\,dy \\
&= \int_0^1 \left[\left(xy + \frac{y^2}{2} \right) \Big|_{y=1}^{2} \right] dx \\
&= \int_0^1 \left(x + \frac{3}{2} \right) dx = \left(\frac{x^2}{2} + \frac{3x}{2} \right) \Big|_0^1 = 2.
\end{aligned}$$

Remark: If $\int_c^d dy \int_a^b f(x,y)\,dx$ exists then so does $\int_R f(x,y)\,d(x,y)$. However, this need not to be true.

Example: If f is defined on $R = [0,1] \times [0,1]$ by

$$f(x,y) = \begin{cases} 2xy & \text{if } y \text{ is rational,} \\ y & \text{if } y \text{ is irrational,} \end{cases}$$

then

$$\int_0^1 f(x,y)\,dx = y, \quad 0 \le y \le 1,$$

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and

$$\int_0^1 dy \int_0^1 f(x,y)\,dx = \int_0^1 y\,dy = \frac{1}{2}.$$

However, f is not integrable on R.

Theorem: Let I_1, I_2, ..., I_n be closed intervals and suppose that f is integrable on $R = I_1 \times I_2 \times \cdots \times I_n$.

Suppose that there is an integer p in $\{1, 2, \ldots, n-1\}$ such that

$$F_p(x_{p+1}, x_{p+2}, \ldots, x_n) = \int_{I_1 \times I_2 \times \cdots \times I_p} f(x_1, x_2, \ldots, x_n)\,d(x_1, x_2, \ldots, x_p)$$

exists for each $(x_{p+1}, x_{p+2}, \ldots, x_n)$ in $I_{p+1} \times I_{p+2} \times \cdots \times I_n$.

Then

$$\int_{I_{p+1} \times I_{p+2} \times \cdots \times I_n} F_p(x_{p+1}, x_{p+2}, \ldots, x_n)\,d(x_{p+1}, x_{p+2}, \ldots, x_n)$$

exists and equals $\int_R f(\mathbf{X})\,d\mathbf{X}$.

Theorem: Let $I_j = [a_j, b_j]$, $1 \le j \le n$, and suppose that f is integrable on $R = I_1 \times I_2 \times \cdots \times I_n$.

Suppose also that the integrals

$$F_p(x_{p+1}, \ldots, x_n) = \int_{I_1 \times I_2 \cdots \times I_p} f(\mathbf{X})\,d(x_1, x_2, \ldots, x_p), \quad 1 \le p \le n-1,$$

exist for all

$$(x_{p+1}, \ldots, x_n) \quad \text{in} \quad I_{p+1} \times \cdots \times I_n.$$

Then the iterated integral

$$\int_{a_n}^{b_n} dx_n \int_{a_{n-1}}^{b_{n-1}} dx_{n-1} \cdots \int_{a_2}^{b_2} dx_2 \int_{a_1}^{b_1} f(\mathbf{X})\,dx_1$$

exists and equals $\int_R f(\mathbf{X})\,d\mathbf{X}$.

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